



ISSN 2536-569X | eISSN 2536-5703

Journal of Anthropology of Sport and Physical Education

www.jaspe.ac.me



OCTOBER 2019

VOL.3
No.4



Journal of Anthropology of Sport and Physical Education

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Art Grafika | Nikšić

Print run

1500

MontenegroSport



**JOURNAL OF ANTHROPOLOGY OF SPORT
AND PHYSICAL EDUCATION**
International Scientific Journal

Vol. 3 (2019), No. 4 (1-71)

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ORIGINAL SCIENTIFIC PAPER

Biomechanical Model of Hurdle Clearance in 100m Hurdle Races: A Case Study

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Abstract

The purpose of the study was to identify and analyze the biomechanical structure of the hurdle clearance of the sixth hurdle in the 100-m hurdle race of Sally Pearson, the Olympic and world champion. An analysis of the hurdle clearance technique was carried out at the IAAF World Challenge - Zagreb 2011 international competition. 3-D kinematical analysis was utilized along with Ariel Performance Analysis System (APAS) for data processing. Digitization of the 15-segment model of the athlete's body, which was defined with 17 reference points, was performed. The results of the study show the extreme rationality of the hurdle clearance technique, which is demonstrated in the time interval between takeoff and landing (0.31 seconds), in the short contact times of the takeoff phase (0.10 seconds) before the hurdle and landing following the hurdle clearance (0.09 seconds), in the low flight parabola of the central TT (0.25 m), in the minimal decrease in horizontal velocity (1.5%), and in the efficient transition between the hurdle clearance and the sprint between hurdles. The obtained parameters can serve as orientational model values in the 100-m hurdle running technique training process for athletes.

Key words: Hurdle Race, Technique, Biomechanics, Kinematics, Case Study

Introduction

The biomechanical model of hurdle clearance for the 100-m hurdle race is based on the technique of the Australian athlete Sally Pearson, one of the world's greatest runners of 100-m hurdle events. The greatest achievement of her career thus far has been winning the gold medal in the 100-m hurdle race at the London Olympic Games in 2012. She holds many other top achievements such as the gold medal at the Daegu World Championship in 2011, the gold medal at the 2017 World Championships in London, the silver medal at the 2008 Beijing Olympics, and the silver medal at the 2013 World Championship in Moscow. Her personal record for the 100-m hurdle event is 12.28 seconds, which is the sixth fastest time in the history of the event. All these achievements rank Sally Pearson among the most elite athletes of modern athletics.

High hurdle races are among the most technically demanding athletic disciplines. According to previous studies (Schluter, 1981; Mero & Luhtanen, 1986; La Fortune, 1988;

Bruggemann & Glad, 1990; McDonald & Dapena, 1991; Dapena, 1991; McLean, 1994; Iskra, 1998; Kampmiller, Slamka, & Vanderka, 1999; Čoh, 2001; Blazevidh, 2013) the hurdle clearance technique is one of the key elements that determines a competitive result. From a biomechanics standpoint, the 100-m hurdle race combines the cyclic sprint and the acyclic clearance of 10 hurdles with a height of 0.838 m. The athlete must, therefore, have a high level of sprinting skills, exceptional hip joint mobility (flexibility), fast power, and a high level of technical knowledge. During the hurdle clearance, the loss of horizontal velocity must be kept to a minimum. This ability depends on a number of factors, especially those that define the takeoff before the hurdle, the trajectory of the movement of the CM (CM = center of mass) and the landing after the barrier (Kampmiller, Slamka, & Vanderka, 1999; Amritpal & Shamsher, 2015). In order to achieve rational hurdle clearance, the takeoff point before the hurdle and the landing point following the barrier are essential. The correct position of

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these two points determines the optimal flight trajectory of the CM, which is reflected in the time duration of the airborne phase, which should be as short as possible (Schluter, 1981; Dapena, 1991). In addition to the correct position of the points, the kinematic-dynamic structure of the takeoff and landing also directly influence the velocity of the hurdle clearance (La Fortune, 1988; McLean, 1994).

The aim of the study was to identify and analyze the biomechanical model of the hurdle clearance of the sixth hurdle in an athlete of the highest quality based on 3-D video analysis of kinematic parameters.

Methods

Biomechanical analysis of the hurdle clearance technique of the sixth hurdle by Sally Pearson (25 years old, body height 1.67 m, body weight 60 kg, P.R. 100 hurdles 12.28) was performed at the Mladost track-and-field stadium for the IAAF World Challenge international competition - Zagreb 2011, Croatia. The weather conditions were optimal; the outside temperature was 23 °C, the wind velocity was - 0.4 m.s.⁻¹. Permission to carry out biomechanical measurements was obtained from the Technical Delegate of the European Athletics Federation and the organizing committee of the competition. The lane in the zone of the sixth hurdle was covered by two high-frequency cameras CASIO-DIGITAL CAMERA EX-F1 (Casio Computer Co., Ltd.,

Tokyo, Japan), which were interconnected and synchronized. The cameras had a frequency of 300 Hz with a resolution of 720 x 576 pixels. The zone of the sixth hurdle was calibrated with a reference measuring frame of 2 m x 2 m x 2 m, and eight points of the calibration frame were considered for analysis. The APAS (Ariel Performance Analysis System) computer system for 3-D kinematic analysis was used for data processing. Digitization of the 15 - segment model of the athlete's body, which was defined using 17 reference points (Winter, 2005), was performed. Point coordinates were smoothed using a digital filter with a frequency of 14 Hz. Recording was performed at a frequency of 300 Hz, and point digitization was performed at 100 Hz. The center of mass (CM) was calculated based on anthropometric tables from the digitized points (Winter, 2005).

Results

According to the results of the biomechanical analysis (Table 1, Figure 1), the athlete had an average velocity of 8.58 m/s at the clearance of the sixth hurdle. The total stride length was 3.17 m. The takeoff distance was 2.31 m or 72.9% of the total step length over the barrier. The landing distance was 0.86 m from the hurdle, which represented 27.1% of the total step length over the barrier. The points of takeoff and landing determine an extremely rational flight parabola of the CM in relation to the hurdle position.

Table 1. Kinematic parameters of the sixth hurdle clearance (Sally Pearson, result of the 100-m hurdle race 12.68) - IAAF World Challenge, Zagreb, 2011.

PARAMETERS	UNIT	R
Rhythmic Units (Hurdle 5- 6)	m/s	8,58
TAKE – OFF (braking phase)		
Horizontal velocity of CM	m/s	8,76
Vertical velocity of CM	m/s	-0,46
Velocity resultant of CM	m/s	8,77
Height of CM	m	0,96
Take –off distance	m	2,31
TAKE – OFF (propulsion phase)		
Horizontal velocity of CM	m/s	8,51
Vertical velocity of CM	m/s	1,45
Velocity resultant of CM	m/s	8,63
Height of CM	m	1,03
Push-off angle	°	81,3
Contact time	s	0,10
FLIGHT		
Flight time	s	0,31
Height of TT above the hurdle	m	0,25
Maximal height CM	m	1,16
LANDING (breaking phase)		
Horizontal velocity of CM	m/s	8,53
Vertical velocity of CM	m/s	-0,93
Velocity resultant of CM	m/s	8,58
Height of CM	m	1,08
Landing distance	m	0,86
LANDING (propulsion phase)		
Horizontal velocity of CM	m/s	8,37
Vertical velocity of CM	m/s	-1,03
Velocity resultant of CM	m/s	8,38
Height of CM	m	1,05
Contact time	s	0,09

Note: m/s - meters per second; s - seconds; ° - degree

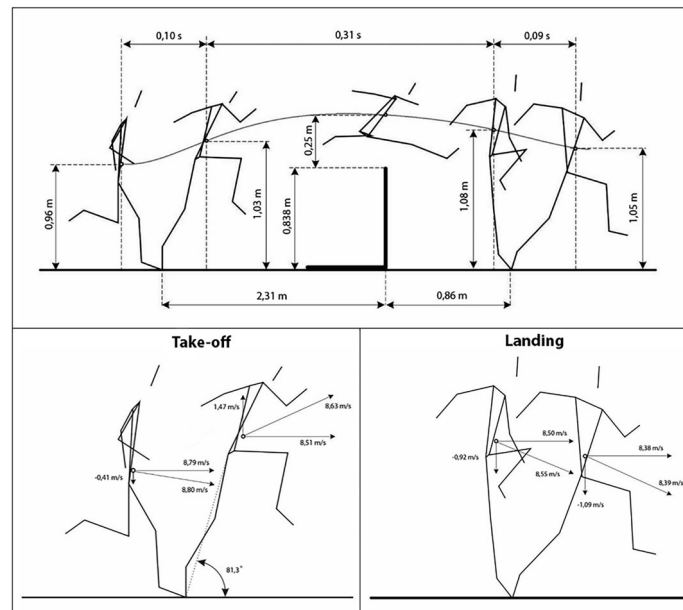


Figure 1. Biomechanical analysis of the sixth hurdle clearance (Sally Pearson, R: 12.68 s).

The highest point of the body's center of mass (CM) was 0.25 m during the hurdle step, while the highest point of the flight parabola was 1.16 m. The difference between the lowest TT point, in the eccentric phase of the takeoff (0.96 m), before the hurdle and the highest CM point, in the flight phase (1.16 m), was 0.20 m. The rationality of the hurdle clearance was also evident in the short flight time, which was 0.31 s.

The push-off angle in the concentric phase of takeoff was 81.3 degrees. The height of the CM at the braking phase was 0.96 m, and 1.03 m at the end of the propulsion phase of the push-off. The path length of CM in the Y - vertical axis was

0.07 m. These parameters provided optimal conditions for the development of velocity of the CM during takeoff.

The horizontal velocity of the TT at the preparatory step of the push-off was 8.79 m / s, and 8.51 m / s at the end of takeoff (Figure 2). The horizontal velocity of the CM was therefore decreased by 0.28 m / s (3.2%) during takeoff. An important parameter was also the vertical velocity of the CM during takeoff, which was 1.47 m / s. The horizontal and vertical velocities of the CM determine the propulsion phase resultant velocity at the hurdle, which was measured at 8.63 m / s and was directed at an angle of 9.6 degrees.

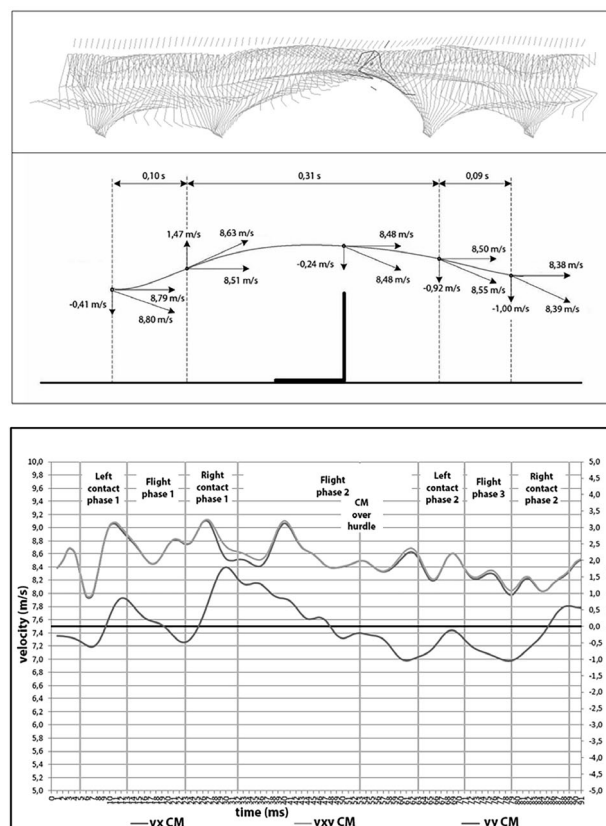


Figure 2. Analysis of the velocity of the central point of the body's center of gravity (TT) during the clearance of the sixth hurdle (Sally Pearson, R: 12.68 s).

Discussion

The transition from the hurdle clearance to the sprint between hurdles depends on the landing phase. In the case of Sally Pearson, the horizontal velocity of the CM after hurdle clearance was 8.38 m/s, which means that when the hurdle was cleared, the velocity was decreased by 0.13 m / s (1.5%). According to previous studies (Mero & Luhtanen, 1986), this represents a very small decrease in velocity at hurdle clearance and is one of the decisive technical factors that determines this elite athlete.

At the moment of initial contact, following the hurdle clearance, the height of the CM was 1.08 m. During the landing phase, the height of the CM was decreased by 0.03 m, and at the moment of push-off this height value was 1.05 m. The small decrease in the height of the CM (2.8%) during the landing phase and the short duration of this phase (0.09 s) indicates an efficient and rational transition into the sprint between hurdles. To maintain horizontal velocity after the hurdle clearance, the contact time of the landing is very crucial, and must last less than 100 milliseconds (La Fortune, 1988; McLean, 1994).

In addition to the correct landing technique following the hurdle clearance, the ability of the muscular system to resist rapid stretching or stiffness is important. Stiffness, as a neural mechanism of muscle activity, depends primarily on muscle preactivation and reflexes: the myotatic and Golgi tendon reflexes (Gollhofer & Kyrolainen, 1991). Depending on the biomechanical characteristics of the landing step, it is characterized by short-range elastic stiffness, where an immediate mechanical response of the activated muscle to eccentric contraction at landing takes place. The parameters mentioned above, in particular, the short contact time of the landing step (90 milliseconds) after hurdle clearance and the extremely small decrease in horizontal velocity after clearing the barrier, indicate an efficient and biomechanically rational hurdling technique.

The present study examined the rationality of the hurdle clearance technique of Sally Pearson, the Olympic winner of the 100-m hurdle race, by using modern diagnostic technology for biomechanical analysis. Technique analysis highlighted the small difference between the height of the CM and the height of the hurdle at the moment of hurdle clearance and the exceptionally small decrease in velocity at the transition from hurdle clearance to the sprint between barriers. The contact phases before and after the hurdle clearance were extremely short, indicating the athlete's high capacity for reactive power. This reactive power makes it possible for the competitor to maintain a high sprint velocity between hurdles, which contributes to her top achievements in running 100-meter hurdle races and has established her as one of the best competitors in this discipline in the history of track-and-field.

Acknowledgements

Pictorial and graphics software is the copyrighted work of Stanko Štuhec.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 7 June 2019 | **Accepted:** 4 July 2019 | **Published:** 11 October 2019

References

- Amritpal, S., & Shamsheer, S. (2015). Relationship Among the Technique of Hurdle Clearance Over the Different Hurdles in 110m Race. *International Journal of Science and Research*, 4(10), 1591-1594.
- Blazevich, A. (2013). *Sports biomechanics: the basics: Optimising human performance*. London: A&C Black Publishers Ltd.
- Bruggemann, G., & Glad B. (1990). Time analysis of the sprint events. In Bruggemann, G., & Glad, B. (Eds.) *IAAF Scientific Research Project at the Games of the XXXIV Olympiad Seoul 1998: Final report* (10-45). Florence: Arti Grafiche Danesi.
- Čoh, M. (2001). *Biomehanika atletike*. Ljubljana: Fakulteta za šport.
- Dapena, J. (1991). Hurdle clearance technique. *Track and Field. Quarterly Review*, 116(3), 710-712.
- Gollhofer, A., & Kyroläinen, H. (1991). Neuromuscular Control of the Human Leg Extensor Muscles in Jump Exercises Under Various Stretch-Load Conditions. *International Journal of Sports Medicine*, 12(01), 34-40. doi:10.1055/s-2007-1024652
- Grimshaw, P. N., Marar, L., Salo, A., Knight, G., & Vernon, J. (1995). A kinematic analysis of sprint hurdles training strategies (isolation drills). *Athletics Coach*, 29(4), 24-28.
- Iskra, J. (1998). *Bieg przez plotki – Teoretyczne podstawy i praktyczne rozwiązania treningowe*. Katowice: Akademii Wychowania Fizycznego.
- Jarver, J. (1997). *The hurdles: Contemporary theory, technique and training*. Mountain View, California: Tafnews Press.
- Kampmiller, T., Slamka, M., & Vanderka, M. (1991). Comparative biomechanical analysis of 110 m hurdles of Igor Kovač and Peter Nedelicky. *Kinesiology Slovenica*, 5(1-2), 26-30.
- La Fortune, M. (1988). Biomechanical analysis of 110 m hurdles. *Track and Field News*, 105, 3355-3365.
- McFarlane B. (1994). Hurdles: a basic and advanced technical model. *Track Technique*, 128(1), 4073-4079.
- McDonald, C., & Dapena, J. (1991). Linear kinematics of the men's and woman's hurdles races. *Medicine and Science in Sports Exercise*, 23(12), 1382-1402.
- McLean, B. (1994). The biomechanics of hurdling: Force plate analysis to assess hurdling technique. *New Studies in Athletics*, 4, 55-58.
- Mero, A., & Luhtanen, P. (1986). Biomechanische Untersuchung des Hurdlenlaufs während der Weltmeisterschaften in Helsinki. *Leistungssport*, 1, 42-43.
- Salo, A., & Grimshaw, P. N. (1998). An Examination of Kinematic Variability of Motion Analysis in Sprint Hurdles. *Journal of Applied Biomechanics*, 14(2), 211-222. doi:10.1123/jab.14.2.211
- Salo, A.I.T., Grimshaw, P.N., & Marar, L. (1997). 3-D biomechanical analysis of sprint hurdles at different competitive levels. *Medicine & Science in Sports & Exercise*, 29(2), 231-237.
- Schluter, W. (1981). Kinematische Merkmale der 110-m Hurdentechnik. *Leistungssport*, 2, 118-127.
- Sidbu, A.S., & Bahadur, S.S.J. (2015). Relationship among the Technique of Hurdle Clearance Over the Different Hurdles in 110m Race. *International Journal of Science and Research*, 4(10), 1591-1594.
- Winter, D. (2005). *Biomechanics and Motor Control of Human Movement*. Hoboken, New Jersey: John Wiley and Sons INC.

ORIGINAL SCIENTIFIC PAPER

The Impact of Physical Activities on Social Inclusion of Elderly People in Montenegro

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Abstract

The purpose of this study was to determine whether there are differences in the level of social inclusion between respondents who are not at all engaged in sports activities and respondents who engage in sports and recreational activities once or twice a week. The sample included 60 subjects 55 to 70 years old divided into two subsamples, 43 respondents who are not at all involved in sports activities and 17 respondents who practice sports and recreational activities once or twice a week. Data were collected using the standardized Social Inclusion Scale questionnaire which respondents are voluntarily applied. To determine the significance of differences in the degree of social involvement between groups, was used t-test at the level of significance of $p < 0.05$. It was concluded, based on these results, that significant difference is not found neither for one pairs in the system of comparable parameters.

Key words: *Inclusion in Society, Sports Activity, Social Inclusion Scale, Elderly People*

Introduction

U kojem god dijelu svijeta da se se zateknemo, primijetimo da ni u jednom društvu ne vlada potpuna ravnopravnost, premda se teži tom idealu koji se temelji na poboljšanju i prilagođavanju. Da bi se omogućilo ostvarivanje slobode, jednakosti, bratstva, preduzima se sve što je moguće da svi članovi društva budu podjednako uključeni u funkcionisanje društvene zajednice. To jest, teži se stvaranju mogućnosti i kapaciteta da svi potpuno i produktivno učestvuju u ekonomskom, društvenom i kulturnom životu, te da svi uživaju životni standard, odnosno blagostanje koje se smatra normalnim u razvijenim društvenim zajednicama (Commission of the European Communities, 2000). Međutim, smatra se da oko 11% odrasle populacije nije dovoljno integrisano u sisteme funkcionisanja društvene zajednice, tj. da nije socijalno uključeno (Jehoel-Gijsbers & Vrooman, 2007), što cjelokupnu društvenu zajednicu unazađuje.

Nekad su razlozi nedovoljne društvene uključenosti nezaposlenost, neosposobljenost, niski prihodi, neadekvatni uslovi života, slabije zdravstveno stanje, visoka stopa kriminala i narušeni odnosi u porodici (Social Exclusion Unit, 2001).

S druge strane, Davidson, Stayner, Nickou, Styron, & Chinman (2001) smatraju da je društveno uključivanje povezano sa prijateljstvom, osjećajem lične vrijednosti i optimizmom, što nas navodi na zaključak da čovjek ne može biti smatran društveno uključenim ukoliko u određenom stepenu nije sposoban ostvariti iskustvo prijateljstva, obavljati korisne i smislene aktivnosti u društvu i nadati se još boljoj budućnosti.

Kako bi se ovaj problem riješio neophodno je sprovesti mjere sprječavanja, ali takođe je potrebno djelovati kako bi se pomoglo onima koji su već isključeni, jer stvaranje puta natrag u društvo osnova je novog pristupa (Social Exclusion Unit, 2001). Aktivacija ljudi u bilo koji vid aktivnosti i saradnje omogućava oporavak od mentalnih problema, jača mentalno zdravlje i povećava mentalnu čvrstinu (Wilson & Seckter, 2015). Potencijal sporta za poboljšanje svih ovih dimenzija prema rezultatima prethodnih studija je veliki (Biddle & Asare, 2011; Mašanović, Popović, Bjelica, Vukotić, & Zorić, 2018; Masanovic, Popovic, & Vukotic, 2019; Popovic, Bjelica, Masanovic, & Gardasevic, 2019), pa ga zbog toga moramo iskoristiti. Aktivna osoba ima manje mogućnosti da misli o negativnim aspektima života pa ne može biti društveno isključena, jer

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se pojedinac ne može smatrati društveno isključenim ako se ne osjeća društveno isključenim (Onken, Craig, Ridgway, Ralph, & Cook, 2007; Pinfold, 2000).

Cilj ovog istraživanja bio je da se utvrdi postoje li razlike u stepenu društvene uključenosti između ispitanika koji se uopšte ne bave sportskim aktivnostima i ispitanika koji se sportsko-rekreativnim aktivnostima bave jednom ili dva puta nedjeljno.

Metod

Ovim istraživanjem obuhvaćeno je 60 ispitanika starosne dobi između 55 i 70 godina, podijeljenih na dva subuzorka. Prvi subuzorak čine 43 ispitanika koji se uopšte ne bave sportskim aktivnostima, dok drugi subuzorak čine 17 ispitanika koji se sportsko-rekreativnim aktivnostima bave jednom ili dva puta nedjeljno. Kriterijum za učestvovanje u eksperimentu bio je da ispitanici nemaju zdravstvenih problema.

Eksperimentalni program vježbanja bio je prilagođen godinama i sposobnostima ispitanika. Step en društvene uključenosti je utvrđen pomoću ankete u okviru koje je korišćen standardni upitnik pod nazivom Social Inclusion Scale koji se sastoji od 17 pitanja podijeljenih u tri poglavlja, odnosno čija skala ima tri podskale koje mjere društvenu izolaciju, društvene odnose i društveno prihvatanje (Wilson & Secker, 2015). Skala se sastoji od izjava u kojima ispitanici biraju jednu od opcija na Likertovoj skali (1: "Ne uopšte", 2: "Ne naročito", 3: "Da, malo" i 4: "Da, definitivno") koja najbolje opisuje njihov odnos sa drugim ljudima u posljednjih mjesec dana.

Popunjavanje upitnika trajalo je u prosjeku 10 minuta, a ispitanici su u istraživanju učestvovali dobrovoljno. Prije anke-

tiranja ispitanici su potpisali saglasnost koja potvrđuje da su u istraživanju učestvovali dobrovoljno. Anketa je bila anonimna, njen sadržaj je bio strogo povjerljiv, a ispitanici su bili u mogućnosti da u svakom momentu opozovu svoje učešće na što se niko od njih nije odlučio.

Empirijski podaci su analizirani putem statističkog paketa za društvene nauke SPSS 20.0 (Chicago, IL, USA). Za svaku varijablu prikazani su aritmetička sredina i standardna devijacija, a za određivanje značajnosti razlika u stepenu društvene uključenosti između grupa korišten je t-test za male nezavisne uzorke, sa statističkom značajnošću od $p < 0.05$.

Rezultati

U ovom dijelu rada prikazani su osnovni deskriptivni statistički parametri za obje grupe ispitanika, kao i rezultati t-testa za male nezavisne uzorke, raspoređeni. Posmatrajući prosječne vrijednosti rezultata dobijenih pomoću Skale društvene uključenosti (Tabela 1), u prosjeku pozitivne odgovore vidimo kod 14 od ukupno 17 pitanja kada je riječ o grupi ispitanika koji se uopšte ne bave sportskim aktivnostima. Kod grupe ispitanika koji se sportsko-rekreativnim aktivnostima bave jednom nedjeljno ili dva puta nedjeljno situacija je identična, pa u prosjeku pozitivne odgovore vidimo takođe kod 14 od ukupno 17 pitanja. Takođe, možemo primijetiti da grupa ispitanika koji se uopšte ne bave sportskim aktivnostima ima u prosjeku pozitivnije odgovore na pet od ukupno 17 pitanja. Grupa ispitanika koji se sportsko-rekreativnim aktivnostima bave jednom ili dva puta nedjeljno u prosjeku pozitivnije odgovore dala je na 12 pitanja od ukupno 17.

Tabela 1. Deskriptivni statistički parametri za procjenu stepena društvene uključenosti obje grupe ispitanika (neaktivnih i aktivnih)

Podskala	Pitanje	Neaktivni Mean±SD	Aktivni Mean±SD
Društvena izolacija	Osjećam se užasno usamljeno i izolovano	1.26±0.49	1.12±0.33
	Osjećam se prihvaćenim od strane mojih prijatelja	3.74±0.62	3.76±0.44
	Družim se sa prijateljima	3.72±0.50	3.65±0.61
	Imam osjećaj da igram korisnu ulogu u društvu*	3.40±0.62	3.59±0.51
	Imam prijatelje sa kojima se gledam ili pričam svake nedjelje*	3.84±0.43	3.88±0.33
Društveni odnosi	Imam osjećaj da igram korisnu ulogu u društvu*	3.53±0.63	3.47±0.51
	Osjećam da ono čime se bavim drugi cijene	3.56±0.63	3.47±0.62
	Bio/la sam na novim mjestima	3.21±0.86	3.41±0.94
	Naučio/la sam nešto o drugim kulturama	3.00±0.79	3.29±0.92
	Bavio/la sam se nekim kulturološkim aktivnostima	2.47±0.88	2.71±0.99
	Osjećam da me ljudi podozrivo gledaju zbog toga kakav/kakva sam	1.65±0.95	1.29±0.47
	Osjećam se nesigurno da šetam sam/a u mom komšiluku po danu	1.47±0.91	1.53±1.07
Društveno prihvatanje	Imam prijatelje sa kojima se gledam ili pričam svake nedjelje *	3.86±0.35	3.88±0.33
	Osjećam se prihvaćeno od strane mojih komšija	3.72±0.55	3.76±0.44
	Osjećam se prihvaćeno od strane moje porodice	3.98±0.15	4.00±0.00
	Sve što se tiče mojih prava mi je jasno	3.79±0.56	3.88±0.33
	Imam slobodu da izrazim moja uvjerenja	3.81±0.45	3.94±0.24

Legenda: Mean – Aritmetička sredina; SD – Standardna devijacija; * - Stavke u više od jedne kolone

Na osnovu rezultata prikazanih u Tabeli 2 jasno se vidi da niti kod jednog para u sistemu upoređivanih parametara ne postoji statistički značajna razlika, tj. da između ispitanika koji

se uopšte ne bave sportskim aktivnostima i ispitanika koji se sportsko-rekreativnim aktivnostima bave jednom ili dva puta nedjeljno nema značajne razlike.

Tabela 2. Razlike u stepenu društvene uključenosti između neaktivnih i aktivnih ispitanika

Podskala	Pitanja	Aktivnost	Mean	SD	t	Sig
Društvena izolacija	Osjećam se užasno usamljeno i izolovano	Neaktivni	1.26	.492	1.1	.290
		Aktivni	1.12	.332		
	Osjećam se prihvaćenim od strane mojih prijatelja	Neaktivni	3.74	.621	-0.1	.900
		Aktivni	3.76	.437		
	Družim se sa prijateljima	Neaktivni	3.72	.504	0.5	.630
		Aktivni	3.65	.606		
	Imam osjećaj da igram korisnu ulogu u društvu*	Neaktivni	3.40	.623	-1.1	.260
		Aktivni	3.59	.507		
Društveni odnosi	Imam osjećaj da igram korisnu ulogu u društvu*	Neaktivni	3.84	.433	-0.4	.700
		Aktivni	3.88	.332		
	Osjećam da ono čime se bavim drugi cijene	Neaktivni	3.53	.631	0.4	.710
		Aktivni	3.47	.514		
	Bio/la sam na novim mjestima	Neaktivni	3.56	.629	0.5	.630
		Aktivni	3.47	.624		
	Naučio/la sam nešto o drugim kulturama	Neaktivni	3.21	.861	-0.8	.430
		Aktivni	3.41	.939		
	Bavio/la sam se nekim kulturološkim aktivnostima	Neaktivni	3.00	.787	-1.2	.220
		Aktivni	3.29	.920		
	Osjećam da me ljudi podozriivo gledaju zbog toga kakav/kakva sam	Neaktivni	2.47	.882	-0.9	.360
		Aktivni	2.71	.985		
	Osjećam se nesigurno da šetam sam/a u mom komšiluku po danu	Neaktivni	1.65	.948	1.5	.150
		Aktivni	1.29	.470		
		Neaktivni	1.47	.909	-0.2	.820
		Aktivni	1.53	1.068		
Društveno prihvatanje	Imam prijatelje sa kojima se gledam ili pričam svake nedjelje *	Neaktivni	3.86	.351	-0.2	.830
		Aktivni	3.88	.332		
	Osjećam se prihvaćeno od strane mojih komšija	Neaktivni	3.72	.549	-0.3	.770
		Aktivni	3.76	.437		
	Osjećam se prihvaćeno od strane moje porodice	Neaktivni	3.98	.152	-0.6	.530
		Aktivni	4.00	.000		
	Sve što se tiče mojih prava mi je jasno	Neaktivni	3.79	.559	-0.6	.530
		Aktivni	3.88	.332		
	Imam slobodu da izrazim moja uvjerenja	Neaktivni	3.81	.450	-1.1	.280
		Aktivni	3.94	.243		

Legenda: Mean – Aritmetička sredina; SD – Standardna devijacija; t – Vrijednost testiranja razlike varijansi; Sig – Značajnost razlike aritmetičkih sredina; * - Stavke u više od jedne kolone

Diskusija

Rezultati ovog istraživanja pokazuju da fizička aktivnost pomenutog obima nije u pozitivnoj korelaciji sa stepenom društvene uključenosti Crnogoraca starije životne dobi. Kada su u pitanju dosadašnja istraživanja, rezultati su dosta raznovrsni. Mogu se pronaći studije čiji su rezultati slični rezultatima ove studije (Masanovic, 2019), nadalje, mogu se pronaći istraživanja koja negiraju statistički značajan sveukupan uticaj sporta na društvenu uključenost, ali govore da su neki aspekti društvene uključenosti ipak pozitivniji poslije učešća u organizovanim sportsko-rekreativnim aktivnostima (Masanovic, 2019a). Takođe, mnogo je istraživanja koja jasno ukazuju na pozitivan uticaj aktivnosti kao što su sport i rekreacija na društvenu uključenost (Bailey, 2005; Skinner, Zakus, & Cowell, 2008; Kelly, 2011; Bjelica, Popovic, Masanovic, & Gardasevic, 2019; Popovic & Masanovic, 2019), ali uglavnom ne pružaju tačne podatke o tome kolika je statistička značajnost uticaja sporta, pa se njihovi rezultati moraju prihvatiti sa dozom rezerve. Na kraju, interesantno je pomenuti i grupu istraživanja koja ukazuju na to da svaka organizovana aktivnost

koja eliminiše usamljenost, pozitivno utiče na društveno uključivanje pojedinca (Perkins, 2010), bez obzira da li je u pitanju sport ili bilo koje drugo društveno angažovanje. Većina stručnjaka iz oblasti sportskih nauka daće ipak prednost sportu, jer se njegova prednost ogleda u tome što osim podizanja društvene uključenosti učesnici dobijaju i pozitivan uticaj na zdravlje i kvalitet života, što bi opet značilo da su jednim potezom neutralisana dva problema (personal communication, S. Popovic). Nedorečenosti i suprotnosti koje su evidentne u prethodnim studijama upućuju na potrebu da se ova oblast dodatno ispita.

Prvo ograničenje ovog istraživanja je nedovoljno obiman uzorak ispitanika koji ne dozvoljava generalizaciju zaključka, stoga bi sljedeća studija trebalo da bude sprovedena na većem uzorku ispitanika. Sljedeće ograničenje je nedovoljan obim fizičke aktivnosti koji je utvrđen kod aktivne grupe jer je vrlo vjerovatno da jedan ili dva časa fizičke aktivnosti čiji intenzitet nije ispitan ne mogu napraviti adekvatnu razliku između grupa, pa bi u sljedećoj studiji trebalo ispitati populaciju koja se svakodnevno bavi sportsko-rekreativnim aktivnostima. Ovakav

pristup bi doprinio kvalitetu dobijenih rezultata, i pružio bi reprezentativnije podatke na osnovu kojih bi se stekao tačan uvid u uticaj fizičke aktivnosti na društveno uključivanje osoba starije životne dobi. Sve ovo ipak, ne umanjuje doprinos ove preliminarne studije.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 10 August 2019 | **Accepted:** 10 September 2019 | **Published:** 11 October 2019

References

- Bailey, R. (2005). Evaluating the relationship between physical education, sport and social inclusion. *Educational Review*, 57(1), 71-90. doi: 10.1080/0013191042000274196
- Biddle, S.J.H., & Asare, M. (2011). Physical activity and mental health in children and adolescents: a review of reviews. *British Journal of Sports Medicine*, 45(11), 886-895. doi: 10.1136/bjsports-2011-090185
- Bjelica, D., Popovic, S., Masanovic, B., & Gardasevic, J. (2019). Impact of professional sports engagement on the extension of life: a systematic review. In *BMC Sports Science, Medicine and Rehabilitation Supplement of 5th International Scientific Conference on Exercise and Quality of Life*, 11(P9, Suppl 1), 18, Novi Sad: Faculty of Sport and Physical Education, University of Novi; doi: 10.1186/s13102-019-0119-7.
- Commission of the European Communities. (2000). *Social Policy Agenda: Communication from the Commission to the Council, the European Parliament, the Economic and Social Committee and the Committee of the Regions*. Brussels: Commission of the European Communities.
- Davidson, L., Stayner, D., Nickou, T., Styron, M., & Chinman, M. (2001). Simply to be let in: Inclusion as a basis for recovery. *Psychiatric Rehabilitation Journal*, 24(4), 375-388. doi: 10.1037/h0095067
- Jehoel-Gijsbers, G., & Vrooman, C. (2007). *Explaining Social Exclusion*. Hague: The Netherlands Institute for Social Research/SCP.
- Kelly, L. (2011). Social inclusion through sports-based interventions? *Critical Social Policy*, 31(1), 126-150. doi: 10.1177/0261018310385442
- Masanovic, B. (2019). The effects of sports-recreational activities on the inclusion of young montenegrians in society. *Journal of Anthropology of Sport and Physical Education*, 3(3), 21-24. doi: 10.26773/jaspe.190704
- Masanovic, B. (2019a). Impact of Physical Exercise Programs and Programs of Social Activity on Public Health and Social Inclusion of Young People. *Iranian Journal of Public Health*, 48(6), 1180-1181.
- Masanovic, B., Popovic, S., & Vukotic, M. (2019). Working ability factors as a protective mechanism against social exclusion of older people: A systematic review. In *Book of Abstracts of the 16th Annual Scientific Conference of Montenegrin Sports Academy "Sport, Physical Activity and Health: Contemporary Perspectives"* (95), Cavtat: Montenegrin Sports Academy.
- Masanovic, B., Popovic, S., Bjelica, D., Vukotic, M., & Zoric, G. (2018). The effects of physical activity on depressive symptoms among elderly people: a systematic review. In *Proceedings World Congress of Performance Analysis of Sport XII* (423). Opatija: International Society of Performance Analysis of Sport.
- Onken, S., Craig, C., Ridgway, P., Ralph, R., & Cook, J. (2007). An analysis of the definitions and elements of recovery: A review of the literature. *Psychiatric Rehabilitation Journal*, 31(1), 9-22. doi: 10.2975/31.1.2007.9.22
- Perkins, D. (2010). Activation and social inclusion: challenges and possibilities. *Australian Journal of Social Issues*, 45(2), 267-287. doi: 10.1002/j.1839-4655.2010.tb00178.x
- Pinfold, V. (2000). "Building up safe havens...all around the world": Users' experiences of living in the community with mental health problems. *Health and Place*, 6(3), 201-212. doi: 10.1016/S1353-8292(00)00023-X
- Popovic, S., & Masanovic, B. (2019). Effects of Physical Activity and Social Activity on Physical Health and Social Inclusion of Elderly People. *Iranian Journal of Public Health*, 48(10).
- Popovic, S., Bjelica, D., Masanovic, B., & Gardasevic, J. (2019). Exercise for cognitive health as a proxy for the social inclusion of older people: a systematic review. In *BMC Sports Science, Medicine and Rehabilitation Supplement of 5th International Scientific Conference on Exercise and Quality of Life*, 11(P8, Suppl 1), 17-18, Novi Sad: Faculty of Sport and Physical Education, University of Novi; doi: 10.1186/s13102-019-0119-7.
- Skinner, J., Zakus, D. H., & Cowell, J. (2008). Development through Sport: Building Social Capital in Disadvantaged Communities. *Sport Management Review*, 11(3), 253-275. doi: 10.1016/s1441-3523(08)70112-8
- Social Exclusion Unit. (2001). *Preventing social exclusion*. London: Office of the Deputy Prime Minister.
- Wilson, C., & Secker, J. (2015). Validation of the Social Inclusion Scale with Students. *Social Inclusion*, 3(4), 52-62. doi: http://dx.doi.org/10.17645/si.v3i4.121

ORIGINAL SCIENTIFIC PAPER

The Effects of Preparation Period on Motor Skills of Wheelchair Basketball Players: A Pilot Study

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Abstract

The aim of this research is the determination of the preparation period effects on motor skills of wheelchair basketball players. The sample of six wheelchair basketball players in Serbia, aged 20 to 47, with at least five years of sports experience at the national level. The following tests of motor skills were carried out: Maximal pass, Hand dynamometry (for both hands), 5 m sprint test, 20 m sprint test, 5 m sprint with the ball, 20m sprint with the ball, Slalom without a ball and Slam with a ball. To determine the difference in the motor skills tests between the initial and the final measurement, the Student's t-test for small dependent samples was used. Experimental treatment includes the preparation period of the Serbian national team in wheelchair basketball in the C Division European Championship. An experimental treatment of 15 days duration containing 20 training sessions (2 h duration each) was insufficient for the statistically significant transformation motor skills of wheelchair basketball players. Missing of this research is the small number of examinees and the inhomogeneity of the examined sample. Future studies should estimate the impact of similar training on the motor skills of wheelchair basketball players, on a larger sample and in the function of injuries, impairment and functional classes of players.

Key words: *Preparation Period, Wheelchair Basketball, Motor Skills*

Uvod

Košarka u kolicima je sport koji se igra još od četrdesetih godina prošlog vijeka (Kasum, 2015). Ser Ludvig Gutman je najzaslužniji za razvoj košarke u kolicima. Kao ljekar radio je u bolnici u Ajlsberiju u Stouk Mandevilu (DePauw & Gavron, 2005), i tamo adaptirao košarku da je mogu igrati osobe koje su u ratu zadobile povrede kičmene moždine (Silver, 2012). Na taj način se ubrzavala rehabilitacija – fizička i psihička (Goosey-Tolfrey, 2010). Košarka u kolicima je paraolimpijski sport od prvih igara koje su održane 1960. godine u Rimu (Milenković & Živanović, 2010).

Osnovna pravila košarke u kolicima veoma su slična klasičnoj košarci (visina koša, veličina terena, linija slobodnih bacanja, linija za tri poena su jednakih dimenzija). Igraju dva tima sa po pet igrača na terenu. Vrijeme trajanja utakmice je četiri perioda po deset minuta (International Wheelchair Basketball Federation, 2017). Jedina razlika je u tome što igrači, iz razumljivih razloga, sjede u kolicima.

Da bi dvije ekipe košarku u kolicima mogle ravnopravno da igraju, uvedena je klasifikacija. Klasifikacija je proces kojim se ukupni timski potencijal na terenu izjednačava sa timskim potencijalom protivnika. Tu se na osnovu funkcionalne sposobnosti igrača, dodeljuje bodovna vrijednost svakom igraču posebno. Međunarodni raspon za klasifikaciju je od 1,0 do 4,5 bodova, a za petorku na terenu 14 bodova (IWBF Player Classification Commission, 2014).

Danas se košarka u kolicima igra na svim kontinentima, a najveća takmičenja su Paraolimpijske igre, svjetska i kontinentalna prvenstva. U Evropi postoji tri ranga takmičenja između 34 nacionalne selekcije (A, B i C divizija). U svakom rangi se takmiči po 10 do 12 selekcija, po dvije ispadaju u niži rang i napreduju u viši (International Wheelchair Basketball Federation, 2018).

Košarka u kolicima je aerobno-anaerobni sport, intermitentnog karaktera, koji zahtijeva od igrača visok nivo fizičke i tehničke sprema (Bloxhamet et al., 2001). Ovaj sport zahtijeva

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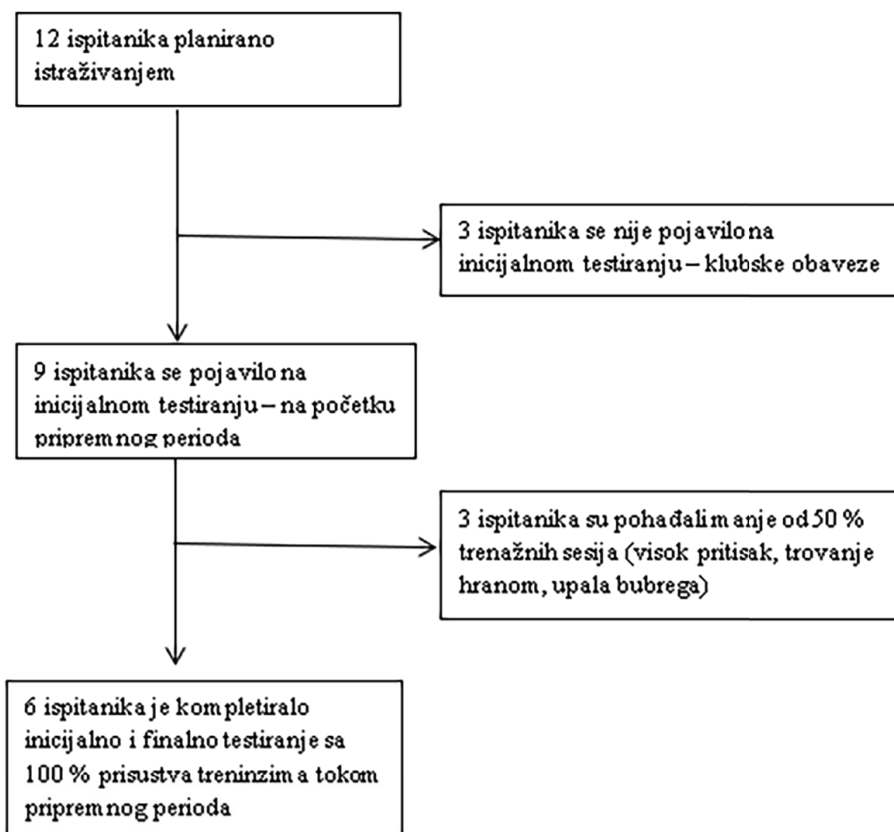
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trening visokog intenziteta, za razliku od drugih sportova u kolicima (Croft et al., 2010). Cilj ovog istraživanja je utvrditi efekte pripremnog perioda na motoriku košarkaša u kolicima.

Metod

Istraživanjem je planiran uzorak ispitanika od 12 košarkaša u kolicima Srbije, uzrasta od 20 do 47 godina, sa najmanje

pet godina sportskog staža na nacionalnom nivou (Grafik 1.). Na inicijalnom testiranju se pojavilo devet ispitanika, jer su tri igrača imala profesionalne obaveze prema svom klubu. Tokom pripremnog perioda (eksperimentalnog tretmana) se razbolio jedan ispitanik (upala bubrega). Na finalnom testiranju su se pojavili šest ispitanika (koja su testirani na inicijalnom mjerenju) i neizostavno prisustvovali svakoj trenažnoj sesiji tokom pripremnog perioda (eksperimentalnog tretmana).



Grafikon 1. Dizajn i protok ispitanika tokom studije.

Prije testiranja su ispitanici anketirani za dobijanje informacija o: sportskom stažu, klasi, povredi, invaliditetu, učestalosti treninga. Isto tako, prije testiranja antropometrijskih karakteristika i motoričkih sposobnosti testirani su puls u miru i krvni pritisak. Samo oni ispitanici koji su imali zadovoljavajuće vrijednosti pulsa u miru i srčane tenzije, mogli su pristupiti mjerenjima.

Eksperimentalni tretman je obuhvatao pripreme reprezentacije Srbije u košarci u kolicima za Evropsko prvenstvo C divizije koje je održano od 25.07. do 28.07.2017. godine u Brnu. Pripremni period je trajao od 01.07. do 15.07.2017. godine u Kragujevcu. U tom periodu je održano 20 trenažnih sesija. Svaka trenažna sesija je trajala 120 minuta i poštovali su se svi principi strukture treninga. Svaka sesija je dominantno sadržala trenažna sredstva tehnike i taktike, sa visokom intenzifikacijom za svakog učesnika u treningu. Za procjenu antropometrijskih karakteristika ispitanika su izmjereni sljedeći parametri (Đurašković, 2001): sjedeća visina, tjelesna masa i starost. Za procjenu motorike ispitanika su izmjereni sljedeći parametri

(Gil et al., 2015): maksimalan pas, dinamometrija lijeve ruke, dinamometrija desne ruke, vrijeme na 5 m, vrijeme na 20 m, vrijeme na 5 m sa loptom, vrijeme na 20 m sa loptom, slalom bez lopte i slalom sa loptom.

Za svaku varijablu su utvrđeni osnovni statistički parametri. Za utvrđivanje razlike u testovima motoričkih sposobnosti između inicijalnog i finalnog mjerenja korišćen je Studentov t-test za male zavisne uzroke.

Rezultati

U Tabeli 1. su prikazani osnovni deskriptivni parametri uzorka ispitanika košarkaša u kolicima. Vrijednosti sjedeće visine, relevantan faktor longitudinalne dimenzionalnosti za košarkaše u kolicima, pokazuje da je uzorak ispitanika homogen, tj. ujednačen (od 91.00 cm do 100.50 cm). To se ne može reći za vrijednosti tjelesne mase, jer je veliki raspon rezultata (od 65.00 cm do 88.00 cm). Uočljiva je i nehomogenost uzorka u godinama starosti ispitanika, jer je raspon između najstarijeg i najmlađeg igrača čak 27 godina (od 20 do 47 godina).

Tabela 1. Osnovni deskriptivni parametri uzorka ispitanika

	MEAN	MIN	MAX	RANGE	SD
Visina tijela	94.53	91.00	100.50	9.50	3.26
Tjelesna masa	75.83	65.00	88.00	23.00	8.89
Godine starosti	35.50	20.00	47.00	27.00	9.50

Legenda: MEAN - aritmetička sredina, MIN - minimalna vrijednost, MAX - maksimalna vrijednost, RANGE - raspon, SD - standardna devijacija

U Tabeli 2. su prikazani neki pokazatelji deskriptivne statistike varijabli motorike na inicijalnom i finalnom mjerenju (MEAN, SD), kao i razlike između pomenuta dva mjerenja izračunate pomoću Studentovog t-testa za male zavisne uzorke. Testovi snage ruku pokazuju da postoji diskretan napredak u numeričkim pokazateljima vrijednosti testova koji predstavlja ovu dimenziju. Međutim, nije ustanovljena statistički značajna razlika između mjerenja kod varijabli maksimalan pas

($p=0.17$), dinamometrija lijeve ruke ($p=0.39$) i dinamometrija desne ruke ($p=0.24$). Diskretan, ali ne i statistički značajan napredak je uočen kod tri varijable motorike: vrijeme na 20 m sa loptom ($p=0.06$), slalom sa loptom ($p=0.39$) i slalom bez lopte ($p=0.71$). Isto tako, kod ostale tri varijable motorike je uočeno diskretno pogoršavanje, ali statistički nije značajno: vrijeme na 5 m ($p=0.57$), vrijeme na 20 m ($p=0.32$) i vrijeme na 5 m sa loptom ($p=0.13$).

Tabela 2. Rezultati studentovog t-testa za male zavisne razlike između inicijalnog i finalnog mjerenja

VARIABLES	MEAN	SD	MEAN	SD	t	P
Maksimalni pas (I)	11.50	1.89	-0.26	0.41	-1.59	0.17
Maksimalni pas (F)	11.76	1.94				
Dinamometrija lijeve ruke (I)	41.33	8.52	-1.33	3.44	-0.95	0.39
Dinamometrija lijeve ruke (F)	42.67	7.97				
Dinamometrija desne ruke (I)	44.00	9.65	-2.17	3.97	-1.34	0.24
Dinamometrija desne ruke (F)	46.17	10.82				
Vrijeme na 5 m (I)	1.76	0.09	-0.04	0.17	-0.61	0.57
Vrijeme na 5 m (F)	1.78	0.13				
Vrijeme na 20 m (I)	5.33	0.28	-0.09	0.21	-1.09	0.32
Vrijeme na 20 m (F)	5.42	0.18				
Vrijeme na 5 m sa loptom (I)	1.79	0.13	-0.07	0.092	-1.81	0.13
Vrijeme na 5 m sa loptom (F)	1.86	0.06				
Vrijeme na 20 m sa loptom (I)	6.22	0.26	0.29	0.28	2.47	0.06
Vrijeme na 20 m sa loptom (F)	5.94	0.13				
Slalom bez lopte (I)	9.47	0.61	0.08	0.48	0.39	0.71
Slalom bez lopte (F)	9.40	0.30				
Slalom sa loptom (I)	11.22	1.10	0.39	1.01	0.94	0.39
Slalom sa loptom (F)	10.84	0.58				

Legenda: (I) - inicijalno mjerenje, (F) - finalno mjerenje, MEAN - aritmetička sredina, SD - standardna devijacija, t - vrijednost Studentovog t-testa, P - značajnost.

Diskusija

Iako je projektom predviđen raspon u godinama starosti košarkaša u kolicima od 18 do 40 godina, zbog malog broja ispitanika uvršten je jedan ispitanik sa višedecenijskim igračkim iskustvom u ovom sportu koji ima 47 godina. Inače, uzrast ispitanika u ovom istraživanju (MEAN=35.50) se ne razlikuje od uzrasta vrhunskih košarkaša u kolicima (Granados et al., 2015). Diferencijacija igrača po pozicijama nije izvršena, zbog malog broja uzorka, ali i zbog činjenice da igrači na svim igračkim pozicijama se značajno ne razlikuju u količini kretanja tokom utakmice (de Witte et al., 2016).

Vrijednosti antropometrijskih mjera ispitanika, sjedeće visine i tjelesne mase, takođe ne odstupa od vrijednosti vrhunskih košarkaša (Granados et al., 2015), a značajno su veće od vrhunskih juniora (Bergaminiet al., 2015). Zbog ograničene dostupnosti specijalizovane opreme za ispitivanje u zavisnosti od učesnika u invalidskim kolicima u laboratoriji, usvajanje ispitivanja na terenu postalo je poželjna opcija timskim trenerima sportista u invalidskim kolicima. Očigledna prednost testiranja na terenu je da se za manje grupe sportista može testirati

za manje vremena. Pored toga, ispitanici su testirani u svom prirodnom okruženju (koristeći svoje normalne sportske postavke za invalidska kolica i površinu poda), potencijalno čineći rezultate takvog testiranja bitnije od laboratorijskog testiranja (Goosey-Tolfrey & Leicht, 2013).

Vrijednosti testa Maksimalan pas u ovom istraživanju se statistički ne razlikuju od istih vrijednosti vrhunskih košarkaša u kolicima (Gil et al., 2015). Isto tako, vrijednosti dinamometrije obje šake ispitanika ovog istraživanja se statistički ne razlikuju od istih vrijednosti vrhunskih košarkaša u kolicima (Ferreira et al., 2017), ali se razlikuju između igrača različitih pozicija, povreda i klasa (Gil et al., 2015).

Specifična motorika propulzije kolica je veoma bitan faktor, tj. osnovni preduslov za uspješno bavljenje košarkom u kolicima (Goosey-Tolfrey, 2010; Mason et al., 2013). Vrijednosti testova brzine vožnje kolica (Vrijeme na 5 m, Vrijeme na 20 m, Slalom bez lopte) pokazuju da ispitanici u ovom istraživanju se ne razlikuju od vrhunskih košarkaša u kolicima (Molik et al., 2013).

Za košarku u kolicima, nadgradnja u specifičnoj motorici propulzije kolica je specifična motorika sa loptom (Bergamini

et al., 2015). Rezultati primijenjenih testova sa loptom su pokazali (Vrijeme na 5 m sa loptom, Vrijeme na 20 m sa loptom, Slalom sa loptom) da se ispitivani uzorak značajno ne razlikuje od vrhunskih košarkaša u kolicima (Molik et al., 2010).

Eksperimentalni tretman u trajanju od 15 dana, koji je sadržao 20 dvočasovnih trenažnih sesija sa dominacijom elementata tehničke i taktičke pripreme, nije se pokazao dovoljnim za statistički značajnu transformaciju dinamometrije šaka, eksplozivne snage ruku i specifične motorike košarkaša u kolicima. Očigledno da je u okviru ovog istraživanja, primijenjeni eksperimentalni tretman nedovoljno dug da bi se došlo do transformacije pokazatelja dinamometrije i tehničkih sposobnosti, što je i dokazano u istraživanjima španskih autora (Iturricastillo et al., 2015), koji konstatuju da je za tu pojavu potrebna cijela sezona. Isto tako, postoje istraživanja koja sugeriraju dopunski intenzivni trening osmosedmični trening snage i moći u teretani kao važan za sportove u kojima se koriste kolica (Turbanski & Schmidtbleicher, 2010).

Treneri košarkaša u kolicima treba na taj način da naprave strukturu treninga snage i kondicije da on bude usmjeren na pozitivnu transformaciju i poboljšanje sprinta i agilnosti, što bi bilo od značaja za uspješno igranje ovog sporta (Granados et al., 2015).

Cilj ovog istraživanja je bio utvrditi efekte pripremnog perioda na motoriku košarkaša u kolicima tokom pripremnog perioda. Uzorak ispitanika činilo šest košarkaša u kolicima Srbije, uzrasta od 20 do 47 godina, sa najmanje pet godina sportskog staža na nacionalnom nivou. Eksperimentalni tretman je obuhvatao pripreme reprezentacije Srbije u košarci u kolicima za Evropsko prvenstvo C divizije koje je održano od 25.07. do 28.07.2017. godine u Brnu. Pripremni period je trajao 15 dana, održano je 20 trenažnih sesija u trajanju od 120 minuta. Eksperimentalni tretman se nije pokazao dovoljnim za statistički značajnu transformaciju dinamometrije šaka, eksplozivne snage ruku i specifične motorike košarkaša u kolicima predstavljen sa devet testova.

Zbog malog broja ispitanika i nehomogenosti ispitivanog uzorka ne može se sa sigurnošću tvrditi da dobijeni rezultati mogu da budu prihvaćene kao aksiom. Potrebno je u budućim istraživanjima ispitati i efekte sličnog trenažnog tretmana na ispitivane pokazatelje motorike košarkaša u kolicima, ali na većem uzorku i u funkciji povreda, invaliditeta i funkcioniranih klasa igrača.

Značaj ovog istraživanja se ogleda u tome da se istakne značaj pripremnog perioda košarkaša u kolicima, te prikaže kako i koliko pripremni period utiče na razvoj motorike. Takođe, zbog slabe razvijenosti košarke u kolicima na našim prostorima, treba sprovoditi što više ovakvih istraživanja, sa još više testova motorike i sve to u cilju što boljeg razvoja i poznavanja košarke u kolicima.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 12 May 2019 | **Accepted:** 30 July 2019 | **Published:** 11 October 2019

References

- Bergamini, E., Morelli, F., Marchetti, F., Vannozzi, G., Polidori, L., Paradisi, F., Traballasi, M., Cappozzo, A., & Delussu, A. S. (2015). Wheelchair propulsion biomechanics in junior basketball players: A method for the evaluation of the efficacy of a specific training program. *BioMed Research International*, 2015, 1-10. doi: 10.1155/2015/275965
- Bloxham, L. A., Bell, G. J., Bhambhani, Y., & Steadward, R. D. (2001). Time motion analysis and physiological profile of Canadian world cup wheelchair basketball players. *Sports Medicine, Training and Rehabilitation*, 10(3), 183-198.
- Croft, L., Dybrus, S., Lenton, J., & Goosey-Tolfrey, V. (2010). A comparison of the physiological demands of wheelchair basketball and wheelchair tennis. *International Journal of Sports Physiology and Performance*, 5(3), 301-315.
- de Witte, A. M., Hoozemans, M. J., Berger, M. A., van der Woude, L. H., & Veeager, D. (2016). Do field position and playing standard influence athlete performance in wheelchair basketball?. *Journal of Sports Sciences*, 34(9), 811-820.
- DePauw, K.P., & Gavron, S.J. (2005). *Disability sport*. Champaign, IL: Human Kinetics.
- Đurašković, R. (2001). *Biologija razvoja čoveka sa medicinom sporta*. Niš: SIIC.
- Ferreira, S. A., Souza, W. C. D., Nascimento, M. A. D., Tartaruga, M. P., Portela, B. S., Mascarenhas, L. P. G., & Queiroga, M. R. (2017). Morphological characteristics, muscle strength, and anaerobic power performance of wheelchair basketball players. *Revista Brasileira de Cineantropometria & Desempenho Humano*, 19(3), 343-353.
- Gil, S. M., Yanci, J., Otero, M., Olasagasti, J., Badiola, A., Bidaurreaga-Letona, I., Iturricastillo, A., & Granados, C. (2015). The Functional Classification and Field Test Performance in Wheelchair Basketball Players. *Journal of Human Kinetics*, 46(1), 219-230.
- Goosey-Tolfrey, V. (2010). *Wheelchair sport - A complete guide for athletes, coaches, and teachers*. Champaign, IL: Human Kinetics.
- Goosey-Tolfrey, V. L., & Leicht, C. A. (2013). Field-based physiological testing of wheelchair athletes. *Sports Medicine*, 43(2), 77-91.
- Granados, C., Yanci, J., Badiola, A., Iturricastillo, A., Otero, M., Olasagasti, J., Bidaurreaga-Letona, I., & Gil, S. M. (2015). Anthropometry and performance in wheelchair basketball. *The Journal of Strength & Conditioning Research*, 29(7), 1812-1820.
- International Wheelchair Basketball Federation (2017, December 1). *Official Wheelchair Basketball Rules & Wheelchair Basketball Equipment*. Retrieved from IWBf website: https://iwbf.org/wp-content/uploads/2017/10/2017_IWBF_rules-Ver-1_gray.pdf
- International Wheelchair Basketball Federation (2018, September). *International Wheelchair Basketball Federation Activity Report 2014-2018*. Retrieved from IWBf website: <https://iwbf.org/wp-content/uploads/2018/09/activity-report-18-digital.pdf>
- Iturricastillo, A., Granados, C., & Yanci, J. (2015). Changes in body composition and physical performance in wheelchair basketball players during a competitive season. *Journal of human kinetics*, 48(1), 157-165.
- IWBf Player Classification Commission (2014, June). *Official Player Classification Manual*. Retrieved from Wheelchair Basketball Canada website: http://www.wheelchairbasketball.ca/uploadedFiles/Members/Classifiers/Policies_and_Procedures/CLASSIFICATION%20MANUAL%202014-2018%20ENGLISH%20FINAL.pdf
- Kasum, G. (2015). *Sport osoba sa invaliditetom*. Beograd: Fakultet sporta i fizičkog vaspitanja.
- Mason, B.S., van der Woude, L.H., & Goosey-Tolfrey, V.L. (2013). The ergonomics of wheelchair configuration for optimal performance in the wheelchair court sports. *Sports Medicine*, 43(1), 23-38.
- Milenković, D., & Živanović, N. (2010). *Paraolimpijske igre*. Niš: Panoptikum.
- Molik, B., Kosmol, A., Laskin, J.J., Morgulec-Adamowicz, N., Skucas, K., Dabrowska, A., Gajewski, J., & Ergun, N. (2010). Wheelchair basketball skill tests: differences between athletes functional classification level and disability type. *Fizyoterapi Rehabilitasyon*, 21(1), 11-19.
- Molik, B., Laskin, J.J., Kosmol, A., Marszalek, J., Morgulec-Adamowicz, N., & Frick, T. (2013). Relationships between anaerobic performance, field tests, and functional level of elite female wheelchair basketball athletes. *Human Movement*, 14(4), 366-371.
- Silver, J. R. (2012). Ludwig Guttmann (1899–1980), Stoke Mandeville Hospital and the Paralympic Games. *Journal of Medical Biography*, 20(3), 101-105.
- Turbanski, S., & Schmidtbleicher, D. (2010). Effects of heavy resistance training on strength and power in upper extremities in wheelchair athletes. *The Journal of Strength & Conditioning Research*, 24(1), 8-16.

ORIGINAL SCIENTIFIC PAPER

Differences in Body Composition of Football Players of Two Top Football Clubs

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Abstract

The aim of this research was to determine the differences among the football players of the football club in Bosnia and Herzegovina, CSC Zrinjski Mostar and the football players of the football club in Montenegro FC Sutjeska Niksic, in the anthropometric characteristics and body composition. A sample of 51 subjects was divided into two sub-samples. The first sub-sample of the subjects consisted of 28 football players of CSC Zrinjski Mostar of the average age 24.36 ± 4.14 , the champions of the Bosnia and Herzegovina in the season 2016/17, while the other sub-sample consisted of 23 players of FC Sutjeska Niksic of the average age of 21.69 ± 4.30 , the winner of the Cup of Montenegro in the season 2016/17. Football players were tested immediately after the end of the competition season 2016/17. Anthropometric characteristics in the body composition were evaluated by a battery of 10 variables: body height, body weight, waist circumference, triceps skinfold, biceps skinfold, skinfold of the back, abdominal skinfold, body mass index, fat percentage and muscle mass. The significance of the differences between the football players of the top two football clubs in the Anthropometric characteristics and variables for assessing body composition was determined by a t-test for independent samples. It was found that the football players of the two mentioned clubs have statistically significant differences by the three variables that estimate the waist circumference, triceps skinfold and skinfold of the back, in a favor of FC Sutjeska Niksic.

Key words: Anthropometric Characteristics, Body Composition, Football Players

Introduction

A football game is said to be the most important secondary thing in the world, it gathers huge masses at stadiums and in front of TVs (Gardašević, 2010; Gardašević, Bjelica, Popović, & Milašinović, 2016). It is a highly dynamic and fast team game which, with its richness of movement, falls under category of polystructural sports games (Bjelica, 2005; Gardašević i Goranović, 2011; Gardašević i Bjelica, 2013; Gardašević & Bjelica, 2014a; Gardasevic i Bjelica, 2014b). Football is a sport that is characterized by numerous and various complex and dynamic kinesiological activities which are then characterized by either cyclical (Gardašević, Vasiljević i Bojanić, 2015; Bjelica, Popović, & Gardašević, 2016a; Bjelica, Popović i Gardašević, 2016b; Sermahhaj, Popovic, Bjelica, Gardasevic, & Arifi, 2017; Gardasevic, Bjelica, & Vasiljevic,

2017a; Gardasevic, Bjelica, & Vasiljevic, 2017b) or acyclical movement (Gardasevic, 2015; Gardašević i sar., 2015; Gardašević, Bjelica i Vasiljević, 2016a; Gardašević, Bjelica i Vasiljević, 2016b; Gardasevic, Bjelica, Milasinovic i Vasiljevic, 2016; Gardašević i Vasiljević, 2016; Gardasevic, Popovic, & Bjelica, 2016). In football, top score can be achieved only under conditions of well-programmed training process (Gardašević, Bjelica i Popović, 2015). High quality management of the training process depends on the knowing of the structure of certain anthropological capabilities and player's characteristics, as well as their development (Bjelica i Popović, 2012; Bjelica, 2013). Various researches are to be done in order to establish certain principles and norms for the transformational processes of the anthropological characteristics important for football (Gardašević, Bjelica, Georgiev, & Popović, 2012); with

anthropometric characteristics and body composition among them as expected. Findings regarding anthropometric characteristics and body composition are of crucial importance for complex sports games such as football. The anthropometric space is defined by the longitudinal dimension of the skeleton, the transversal dimensionality of the skeleton, the mass and volume of the body (Bjelica & Fratrić, 2011). The purpose of knowing anthropometric characteristics is to improve skills in many sports (Carter & Heath, 1990). The anthropometric status of top level athletes is relatively homogeneous, depending on the sport, and it can be defined as a model of athletic achievement (Mišigoj-Duraković, Matković, & Medved, 1995). Research on anthropometric characteristics and body composition among athletes of different sports indicates that athletes of different sports have their own specific characteristics. Muscle mass improves performance in activities that require muscular strength and endurance, but also in those that require enviable aerobic ability (Ramadan & Byrd, 1987; Green, 1992; Rico-Sanz, 1998).

Today, football is certainly the number one sport in the world for its view and popularity (Gardašević, Georgiev, & Bjelica, 2012; Vasiljević, Gardašević, & Bojanić, 2013; Gardasevic, Bjelica, Vasiljevic, Arifi, & Sermakhaj, 2019), and the same applies to Bosnia and Herzegovina and Montenegro (Bjelica, Gardasevic, Vasiljevic, Arifi, & Sermakhaj, 2019). The two clubs that are at the top of the Premier League of Bosnia and Herzegovina and of the First Montenegrin Telecom League, in the 2016/17 competitive season, they both have achieved a staggering success, CSC Zrinjski Mostar was the champion of Bosnia and Herzegovina and FC Sutjeska-Niksic was the winner of the Cup of Montenegro. Based on these two trophies that they have won at the end of the competition season, both clubs have acquired the right to play on the international football scene within the framework of UEFA's Champions League qualification and UEFA's Europa League qualification. It became as interesting for researchers to determine the models of anthropometric characteristics and body composition of the players who play for these clubs as to determine the differences among them.

The aim of this research was to determine anthropometric characteristics and body composition of elite football players, players of CSC Zrinjski Mostar who compete in the Telecom Premier League of Bosnia and Herzegovina and players of FC Sutjeska Niksic, who compete in the First Montenegrin Telecom League. After which, compare the variables between these football players and determine the possible differences between them.

Method

The data obtained in the study of anthropometric characteristics and body composition are checked and prepared for processing according to the set goal. Data bases are arranged according to the features and prepared for planned statistical processing. The results obtained by statistical analysis are presented in the tables and analyzed by the corresponding logical units. In general, the results of the research, through gradualness in the explanation of individual relationships, allow seeing differences in the observed anthropometric measures and body composition in accordance with the aim of the research, that is, they contribute to a clearer application of the obtained results in practice. In terms of time constraint, the research is of transversal character, and it consists of a

one-off measurement of the corresponding anthropometric characteristics and body composition of top-level senior football players.

Sample of subjects

A sample of the subjects consists of a total of 51 top-level senior football players who performed in the Premier League of Bosnia and Herzegovina and the First Montenegrin Telecom League, divided into two sub-samples. The first one consists of 28 football players of CSC Zrinjski Mostar, the average age of 24.36 ± 4.14 , Bosnia and Herzegovina's Championship winner in season 2016/17, and the second one that consists of 23 football players of FC Sutjeska Niksic of the average age 21.69 ± 4.30 , the winner of the Cup of Montenegro in the season 2016/17. The football players were tested immediately after the 2016/17 season ended.

Sample of measures

Anthropometric research has been carried out with respect to the basic rules and principles related to the selection of measuring instruments and measurement techniques standardized in accordance with the International Biological Program guidelines. For the purpose of this study, 7 anthropometric measures have been taken: body height, body weight, waist circumference, triceps skinfold, biceps skinfold, skinfold of the back and abdominal skinfold, and 3 body composition assessment variables: body mass index (BMI), fat percentage and muscle mass. Anthropometer, caliper, and measuring tape were used for anthropometric measurements. To evaluate the body composition, Tanita body fat scale - model BC-418MA, was used. The principle of this scale is based on indirect measurement of the body composition; a safe electrical signal is transmitted through the body via electrodes located in the standalone unit. The Tanita Scale, thanks to its athletics mode, enables athletes to closely monitor their body weight, health condition and form with all relevant parameters.

Method of data processing

The data obtained through the research are processed by descriptive and comparative statistical procedures. For each variable, central and dispersion parameters, as well as asymmetry and flattening measures are processed. Differences in anthropometric characteristics and the composition of the body of the football players of these two clubs were determined by using a discriminatory parametric procedure with t-test for small independent samples, with statistical significance of $p < 0.05$.

Results

In Tables 1 and 2, basic descriptive statistical parameters of anthropometric variables and body composition of the football players of the two clubs, where the values of central measurements and dispersion tendencies are calculated, are shown: arithmetic mean (Mean), standard deviation (S.D.), variance, minimal (Min) i maximal (Max) values, coefficient of curvature (Skewness) and elongation (Kurtosis). First, the central and dispersion parameters of the variables were analyzed to evaluate the anthropometric characteristics and body composition of the football players of CSC Zrinjski Mostar (Table 1).

Table 1. Central and dispersion parameters of variables for assessment of anthropometric characteristics and body composition of football players of CSC Zrinjski Mostar (N=28)

Variables	Min	Max	Mean±S.D.	Variance	Skewness	Kurtosis
Body height	170.8	193.0	182.59±4.82	23.27	-.07	.25
Body weight	70.0	90.5	78.85±5.80	33.68	.13	-.92
Waist circumference	77.0	98.0	86.39±4.35	18.91	.34	.95
Triceps skinfold	4.6	13.0	7.59±2.09	4.39	.88	.63
Biceps skinfold	3.3	6.2	4.33±.74	.55	1.07	.59
Skinfold of the back	3.7	13.8	9.23±2.18	4.74	.31	.92
Abdominal skinfold	4.0	15.0	8.02±2.77	7.66	.89	.36
BMI-body mass index	21.4	26.1	23.63±1.14	1.30	.16	-.47
Fat percentage	3.9	14.6	8.79±3.18	10.14	-.05	-.88
Muscle mass	35.5	46.9	40.67±2.67	7.12	.05	-.03

Based on the central and dispersion parameters, the values of the skewness and the kurtosis, it can be noted that all the variables are placed within the normal distribution boundaries. Generally, according to all statistical parameters, it can be concluded that here we have some top football players; that there is a normal distribution in all variables and that the results that prevail are superior to the arithmetic mean, which

is not statistically significant because it is to be expected that regarding football players of a professional football club, there is no too large a span between the results of analyzed variables. Table 2 showed the central and dispersion parameters of the variables were analyzed to evaluate the anthropometric characteristics and body composition of the football players of FC Sutjeska Niksic.

Table 2. Central and dispersion parameters of variables for assessment of anthropometric characteristics and body composition of football players of FC Sutjeska Niksic (N=23)

Variables	Min	Max	Mean±Std.D.	Variance	Skewness	Kurtosis
Body height	166.0	195.0	182.99±6.77	45.83	-.55	.61
Body weight	68.0	91.7	78.33±7.71	59.43	.48	-1.16
Waist circumference	76.0	90.0	83.00±3.82	14.64	.28	-.54
Triceps skinfold	3.4	9.8	6.46±1.66	2.77	.26	-.22
Biceps skinfold	3.1	7.7	4.52±1.22	1.49	1.12	.94
Skinfold of the back	6.4	11.8	8.13±1.42	2.01	1.10	1.15
Abdominal skinfold	5.4	14.0	8.25±2.38	5.65	.85	.03
BMI-body mass index	21.7	25.8	23.36±1.41	1.99	.43	-1.33
Fat percentage	2.6	13.6	8.66±2.92	8.53	-.38	-.51
Muscle mass	34.1	46.8	40.56±3.81	14.51	-.03	-1.19

Based on the central and dispersion parameters, the values of skewness and kurtosis of the football players of FC Sutjeska-Niksic, it can be stated that all the variables are within the normal distribution boundaries and that the values are very similar to those of the football players of CSC Zrinjski Mostar. It can also be stated that the football players of FC Sutjeska-Niksic are younger on average. It can also be concluded that almost all variables of quantitative value are better with football players of FC Sutjeska-Niksic. However, a compara-

tive statistical procedure, t-test (Table 3), will show whether it is statistically significant. By the value of the skewness, it can be noticed that in the variables of the biceps skinfold and skinfold of the back, there was a slight inclination on the side of the lower results, which is good because subcutaneous fat is a disrupting factor for professional athletes. In order to determine whether there are statistically significant differences in the analyzed variables in the top football players of these two clubs, the statistical procedure t-test (Table 3) was applied.

Table 3. T-test values between the arithmetic mean of variables for the evaluation of anthropometric characteristics and body composition of football players of CSC Zrinjski Mostar (N=28) and FC Sutjeska Niksic (N=23)

Variables	Club	Mean±S. D.	Mean Difference	t-test	Sig.
Body Height	CSC Zrinjski	182.59±4.82	-.3984	-.245	.807
	FC Sutjeska	182.99±6.77			
Body Weight	CSC Zrinjski	78.85±5.80	.5239	.277	.783
	FC Sutjeska	78.33±7.71			
Waist Circumference	CSC Zrinjski	86.39±4.35	3.3929	2.925	.005
	FC Sutjeska	83.00±3.82			
Triceps Skinfold	CSC Zrinjski	7.59±2.09	1.1241	2.087	.042
	FC Sutjeska	6.46±1.66			

(continued on next page)

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Variables	Club	Mean±S. D.	Mean Difference	t-test	Sig.
Biceps Skinfold	CSC Zrinjski	4.33±.74	-.1932	-.695	.490
	FC Sutjeska	4.52±1.22			
Skinfold of the Back	CSC Zrinjski	9.23±2.18	1.1017	2.088	.042
	FC Sutjeska	8.13±1.42			
Abdominal Skinfold	CSC Zrinjski	8.02±2.77	-.2300	-.314	.755
	FC Sutjeska	8.25±2.38			
BMI-Body Mass Index	CSC Zrinjski	23.63±1.14	.2677	.749	.458
	FC Sutjeska	23.36±1.41			
Fat Percentage	CSC Zrinjski	8.79±3.18	.1292	.150	.882
	FC Sutjeska	8.66±2.92			
Muscle Mass	CSC Zrinjski	40.67±2.67	.1026	.113	.911
	FC Sutjeska	40.56±3.81			

Based on the obtained values of t-test results, it was found that the football players of the two mentioned clubs have statistically significant differences by the three variables that estimate the the waist circumference, triceps skinfold and skinfold of the back, in a favor of FC Sutjeska-Niksic.

Discussion

The aim of this study was to determine the difference in the anthropometric characteristics and body composition of the top football players of the champions' football club in Bosnia and Herzegovina, CSC Zrinjski Mostar and the top football players of the winner football club in Montenegro FC Sutjeska Niksic. A sample of 51 respondents was divided into two sub-samples. The first sub-sample consisted of the 28 football players of CSC Zrinjski Mostar of 24.36 ± 4.14 age on average, who were a statistically significant older than the 23 football players of FC Sutjeska Niksic, who made the second sub-sample of 21.69 ± 4.30 age on average. The results were obtained by using a battery of 10 tests in the area of anthropometric characteristics and body composition. By looking into the basic descriptive statistical parameters, it can be concluded that we have examined professional sportsmen indeed. It can be noticed that the football players of both clubs are of the approximately similar mean values of the variables analyzed, which is not surprising because these are the top two football clubs in Bosnia and Herzegovina and Montenegro, a state where there are also a great concentration of good football players. The t-test results showed that the football players of the two mentioned clubs have statistically significant differences by the three variables that estimate the waist circumference, triceps skinfold and skinfold of the back, in a favor of FC Sutjeska Niksic. The first variable in which a statistically significant difference has been found is a variable that estimates waist circumference, where the football players of CSC Zrinjski Mostar also have a statistically higher value than the football players of FC Sutjeska Niksic. Also, at the variables of triceps skinfold and skinfold of the back, football players of CSC Zrinjski Mostar have shown statistically better values because a smaller number means a better result when the disrupting factor of subcutaneous fat on playing football is taken into account. Very similar characteristics of football players were obtained in the region, which shows that football players have similar the anthropometric characteristics and body composition in the whole region (Gardasevic, Bjelica, Popovic, Vasiljevic, & Milosevic, 2018; Bjelica, Gardasevic, & Vasiljevic, 2018; Corluka & Vasiljevic, 2018; Bjelica, & Gar-

dasevic, 2019; Gardasevic, Bjelica, & Vasiljevic, 2019; Bjelica, Gardasevic, Vasiljevic, Jeleskovic, & Covic, 2019).

For other variables, some values are better for football players of CSC Zrinjski Mostar and some for football players of FC Sutjeska Niksic, although, insignificantly for statistics, which indicates that these football players have very similar anthropometric parameters and body composition, which is again, not surprising, considering that these two football clubs are the best in their countries in the 2016/17 competitive season. The values obtained in this research can be useful for coaches of these football clubs for making a comparison of their football players with others and formulate their work in a way that enables reduction of those parameters that are not good, and raise those that are good to a higher level. That will surely make their football players even better and more successful. Also, both clubs should turn to other researches and check the functional-motoric status, psychological preparation as well as tactical training of their football players and analyze whether there is room for their improvement. The results obtained in this research can serve as model parameters for the estimated variables for football players of all other football clubs in Bosnia and Herzegovina and Montenegro, because the football players that have been analyzed here, were among the best and the most successful football players in those two countries at the end of the competitive season 2016/17.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 30 July 2019 | **Accepted:** 18 August 2019 | **Published:** 11 October 2019

References

- Bjelica, D. (2005). *Sistematizacija sportskih disciplina i sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2013). *Teorija sportskog treninga*. Podgorica: Univerzitet Crne Gore.
- Bjelica, D., & Fratrić, F. (2011). *Sportski trening: teorija, metodika i dijagnostika*. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Bjelica, D., i Popović, S. (2012). *Fudbal-teorija, tehnika i taktika*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D., Popović, S., i Gardašević, J. (2016a). Modeli fizičke pripreme vrhunskih sportaša i doziranje opterećenja. U *Zborniku radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (185-189), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bjelica, D., Popović, S., i Gardašević, J. (2016b). Opći principi planiranja i programiranja fizičkih priprema sportaša. U *Zborniku radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (190-192), Zagreb: Udruga kondicijskih trenera Hrvatske.

- Bjelica, D., Gardasevic, J., & Vasiljevic, I. (2018). Differences in the Anthropometric characteristics and body composition of soccer players FC Sutjeska and FC Mladost in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(2), 31-35. doi: 10.26773/jaspe.180406
- Bjelica, D., & Gardasevic, J. (2019). Body composition and anthropometric measures of soccer players, champions of Montenegro and Bosnia and Herzegovina. In *Book of Abstracts of the 16th Annual Scientific Conference of Montenegrin Sports Academy "Sport, Physical Activity and Health: Contemporary Perspectives"* (73-74), Podgorica: Montenegrin Sports Academy.
- Bjelica, D., Gardasevic, J., Vasiljevic, I., Arifi, F., & Sermahaj, S. (2019). Anthropometric measures and body composition of soccer players of Montenegro and Montenegro. *Journal of Anthropology of Sport and Physical Education*, 3(2), 29-34. doi: 10.26773/jaspe.190406
- Bjelica, D., Gardasevic, J., Vasiljevic, I., Jeleskovic, E., & Covic, N. (2019). Body Composition and Morphological Characteristics of Soccer Players in Bosnia and Herzegovina. *Kinesiology Slovenica*, 25(1), 5-13.
- Carter, J.E.L., & Heath, B.H. (1990). *Somatotyping—Development and application*. Cambridge, United Kingdom: Cambridge University Press.
- Corluka, M., & Vasiljevic, I. (2018). Differences in the Anthropometric characteristics and body composition of soccer players in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(1), 3-7. doi: 10.26773/jaspe.180101
- Gardašević, J. (2010). *Efekti programiranog rada u pripremnom periodu na transformaciju bazično-motoričkih i situaciono-motoričkih sposobnosti kod fudbalera kadetskog uzrasta*. Neobjavljena magistarska teza. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Gardašević, J., & Goranović, K. (2011). Efekti programiranog rada u pripremnom periodu na transformaciju eksplozivne snage kod fudbalera kadeta. *Sport Mont*, 9(28-30), 55-62.
- Gardašević, J., Georgiev, G., & Bjelica, D. (2012). Qualitative changes of basic motor abilities after completing a six-week training programme. *Acta Kinesiológica*, 6(1), 70-74.
- Gardašević, J., Bjelica, D., Georgiev, G., & Popović, S. (2012). Transformation of situational motor abilities with soccer players—cadets. In *Proceeding book, XVI International Scientific Congress "Olympic Sports and Sport for All" & VI International Scientific Congress, Sport, Stress, Adaptation* (373-377), Soňa: National Sports Academy "Vassil Levski".
- Gardašević, J., i Bjelica, D. (2013). Efekti programiranog trenažnog rada u trajanju od šest nedjelja na transformaciju fleksibilnosti kod fudbalera kadetskog uzrasta. *Sport Mont*, 11(37-39), 212-217.
- Gardašević, J., & Bjelica, D. (2014a). The effects of the training in the preparation period on the dribbling speed with fifteen years old soccer players. In *Book of Abstracts of the 11th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (22-23), Podgorica: Montenegrin Sports Academy.
- Gardasevic, J., i Bjelica, D. (2014b). Efekti rada u pripremnom periodu na brzinu vođenja lopte petnaestogodišnjih fudbalera. *Sport Mont*, 12(40-42), 160-166.
- Gardašević, J., Vasiljević, I., & Bojanić, D. (2015). Six-week preparation period and its effects on coordination transformation with soccer players under 16. In *Book of Abstracts 11th International Scientific Conference Management, Sport, Olympism* (36), Beograd: Fakultet za menadžment u sportu, Alfa univerzitet.
- Gardasevic, J. (2015). The effects of the training in the preparation period on the agility transformation with cadet level soccer players. In *Book of Abstracts of the 12th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (76-77), Podgorica: Montenegrin Sports Academy.
- Gardašević, J., Vasiljević, I., Bojanić, D., Muratović, A., Ljubojević, M., Milašinović, R., & Bujanja, M. (2015). Six-week Preparation Period and its Effects on Transformation Movement Speed with Soccer Players Under 16. In *Book of Abstracts, International Scientific Conference "Effects of Physical Activity Application to Anthropological Status with Children, Youth and Adults"* (148), Belgrade: University of Belgrade: Faculty of Sport and Physical Education.
- Gardašević, J., Bjelica, D., i Popović S. (2015). Efekti programiranog rada tokom pripremnog perioda na transformaciju agilnosti kod fudbalera kadetskog uzrasta. *Sport Mont*, 13(43-45), 355-360.
- Gardašević, J., Bjelica, D., & Vasiljević, I. (2016a). Six-Week Preparation Period and its Effects on Transformation Movement Speed with Soccer Players Under 16. *Sport Mont*, 14(1), 13-16.
- Gardašević, J., Bjelica, D., & Vasiljević, I. (2016b). The Effects of the Training in the Preparation Period on the Repetitive Strength Transformation with Cadet Level Soccer Players. In *Book of Abstracts of the 13th International Scientific Conference on Transformation Processes in Sport "Sport Performance"* (43), Podgorica: Montenegrin Sports Academy.
- Gardasevic, J., Bjelica, D., Milasinovic, R., & Vasiljevic, I. (2016). The Effects of the Training in the Preparation Period on the Repetitive Strength Transformation with Cadet Level Soccer Players. *Sport Mont*, 14(2), 31-33.
- Gardašević, J., & Vasiljević, I. (2016). Effects of Preparation Period on Endurance in U16 Soccer Players. In *Book of Abstracts of the 4TH International Scientific Conference "Exercise and Quality of Life"* (108), Novi Sad: University of Novi Sad, Faculty of Sport and Physical Education.
- Gardašević, J., Bjelica, D., Popović, S., & Milašinović, R. (2016). Preparation Period and its Effects on the Speed of Ball Leading at Players U16. In *Book of Summaries of 11th FIEP European Congress "Anthropological Aspects of Sport, Physical Education and Recreation"* (30-31), Banjaluka: University of Banjaluka, Faculty of Physical Education and Sport.
- Gardasevic, J., Popovic, S., & Bjelica, D. (2016). After preparation period ball shooting accuracy at players U15. In *Abstract Book of the 8th Conference for Youth Sport* (88), Ljubljana: University of Ljubljana, Faculty of Sport.
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2017a). The strength of kicking the ball after preparation period with U15 soccer players. In *Book of Abstracts of the 14th International Scientific Conference on Transformation Processes in Sport "Sport Performance"* (65-66), Podgorica: Montenegrin Sports Academy.
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2017b). The Strength of Kicking the Ball after Preparation Period with U15 Soccer Players. *Sport Mont*, 15(2), 39-42.
- Gardasevic, J., Bjelica, D., Popovic, S., Vasiljevic, I., & Milosevic, Z. (2018). Differences in the Anthropometric characteristics and body composition of soccer players FC Buducnost and FC Mladost in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(1), 51-55. doi: 10.26773/jaspe.180109
- Gardasevic, J., Bjelica, D., Vasiljevic, I., Arifi, F., & Sermahaj, S. (2019). Differences in anthropometric measures of soccerers, cup winners of Montenegro and Montenegro. *Journal of Anthropology of Sport and Physical Education*, 3(1), 23-27. doi: 10.26773/jaspe.190105
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2019). Body composition and anthropometric measures of soccer players, champions of Bosnia and Herzegovina and Montenegro. In *Book of Abstracts of the 16th Annual Scientific Conference of Montenegrin Sports Academy "Sport, Physical Activity and Health: Contemporary Perspectives"* (74-75), Podgorica: Montenegrin Sports Academy.
- Green, S. (1992). Anthropometric and physiological characteristics of south Australian soccer players. *Australian Journal of Science and Medicine in Sport*, 24, 3-7.
- Mišigoj-Duraković, M., Matković, B., & Medved, R. (1995). *Morfološka antropometrija u športu*. Anthropometricanthropometry in sports. Zagreb, Croatia: Fakultet za fizičku kulturu.
- Ramadan, J., & Byrd, R. (1987). Physical characteristics of elite soccer players. *Journal of Sports Medicine and Physical Fitness*, 27, 424-428.
- Rico-Sanz, J. (1998). Body composition and nutritional assessments in soccer. *International Journal of Sport Nutrition*, 8, 113-123.
- Sermahaj, S., Popovic, S., Bjelica, D., Gardasevic, J., & Arifi, F. (2017). Effect of recuperation with static stretching in isokinetic force of young soccer players. *Journal of Physical Education and Sport*, 17(3), 1948-1953. doi: 10.7752/jpes.2017.03191
- Vasiljević, I., Gardašević, J., i Bojanić, D. (2013). Uporedna analiza motoričkog prostora između aktivnih fudbalera kadetskog uzrasta i učenika srednje škole. U *Zborniku naučnih i stručnih radova VI međunarodni simpozijum "Sport i zdravlje"* (212-215), Tuzla: Fakultet za tjelesni odgoj i sport.

ORIGINAL SCIENTIFIC PAPER

Impact of Physical Activity on the Aggressiveness, Deviant Behavior and Self-esteem with School Children Aged 11-15

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Abstract

Comprehension of factors responsible for reduction of physical activities with adolescents is of a primary importance for an effective development of programs and strategies in promoting the physical activity among this population group. The aim of the present research is to explore the impact of physical and sports activities on the aggressive, deviant behavior and self-confidence among schoolchildren aged from 11 to 15. The research was conducted on a sample of 177 Macedonian adolescents from several primary schools in the city of Skopje. More scales of assessing physical activity, sedentary habits, self-confidence and behaving aggressively and in a deviant way were applied for the purposes of the present research. The results obtained from the research are statistically processed and each variable is with individually calculated basic descriptive statistic, while the relation between the applied variables is determined by the use of Pearson's coefficient of correlation (R). On the base of the obtained data a conclusion can be drawn that there is no statistically significant relation between physical activity and the total or individual types of aggressiveness, deviant behavior and self-confidence. There is a statistically significant positively low correlation between the times spent at the computer desk on work days and physically manifested aggressiveness, and there is statistically significant negative low correlation between the time spent studying and deviant behavior within the whole group of respondents.

Key words: *Physical Activity, Aggressiveness, Deviant Behavior, Self-Confidence, Adolescents*

Introduction

An attitude exists that if the game is recognized as an activity of social and psychological significance in the process of human development, it means that this type of activity is an ancestor of the very culture (Huizinga, 1970), i.e. an ancestor of civilization itself (Mandell, 1984). Sport, as a segment of more broad space of playing, being significant in the life and development of the individual person as well as the society the person belongs to, is recognized as far back as the time of Ancient Greece (Hyland, 1990). Pieces of evidence about the existence of sports activities have reached to us from other archaic cultures as well, and not only from the ancestors of western civilization (Mandell, 1984). Those particular authors mention also about existence of sports

activities and games and their importance in ancient Chinese and Indian civilizations.

If we take into consideration the short history of scientific attention towards physical activity and sport, we should not be surprised by the fact that the interdisciplinary subjects relevant to sport were either discussed or not discussed at all. One particular issue that have never been focused for a profound research is the impact of physical and sports activity on the self-esteem, aggressiveness and deviant behavior among children and young people (Stark, Kent, & Finke, 1987; Stevenson, 1985).

For a considerable amount of time, recreation has been seen as an effective approach of recreation (Aguilar, 1986, 1987; Collingwood & Engelsgerd, 1977; Hormachea & Hormachea,

1983; Krichbaum & Alston, 1991; Munson, 1991; Segrave, 1982, 1983). At least two special scientific journal editions in the fields of examining the leisure time and recreation are dedicated to the topic of therapeutic recreation in juvenile reformatory society (Jewell, 1977; Orton, 1977). The articles contain too much theoretical speculations and insufficiency of empiric and scientific data and assessments of actual programs and activities.

When it comes to the effectiveness of rehabilitation, there can be pointed that the existing literature on recreation and physical activity is not of help to explain the relatively unconvincingly thin results related to the dependence between taking part in sport and the process of socialization. Yet in literature – both theoretical and empiric – exists a common consent that recreational-reformatory programs are necessary and useful both for preventive and rehabilitation purposes. If the recreational process is able to achieve its ultimate purpose – to fill up the leisure time in a productive way – then we possess a massive option of working on preventing delinquents and potential delinquents from misusing their free time.

Concerning the prevention and the proof about the effectiveness of community recreational programs in urban environments with quite frequent situations of deviant/delinquent behavior (Elliott, 1963; Lutzin & Orem, 1967; Todd, 1941; United States Department of Justice, 1981; Wylie, 1960, 1965), there is a certain amount of positive evidence (Burns, 1907; Shanas, 1942; Truxal, 1929).

The aim of the present research is to explore the impact of physical and sports activities on the aggressive, deviant behavior and self-confidence among schoolchildren aged from 11 to 15.

Methods

Subjects

The research was conducted on a sample of 177 Macedonian adolescents, selected by random choice from several primary schools in the city of Skopje. The average age of the respondents was 12.4 years.

The research was conducted from mid-April to mid-May 2016. Previously the principals of all primary schools that were involved in the research had been sent a letter introducing the objectives of the research. The research was carried out in cooperation with professors of physical education and psychological-pedagogical offices, in schools where they existed. The surveying was carried out in classrooms at special classes using proper organization of work appropriate for such researches. The study protocol was performed following the ethical principles by the guidelines of the Declaration of Helsinki (World Medical Association, 2001).

Instrument for Assessing

Within the research the time spent in the physical activity is rated by means of the instrument for assessing the physical activity – Physical Activity Questionnaire (Elementary School), constructed by Kowalski and Crocker (2001).

Sedentary habits are assessed according to the scale that is a segment of Health Behavior in School-aged Children (HBSC) study protocol (Currie et al., 2000). The respondents are asked to answer the question about the length of time they spend with sedentary activities. The questions were asked for a usual weekday and weekend day separately. Aggressiveness measure from the Questionnaire A – 87 (Žužul, 1989) consists of 15 items of different situations with five possible responses. The possible responses or reactions are the five most frequent forms of aggressive responses: a) verbal manifest aggression (VM); b) physical manifest aggression (PHM); c) indirect aggression (IND); d) verbal latent aggression (VL), and e) physical latent aggression (PHL). Deviant Behavior measure from the Denver Youth Survey is a seven-item scale (Institute of Behavioral Sciences, 1987), which asks participants to report how many of their friends have engaged in delinquent behaviors (e.g., hitting or threatening to hit someone, damaging or destroying property, drinking alcohol) in the past year. Self-esteem was measured with five items from the Rosenberg Self Esteem Scale (Rosenberg, 1965).

Statistical Analysis

The analysis was performed by using the Statistical Package for Social Sciences (SPSS) version 20.0. The results obtained from the research are statistically processed and each variable is with individually calculated basic descriptive statistic parameters: arithmetical mean, standard deviation, a bottom and top limit of the results (min – max), coefficient of symmetry – skewness, coefficient of flatness – kurtosis. The relation between the applied variables is determined by the use of Pearson's coefficient of correlation (R).

Results

Examination of Table 1 denotes the fact that the respondents' results of Skewness demonstrate positive asymmetry with the variables of physical manifest aggression (PHM), indirect aggressiveness (IND), verbal latent aggressiveness (VL), physical latent aggressiveness (PHL), summary aggressiveness (Total aggressiveness) and deviant behavior. With the rest of variables Skewness varies within the acceptable limits (-1 to +1). The values of Kurtosis show that most variables demonstrate platykurtic (flatness of the curve), namely there is dispersion of results with most of variables. It is the variable of deviant behavior alone that shows a marked leptokurtic.

Aiming to establish the existence of statistically significant correlation between the physical activity and total and individual types of self-esteem, aggressiveness and deviant behavior, there are calculated Pearson's coefficients of correlation.

Table 1. Basic descriptive statistic parameters of variables for assessing physical activity, sedentary habits, aggressiveness A-87, delinquent behavior and self-confidence

	Minimum	Maximum	Mean	SD	Skewness	Kurtosis
Physical activity	1,40	4,80	3,44	0,68	-0,07	-0,61
Weekday television	1,00	5,00	1,67	0,75	0,97	1,02
Weekend television	1,00	4,00	1,87	0,73	0,31	-0,79
Weekday PC	1,00	3,00	1,55	0,71	0,91	-0,47
Weekend PC	1,00	3,00	1,91	0,78	0,17	-1,35
Time spent in learning	1,00	5,00	2,14	0,80	0,81	1,70
VM aggressiveness	15,00	70,00	31,48	12,32	0,76	0,01

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	Minimum	Maximum	Mean	SD	Skewness	Kurtosis
PHM aggressiveness	15,00	62,00	23,97	10,37	1,44	1,47
IND aggressiveness	15,00	61,00	23,32	9,04	1,66	3,10
VL aggressiveness	15,00	63,00	27,65	11,35	1,08	0,64
PHL aggressiveness	15,00	66,00	23,62	10,10	1,53	2,76
Total aggressiveness	14,80	59,80	25,67	9,45	1,26	1,49
Deviant Behavior Scale	8,00	35,00	11,40	4,00	2,25	7,63
Self-esteem	1,00	5,00	3,87	0,73	-0,98	1,61

Note: VM - verbal manifest, PHM - physical manifest, IND - indirect, VL - verbal latent, PHL - physical latent

Examining Table 2 shows that the calculated coefficients of correlation are explicitly low and there is no statistically significant correlation of the physical activity with the total and individual types of aggressiveness, deviant behavior and self-confidence.

Aiming to establish the existence of statistically significant correlation of sedentary habits with the total and individual types of aggressiveness, deviant behavior and self-confidence, there are additionally calculated Pearson's coefficients of correlation.

Table 2. Coefficients of correlation between the physical activity and Aggressiveness, Deviant Behavior and Self-esteem

	VM	PHM	IND	VL	PHL	Total	DBVS	Self-esteem
Physical activity	-0,032	0,089	0,034	0,017	0,101	0,042	0,019	0,012

Note: VM - verbal manifest aggressiveness; PHM - physical manifest aggressiveness; IND - indirect aggressiveness; VL - verbal latent aggressiveness; PHL - physical latent aggressiveness; Total - total aggressiveness; DBVS - deviant behavior scale

Observing Table 3, we can see that there is a statistically significant positively low correlation of the time spent working on a computer in working days with physically manifested aggressiveness ($r = 0,218$; $p < 0,05$), and statistically significant negative correlation is established between the time spent studying and

deviant behavior ($r = -0,212$; $p < 0,05$).

Aiming to establish the existence of statistically significant correlation between the deviant behavior and self-confidence on one hand, and aggressive behavior on the other hand, Pearson's coefficients of correlation are calculated.

Table 3. Coefficients of cross-correlation between sedentary habits and total and individual types of aggressiveness, deviant behavior and self-confidence

	VM	PHM	IND	VL	PHL	Total	DBVS	Self-esteem
Weekday television	0,074	0,162	0,009	0,051	0,091	0,088	0,177	0,063
Weekend television	0,151	0,095	0,041	0,084	0,035	0,096	0,048	0,015
Weekday PC	0,135	0,218	0,144	0,119	0,164	0,172	0,086	-0,021
Weekend PC	0,162	0,160	0,146	0,112	0,111	0,154	0,126	-0,011
Time spent in learning	-0,080	-0,067	-0,012	-0,004	0,026	-0,033	-0,212	-0,120

Note: VM - verbal manifest aggressiveness; PHM - physical manifest aggressiveness; IND - indirect aggressiveness; VL - verbal latent aggressiveness; PHL - physical latent aggressiveness; Total - total aggressiveness; DBVS - deviant behavior scale

Examining Table 4, it is obvious that there is a statistically significant positive correlation between deviant behavior and verbal manifest aggressiveness ($r=0,551$; $p<0,00$), deviant behavior and verbal latent aggressiveness ($r=0,428$; $p<0,00$). The highest correlation demonstrates the deviant behavior and indirect aggressiveness.

On the other hand, there can be noticed that, in this sample of respondents, there is no statistically significant correlation between self-confidence and total and individual types of aggressiveness.

Table 4. Coefficients of cross-correlation between the deviant behavior and self-confidence on one hand, and aggressive behavior on the other hand

	VM	PHM	IND	VL	PHL	Total
DBVS	0,455	0,551	0,558	0,428	0,431	0,535
Self-esteem	-0,101	-0,084	-0,136	-0,136	-0,141	-0,133

Note: VM - verbal manifest aggressiveness; PHM - physical manifest aggressiveness; IND - indirect aggressiveness; VL - verbal latent aggressiveness; PHL - physical latent aggressiveness; Total - total aggressiveness; DBVS - deviant behavior scale

Discussion

The ruining effect that physical non-activity has on health brings forth the need of better understanding the models and determinants that would stimulate boosting physical activities among schoolchildren. A considerable number of study papers point at the decrease in physical activities among adolescents with further decline in the years of early maturity. A large number of studies on that topic are completed in many countries worldwide, but the question is whether the results obtained from these researches can be generally applied to the Macedonian population of adolescents (Gordon-Larsen, McMurray, & Popkin, 2000). In fact, the comprehension of factors involving reduction in physical activity among adolescents is of a vital importance for the development of effective programs and strategies in promoting the physical activity among that group of population.

The present research, carried on the described sample of respondents and the given age, does not establish a presence of statistically significant correlation between physical activity, on the one hand, and the total and individual types of aggressiveness, deviant behavior and self-confident, on the other hand. Further, the results of the research provide indications of correlation existing between the time spent working on a computer in school days and aggressiveness (physically manifested aggressiveness) and deviant behavior. Longitudinal research exploring excessive TV watching in early childhood has suggested that large amounts of exposure to electronic media in early childhood may be related to symptoms/diagnosis of attention deficit/ hyperactivity disorder (ADHD)—see Barlett, Anderson, & Swing (2009) for a review. In turn, ADHD may be related to delinquency (Pratt, Cullen, Blevins, Daigle, & Unnever, 2002); however, some research has suggested otherwise (Stevens & Mulsow, 2006). ADHD may be also indirectly related to delinquency through other potentially enabling factors, such as personal strains (Johnson & Kercher, 2007). This finding also supports research in that excessive TV viewing may be associated with risky behavior; however, there is evidence to suggest that parental regulation may be an important factor in this relationship (Nelson & Gordon-Larsen, 2006). Schoolchildren of both genders who spend more time working on a computer show greater aggressiveness and deviant behavior. Also, there is a positive correlation between the deviant behavior and the total and individual types of aggressiveness, whereas statistically significant correlation is not established between the deviant behavior and self-confidence.

The worrying moment is the fact that schoolchildren of both genders spend a great amount of their free time watching TV, or working on computers. World Health Organization (De Onis et al., 2007) strongly recommend that “children and young people should not spend more than two hours daily in watching TV or working on a computer (surfing, chatting or playing video games)”.

The characteristic feature of the present research is the application of A-87 questionnaire. The results obtained from the research indicate that the average discourse in aggressiveness within that sample of respondents is about half a standard deviation of the existing norms, both in total and individual types of aggressive behavior, which suggests that adolescents have certain inclination towards aggressive behavior (Mitrofan, Paul, & Spencer, 2009; van Egmond-Frohlich, Weghuber, & de Zwaan, 2012; Robertson, McAnally, & Hancox, 2013). In support of that stands the arch of the curve showing positive asymmetry, a greater number of the results being in the zone of higher ones. The

higher aggressiveness obtained with that sample of respondents in contrast to matured people can be explained with the approach of the respondents into the adolescent period, when young people always look for being themselves and try to build their identity, so aggressiveness is quite often a tool to struggle for one's position. Despite the fact that adolescents possess a greater range of strategies in solving conflicts as compared to children, yet they are not so socially skilled as compared to adults. Moreover, the civil law tolerates to some extent the aggressiveness with persons younger than the age of 18 years, whereas legal penalty measures against adults are far more restrictive, so they are willing to control aggressive expressions.

On the base of the obtained results, the following conclusions can be drawn: there is no statistically significant correlation between the physical activity and the total and individual types of aggressiveness, deviant behavior and self-confidence. There is a statistically significant positively low correlation between the times spent working on the computer at work and physically manifested aggressiveness; and statistically significant negative low correlation between the times spent studying and deviant behavior.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 18 May 2019 | **Accepted:** 20 August 2019 | **Published:** 11 October 2019

References

- Aguilar, T.E. (1986). Recreation: An untapped resource. *Corrections Today*, 48(2), 173-178.
- Aguilar, T.E. (1987). Effects of a leisure education program on expressed attitudes of delinquent adolescents. *Therapeutic Recreation Journal*, 21(4), 43-51.
- Barlett, C.P., Anderson, C.A., & Swing, E.L. (2009). Video Game Effects—Confirmed, Suspected, and Speculative. *Simulation & Gaming*, 40(3), 377-403. doi: 10.1177/1046878108327539
- Burns, T. (1907). Relation of playgrounds to juvenile delinquency. *Charities*, 21, 25-31.
- Collingwood, T.R., & Engelsgerd, M. (1977). Physical Fitness, Physical Activity, and Juvenile Delinquency. *Journal of Physical Education and Recreation*, 48(6), 23-23. doi: 10.1080/00971170.1977.10621069
- Currie, C., Griebler, R., Inchley, J., Theunissen, A., Molcho, M., Samdal, O., & Dür, W. (2010). *Health Behaviour in School-aged Children (HBSC) study protocol: background, methodology and mandatory items for the 2009/10 survey*. Edinburgh: CAHRU.
- De Onis, M., Onyango, A.W., Borghi, E., Siyao, A., Nishida, C., & Siekmann, J. (2007). Development of a WHO growth reference for school-aged children and adolescents. *Bulletin of the World Health Organization*, 85(09), 660-667. doi: 10.2471/BLT.07.043497
- Elliott, M.A. (1963). Group therapy in dealing with juvenile and adult offenders. *Federal Probation*, 27(4), 48-54.
- Gordon-Larsen, P., McMurray, R.G., & Popkin, B. M. (2000). Determinants of adolescent physical activity and inactivity patterns. *Pediatrics*, 105(6), e83. doi:10.1542/peds.105.6.e83
- Hormachea, M., & Hormachea, C. (1983). *Recreation and the youthful and adult offenders*. In Stein, T.A., & Sessoms, H.D. (Eds.), *Recreation and special populations* (105-134), London: Allyn and Bacon, Inc.
- Huizinga, J. (1970). *Homo ludens: A study of the play element in culture*. London: Temple Smith.
- Hyland, D.A. (1990). *Philosophy of sport*. New York: Paragon House.
- Institute of Behavioral Sciences. (1987). *Youth Interview Schedule: Denver Youth Survey*. Boulder, CO: University of Colorado, 1987.
- Jewell, D.L. (1977). Maximum security: Some obstacles to meaningful recreational programming. *Therapeutic Recreational Journal*, 11, 185-188.
- Johnson, M.C., & Kercher, G.A. (2007). ADHD, strain, and criminal behavior: a test of general strain theory. *Deviant Behavior*, 28(2), 131-152. doi: 10.1080/01639620601130992
- Kowalski, K.C., & Crocker, P.R.E. (2001). Development and validation of the Coping Function Questionnaire for adolescents in sport. *Journal of Sport & Exercise Psychology*, 23(2), 136-155. doi: 10.1123/jsep.23.2.136

- Krichbaum, D., & Alston, M.A. (1991). Youth restitution & recreation: A successful mix. *Parks and Recreation*, 26(3), 42-45.
- Lutzn, S.G., & Orem, R.C. (1967). *Prevention through recreation*. In Amos, W.A., & Wellford, C.F. (Eds.), *Delinquency Prevention: Theory and Practice* (150 - 170). Englewood Cliffs, New Jersey: Prentice-Hall, Inc.
- Mandell, R.D. (1984). *Sport: A cultural history*. New York: Columbia University Press.
- Mitrofan, O., Paul, M., & Spencer, N. (2009). Is aggression in children with behavioural and emotional difficulties associated with television viewing and video game playing? A systematic review. *Child: care, health and development*, 35(1), 5-15. doi: 10.1111/j.1365-2214.2008.00912.x
- Munson, W.W. (1991). Juvenile delinquency as a societal problem and social disability: The therapeutic recreator's role as ecological change agent. *Therapeutic Recreation Journal*, 25(2), 19-30.
- Nelson, M. C., & Gordon-Larsen, P. (2006). Physical activity and sedentary behavior patterns are associated with selected adolescent health risk behaviors. *Pediatrics*, 117(4), 1281-1290.
- Orton, D.J. (1977). An investigation of the past, present and future recreation pursuits of adult inmates in two Iowa correctional institutions. *Therapeutic Recreation Journal*, 2, 66-69.
- Pratt, T. C., Cullen, F., Blevins, K., Daigle, L., & Unnever, J. (2002). The relationship of attention deficit hyperactivity disorder to crime and delinquency: A meta-analysis. *International Journal of Police Science and Management*, 4(4), 344-360. doi: 10.1350/ijps.4.4.344.10873
- Robertson, L.A., McAnally, H.M., & Hancox, R.J. (2013). Childhood and adolescent television viewing and antisocial behavior in early adulthood. *Pediatrics*, 131(3), 439-446.
- Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton, New Jersey: Princeton University Press.
- Segrave, J.O. (1982). *An investigation into the relationship between participation in interscholastic athletics and delinquent behavior*. Unpublished Doctoral Dissertation. Tempe, Arizona: Arizona State University.
- Segrave, J.O. (1983). Sport and juvenile delinquency. *Exercise and Sport Sciences Reviews*, 11(1), 181-209.
- Shanas, E. (1942). *Recreation and delinquency: A study of five selected Chicago communities*. Chicago: University of Chicago Press.
- Stark, C.L., Kent, L., & Finke, R. (1987). *Sports and delinquency*. In Gottfredson, M.R., & Hirschi, T. (Eds.), *Positive criminology* (115-124). Newbury Park, California: Sage Publications.
- Stevens, T., & Mulsow, M. (2006). There is no relationship between television exposure and symptoms of attention-deficit/hyperactivity disorder. *Pediatrics*, 117(3), 665-672.
- Stevenson, C.L. (1985). *College athletes and "character": The decline and fall of socialization research*. In Chu, D.B., Segrave, J.O., & Becker, B.J. (Eds.), *Sport and higher education* (249-266). Champaign, Illinois: Human Kinetics Publishers, Inc.
- Todd, A.J. (1941). Recreation and delinquency. *Federal Probation*, 5(4), 19-24.
- Truxal, A.G. (1929). *Outdoor recreation legislation and its effectiveness*. New York: Columbia University Press.
- United States Department of Justice. (1981). *Executive summary of the national evaluation of prevention final report*. San Francisco: National Council on Crime and Delinquency Research Center.
- Van Egmond-Fröhlich, A.W., Weghuber, D., & de Zwaan, M. (2012). Association of symptoms of attention-deficit/hyperactivity disorder with physical activity, media time, and food intake in children and adolescents. *PloS one*, 7(11), e49781.
- World Medical Association. (2001). Declaration of Helsinki: Ethical principles for medical research involving human subjects. *Bulletin of the World Health Organization*, 79(4), 373-374.
- Wylie, J.A. (1960). Recreation and delinquency. *The Journal of Education*, 143, 14-20.
- Wylie, J.A. (1965). Recreation and juvenile delinquency. *Education*, 86, 86-90.
- Žužul, M. (1989). *Agresivno ponašanje: psihologijska analiza*. [Aggressive behavior: Psychological analysis - In Croatian.] Zagreb: Radna zajednica Republičke konferencije Saveza socijalističke omladine Hrvatske.

ORIGINAL SCIENTIFIC PAPER

Attitudes of Young School Aged Childrens to the Teaching of Physical Education

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Abstract

Physical Education as a school subject has a great influence on the development of psycho-physical abilities of children. In view of this, it is important to know what the attitude of students towards the teaching of Physical Education is. The aim of our research was to determine exactly what the attitudes of the junior schoolchildrens toward the teaching of Physical Education are in the function of improving or changing the teaching process. We tried to answer the question using anonymous standardized questionnaires adapted to this research. The research involved 327 students of the 4th and 5th grade primary schools from Podgorica and Niksic. The results showed that almost all students are eager to attend classes of Physical Education, which they are very pleased to consider contributing to their health and physical development, as well as the development of friendship, strength, care, value and endurance. Most students believe that Physical Education is as important as other subjects, and that Physical Education is best performed in a schoolroom. A large number of pupils would also not change anything compared to the previous lessons. As far as favorite sports are concerned boys traditionally prefer to play football, while girls are more oriented towards volleyball, handball, rhythmic and gymnastics.

Key words: Teaching, Physical Education, Student Attitudes, Young School Age

Uvod

Fizičko vaspitanje je specifično vaspitno područje koje doprinosi razvoju čovjeka u cjelini, to nije samo podsticanje rasta i razvoja, razvijanje tjelesnih sposobnosti i unaprjeđenje zdravlja, već i znalačko korišćenje sistema fizičkih vježbi, igara i sportova, kojima se utiče na cjelokupni razvoj čovjeka (Jovanović, 1998, prema Šimleša i Potkonjak, 1989). Sličnu definiciju dali su Krulj, Kačapor i Kulić (2001), prema kojima fizičko vaspitanje predstavlja sistematski i organizovan proces sticanja motoričkih vještina, znanja i sposobnosti, jačanja zdravlja i razvoja psiho-fizičkih snaga i sposobnosti vaspitanika. Na osnovu navedenog, primjećuje se da fizičko vaspitanje predstavlja sistemski proces, koji nema uticaj samo na razvoj tjelesnih sposobnosti djeteta i čovjeka, već na njihov cjelokupni psiho-fizički razvoj (Bjelica i Petković, 2009; Bjelica i Krivokapić, 2010a, Bjelica i Krivokapić, 2019). Kao takvo, Fizičko vaspitanje doprinosi normalnom razvoju organizma,

jačanju zdravlja, snaženju i čeličenju organizma, osposobljavanje za raznovrsnu pokretljivost, stvaranju higijenskih navika i obezbjeđivanju aktivnog odmora (Bakovljević, 1997). Ukoliko se uzme u obzir činjenica da savremeni način života redukuje fizičku aktivnost djece, to još više naglašava važnost Fizičkog vaspitanja kao školskog predmeta (Đorđić, 2005). Za sprovođenje kvalitetne nastave neophodni su materijalni uslovi, kao i kompetentni nastavnici, koji bi trebalo da podstiču pozitivne stavove učenika prema nastavi Fizičkog vaspitanja (N. Zrnzević i J. Zrnzević, 2015; Dragutinovic & Mitrovic, 2019a).

Stavovi predstavljaju trajnu tendenciju da se prema nekom objektu reaguje na određeni, pozitivan ili negativan način (Rot, 1983). Oni omogućavaju sticanje fiksiranih standarda za lično prosuđivanje i shvatanje svojih postupaka. Na taj način lakše se ocjenjuju i klasifikuju objekti i situacije, što omogućava lakše i brže snalaženje i djelovanje. Kako bi nastava Fizičkog vaspitanja imala pozitivan uticaj na cjelokupni razvoj

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**Montenegro
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djeteta, neophodno je da ono razvije pozitivne stavove prema njoj. Važnu ulogu u formiranju stavova djece prema Fizičkom vaspitanju imaju nastavnici, koji bi trebalo da motivišu djecu i upućuju ih na važnost fizičkog vježbanja za razvoj i zdravlje (Dragutinovic & Mitrovic, 2019b). Pozitivne stavove je moguće formirati i kroz razne zanimljive sadržaje u procesu nastave, za šta je takođe neophodna kreativnost nastavnika. U plan nastave je neophodno uvrstiti što više takmičarskih igara, akrobatike i vježbi, različitih po svojoj strukturi, kako bi se zadovoljile dječje potrebe za kretanjem i fizičkim razvojem, ali i njegove emocionalne potrebe (Bjelica i Krivokapić, 2010b; Bjelica i Krivokapić, 2011a). Ukoliko je nastava jednolična, djeca sigurno neće steći veću naklonost prema ovom predmetu. Sa razvijanjem pozitivnih stavova neophodno je započeti što ranije, kako bi se formirala dobra osnova za dalji razvoj djeteta, kao i stekla navika bavljenja fizičkim aktivnostima i sportom.

S obzirom na važnost ovog školskog predmeta za razvoj

djece, neophodno je ispitati kakve stavove djeca imaju prema nastavi Fizičkog vaspitanja, kako bi utvrdili da li ona odgovara njihovim potrebama. Stoga, osnovni cilj ovog rada je utvrditi stavove djece mlađeg školskog uzrasta prema nastavi Fizičkog vaspitanja. Na osnovu dobijenih rezultata, moguće je preduzeti određene mjere u pojedinim segmentima nastave, kako bi se ona poboljšala i omogućila zadovoljenje učenikovih potreba.

Metod rada

Uzorak ispitanika činilo je 327 učenika oba pola (180 dječaka i 147 djevojčica) četvrtog i petog razreda osnovnih škola "Luka Simonović" iz Nikšića i "Vuk Karadžić" iz Podgorice. Uzorak je podijeljen na četiri subuzorka, prema kriterijumima pola i uzrasta. Prvi uzorak činilo je 110 učenika četvrtog razreda, drugi subuzorak je činilo 70 učenika petog razreda, treći subuzorak je činilo 74 učenika četvrtog razreda i četvrti subuzorak je činilo 73 učenice petog razreda (Tabela 1).

Tabela 1. Uzorak ispitanika

Pol/Uzrast	IV razred	V razred	Ukupno
Dječaci	110	70	180
Djevojčice	74	73	147
Ukupno	184	143	327

U istraživanju je korišćen anonimni standardizovani upitnik (E. Međedović, Murić i A. Međedović, 2005) koji je modifikovan i prilagođen istraživanju. Upitnik se sastojao od 10 pitanja, od kojih je devet bilo zatvorenog tipa, a jedan otvorenog tipa. Učenici su popunjavali upitnik u školi, za šta je saglasnost dao direktor škole. Popunjavanje upitnika je trajalo 15 minuta. Svi učenici su dobrovoljno popunjavali upitnike, i mogli su da odustanu u bilo kom trenutku, ali to niko nije učinio. Priкупljanje podataka je takođe sprovedeno u školi. Nakon pregleda, iz istraživanja su isključeni nepotpuno i nejasno popunjeni

upitnici. Rezultati su prikazani tabelearnno (u procentima) i deskriptivno. Statistički značajne razlike u varijablama s obzirom na pol utvrđene su Pirsonovim H2 (HI kvadrat) testom na nivou značajnosti od $p < 0.05$.

Rezultati

U Tabeli 2 su prikazani rezultati upitnika za svako pitanje i pojedinačne odgovore kod djevojčaka četvrtog i petog razreda. Vrijednosti rezultata su prikazani brojačano i procentualno.

Tabela 2. Stavovi učenika IV i V razreda o Fizičkom vaspitanju

Pitanja	Odgovori	Broj	Procentat (%)
1. Da li sa zadovoljstvom odlaziš na časove fizičkog vaspitanja?	a) Da	161	89
	b) Ne	19	11
2. Šta najviše voliš da radiš na časovima fizičkog vaspitanja?	a) Trčim	16	9
	b) Igram igre	136	75
	c) Vježbam	28	16
3. Da li ste zadovoljni nastavom fizičkog vaspitanja u vašoj školi?	a) Potpuno	154	85
	b) Djelimično	10	6
	c) Nezadovoljan	16	9
4. Da li vježbanje i igranje na časovima fizičkog vaspitanja pozitivno utiče na zdravstveno stanje i razvitak učenika?	a) Da	163	89
	b) Ne	9	6
	c) Šteti zdravlju	8	5
5. Da li bi više volio/voljela da imaš:	a) Lijepo i moderno odijelo	21	12
	b) Zdravo i razvijeno tijelo	159	88
6. Zaokruži tri sposobnosti koje se po tvom mišljenju naročito mogu razviti na časovima fizičkog vaspitanja:	a) Pamćenje,	23	4
	b) Snaga,	95	18
	c) Drugarstvo,	78	14
	d) Brzina,	136	25
	e) Vrednoća,	44	8
	f) Izdržljivost,	156	29
	g) Muzikalnost	8	2

(nastavak je na sledećoj strani)

(nastavak sa prethodne strane)

Pitanja	Odgovori	Broj	Procenat (%)
7. Šta je za učenika najbolje?	a) Da pravilno igra	43	24
	b) Da rješava zadatke iz matematike	14	8
	c) I jedno i drugo	123	68
8. Od igara koje imamo meni se posebno dopadaju:	a) Igre u učionici	8	5
	b) Igre u dvorištu	29	16
	c) Igre u sali	143	79
9. Kada bi ste bili u prilici da nešto mijenjate u nastavi fizičkog vaspitanja, mijenjali bi ste?	a) Ništa	151	84
	b) Uslove	7	4
	c) Program	22	12

Posmatrajući Tabelu 2, uviđamo da su dječaci, prilikom odovaranja na prvo pitanje, jasno stavili do znanja da većina njih 89% voli da prisustvuje časovima Fizičkog vaspitanja, te da im ti časovi predstavljaju zadovoljstvo. Ostatak dječaka je izrazio negativan stav prema ovom predmetu (11%).

Ono što najviše vole na časovima jesu igre, jer je njih 75% odgovorilo da su im to omiljene aktivnosti, dok su vježbanje (16%) i trčanje (9%) aktivnosti koje interesuju manji broj dječaka. Shodno tome, proističu rezultati sljedećeg pitanja, a koji se tiču samog procesa organizacije nastave. Dječaci su većim dijelom zadovoljni organizacijom ovih časova (85%), ali postoje i oni koji su djelimično zadovoljni (6%) i koji bi mijenjali cjelokupnu nastavu (9%).

Koliko su djeca ovog uzrasta svjesna o važnosti ovog predmeta za zdravlje i razvoj, ukazuju odgovori koje su dječaci dali na četvrto i peto pitanje. Njih 89% je svjesno toga da nastava Fizičkog vaspitanja doprinosi zdravlju, a neki mali procenat se izjasnio da nastava ovog predmeta ne doprinosi zdravlju (6%) ili čak da šteti istom (5%). Sljedeće pitanje se samo nadovezuje, i prosto daje jasniju sliku, koliko su zapravo djeca svjesna važnosti pravilnog razvoja i rasta, te je tako njih 88% odgovorilo da želi zdravo i razvijeno tijelo, dok je njih 12% stavilo zdravlje po strani.

Tabela 3. Stavovi učenika IV i V razreda o Fizičkom vaspitanju

Pitanja	Odgovori	Broj	Procenat (%)
1. Da li sa zadovoljstvom odlaziš na časove fizičkog vaspitanja?	a) Da	139	95
	b) Ne	8	5
2. Šta najviše voliš da radiš na časovima fizičkog vaspitanja?	a) Trčim	23	16
	b) Igram igre	93	63
	c) Vježbam	31	21
3. Da li ste zadovoljni nastavom fizičkog vaspitanja u vašoj školi?	a) Potpuno	134	91
	b) Djelimično	10	7
	c) Nezadovoljan	3	2
4. Da li vježbanje i igranje na časovima fizičkog vaspitanja pozitivno utiče na zdravstveno stanje i razvitak učenika?	a) Da	108	73
	b) Ne	36	24
	c) Šteti zdravlju	3	3
5. Da li bi više volio/voljela da imaš:	a) Lijepo i moderno odijelo	6	4
	b) Zdravo i razvijeno tijelo	141	96
6. Zaokruži tri sposobnosti koje se po tvom mišljenju naročito mogu razviti na časovima fizičkog vaspitanja	a) Pamćenje,	26	6
	b) Snaga,	101	23
	c) Drugarstvo,	82	19
	d) Brzina,	103	23
	e) Vrednoća,	63	14
	f) Izdržljivost,	53	12
	g) Muzikalnost	13	3

(nastavak je na sledećoj strani)

(nastavak sa prethodne strane)

Pitanja	Odgovori	Broj	Procenat (%)
7. Šta je za učenika najbolje?	a) Da pravilno igra	14	9
	b) Da rješava zadatke iz matematike	20	14
	c) I jedno i drugo	113	77
8. Od igara koje imamo meni se posebno dopadaju:	a) Igre u učionici	9	6
	b) Igre u dvorištu	16	11
	c) Igre u sali	122	83
9. Kada bi ste bili u prilici da nešto mijenjate u nastavi fizičkog vaspitanja, mijenjali bi ste?	a) Ništa	131	90
	b) Uslove	8	5
	c) Program	8	5

Djevojčice su, kod prvog pitanja, jasno stavile do znanja koliko vole časove Fizičkog vaspitanja (95%), samo njih 5% je odgovorilo negativno. Kao i kod dječaka, najviše vole igre tokom ovih časova (63%), dok vježbanje (21%) i trčanje (16%) nešto manje. Odgovorom na treće pitanje, izjasnile su se da su zadovoljne organizacijom nastave ovog predmeta, u potpunosti njih 91%, dok je djelimično bilo 7% djevojčica i samo 3% nezadovoljnih.

Njihovi odgovori vezani za uticaj ovog predmeta na zdravlje i razvoj, dali su uvid u to koliko djevojčice zapravo znaju benefite koje on donosi sa sobom. U prilog tome govori činjenica da je na četvrtom pitanju njih 73% odgovorilo pozitivno, dok je ostatak neupućen. One su i kod petog pitanja, gdje su trebale da se odluče za zdravlje, u većini odgovorile pozitivno (96%).

S obzirom da su dječaci većinom zaokruživali brzinu, snagu

izdržljivost, kao sposobnosti koje se na ovim časovima razvijaju, djevojčice su na tom pitanju dale različite odgovore. Takođe, su ove tri sposobnosti bile u većini, ali su drugarstvo (19%) i vrednoća (14%), bile nešto izraženije kod njih.

Između predmeta nisu pravile razliku, te su tako kod Matematike i Fizičkog vaspitanja, odlučile se za oba predmeta (77%), ali je 14% njih odgovorilo da su ipak važniji zadaci iz Matematike. Kao najbolji prostor za izvođenje nastave i one su odabrale fiskulturnu salu 83%, zatim igre u dvorištu 11% i naposljetku u učionici 6%.

Svoje zadovoljstvo nastavom iskazale su visokim procentom (90%), ali su podijeljena mišljenja kod mijenjanja programa 5% i uslova 5%. Kod omiljenih sportova, djevojčice su se prije odlučivale za rukomet, odbojku, ritmiku i gimnastiku.

Tabela 4. HI kvadrat test razlika stavova o Fizičkom vaspitanju između dječaka i djevojčica

Variables	Sig.
Da li sa zadovoljstvom odlaziš na časove fizičkog vaspitanja?	.017
Šta najviše voliš da radiš na časovima fizičkog vaspitanja?	.062
Da li ste zadovoljni nastavom fizičkog vaspitanja u vašoj školi?	.413
Da li vježbanje i igranje na časovima fizičkog vaspitanja pozitivno utiče na zdravstveno stanje i razvitak učenika?	.082
Da li bi više volio/voljela da imaš:	.013
Zaokruži tri sposobnosti koje se po tvom mišljenju naročito mogu razviti na časovima fizičkog vaspitanja: a) snaga	.078
b) brzina	.000
c) izdržljivost	.000
Šta je za učenika najbolje?	.001
Od igara koje imamo meni se posebno dopadaju:	.390
Kada bi ste bili u prilici da nešto mijenjate u nastavi fizičkog vaspitanja, mijenjali bi ste?	.079

Note: Sig. – statistička značajnost

Prikaz Tabele 4, nakon sprovedenog HI kvadrat testa, pokazuje gdje su postojale razlike u stavovima djevojčica i dječaka, kada je u pitanju Fizičko vaspitanje. Statistički značajne razlike su se javile kod prvog, petog, šestog i sedmog pitanja. Ono što je značajno, jeste da su kod oba pola preovladavali pozitivni stavovi.

Diskusija

Sam proces organizacije nastave Fizičkog vaspitanja iziskuje mnogo truda i kreativnosti, a pored toga i prilagođavanje nastave djeci različitog pola. Nekada su dječaci pokazivali veću sklonost ka Fizičkom vaspitanju i aktivnostima, što se može objasniti time da dječaci po uzoru na očeve većinom su uključeni u sportske aktivnosti, a djevojčice manje (Džibrić, Bašinc, Biberović i Tuzović, 2014). Međutim, to sada i nije slučaj, što se jasno može vidjeti kroz rezultate ovog istraživanja.

Kada je u pitanju zainteresovanost i zadovoljstvo prilikom

nastave Fizičkog vaspitanja, veći procenat je pripao ženskom polu, odnosno djevojčicama. Baš na tom pitanju se i javila statistički značajna razlika između dječaka i djevojčica, koje su odgovorima potvrdile da zaista uživaju u fizičkim aktivnostima. S tim što tokom nastave, oba pola, radije učestvuju u igrama, nego što vježbaju ili trče. Tu su poprilično ujednačeni u svojim stavovima i nemamo statistički značajnih razlika. Oba pola, su takođe izrazila svoje zadovoljstvo procesom organizacije nastave u njihovim školama. Većina je izrazila pozitivan stav, što znači da je ovaj predmet konačno na pravom putu, gdje će se shvatiti važnost i uvažavati isti kao i svi ostali.

Učenici mlađeg školskog uzrasta su pokazali svjesnost i informisanost o važnosti Fizičkog vaspitanja za integralni razvoj ličnosti, baš kao i u prethodno sprovedenim istraživanjima (Radovanović, Madić i Nikolić, 2003; Radisavljević i Višnjić, 2004; D. Lazarević, Orlić, B. Lazarević i Radisavljević, Jelić, 2015). Zdravlje, kao veoma važan faktor za uspješan i dobar život, pospješuju

fizičke aktivnosti (Bjelica i Krivokapić, 2011b, Bjelica i Krivokapić, 2019). Veoma je važno, još kod mlađih uzrasta da razvijemo naviku i ljubav prema njima, a sve to neće biti moguće ukoliko se djeci ne predoče sve pozitivne strane. Djeca ovog uzrasta su pokazala da znaju koliko aktivnosti doprinose njihovom zdravlju i kod oba pola, u većini, imamo pozitivne stavove. Ono što je drugačije, jeste činjenica, da su se kod djevojčica, prilikom davanja odgovora na peto pitanje, javile statistički značajne razlike u njihovu korist u odnosu na dječake. Tu je veći procenat djevojčica odlučio u korist zdravog i razvijenog tijela.

Šesto pitanje, kod kojeg je bilo dato da djeca zaokruže tri odgovora za koje smatraju da su tačni, dalo je statistički značajne razlike između dječaka i djevojčica. Ono što je za pohvalu, jeste da pored sposobnosti koje su karakteristične za ovaj predmet, djeca prepoznaju i druge koje se razvijaju tokom procesa ove nastave. Kako fizičko vaspitanje i sama fizička aktivnost van škole podstičajno djeluju na djecu, njihovu popularnost u društvu i način za druženje, sve to kod djece stvara pozitivne stavove o nastavi Fizičkog vaspitanja (Švraka, 2012). Djeca koja se bave sportom uvijek su prihvaćena u društvu i imaju osjećaj za timski rad, što dovodi do jednog skladnog života i razvoja socijalne ličnosti (Švraka, 2012). Razlika se javila, zato što su djevojčice istakle da se drugarstvo i vrednoća razvijaju kroz ovaj predmet, što je kod dječaka bilo u manjem procentu.

Ono što se često pominje u školama, a veoma je važno, pogotovo u mlađem školskom uzrastu, jeste odnos prema Fizičkom vaspitanju kao predmetu. Često se smatralo da je to predmet koji je manje bitan i gdje svako može da ima visoku ocjenu, što je dovelo do nezainteresovanosti tokom nastave. Upravo zbog toga se u ovom pitanju upoređivala Matematika i Fizičko vaspitanje, da bi od djece saznali šta zapravo oni misle. Većina je rekla da su oba predmeta bitna, ali onaj procenat koji se odlučio za jedan ili drugi predmet je doveo da statistički značajnih razlika kod ovog pitanja. Dječaci su u većini odlučili da je bitnije pravilno igranje, od pravilnog rješavanja zadataka, što je kod djevojčica bio obrnut slučaj.

Oba pola su bila složna kod odabira prostora za izvođenje nastave Fizičkog vaspitanja, a to je fiskulturna sala, koja se zbog prostora i opremljenosti rekvizitima čini interesantnija i primamljivija za učenje. Takođe, dvorište je pogodno za realizovanje aktivnosti, dok učionica, zbog samih uslova i deficit slobodnog prostora u oba slučaja je imala najmanji procenat. Zadovoljstvo nastavom, koje su negdje kroz prva pitanja iskazali, potvrdili su i u pretposljednem, gdje je većina dječaka i djevojčica odgovorila da ne bi ništa mijenjali.

Deseto pitanje, koje nije prikazano u tabeli, bilo je otvorenog tipa, a djevojčice i dječaci su birali omiljeni sport. Kako je kod nas, pa tako i u svijetu najpopularniji fudbal, tako su se dječaci opredijelili za njega, dok su djevojčice imale različite odgovore gdje su za svoje omiljenje sportove istakle: odbojku, rukomet, gimnastiku i ritmiku. Moglo bi se reći da je to negdje i očekivano zbog genetskih predispozicija, ali i kako je već navedeno, onoga što se plasira u njihovoj okolini.

Mjere za poboljšanje istraženog i predstavljenog stanja u pitanju stavova i poznavanja problematike fizičkog vaspitanja od strane učenika mogu se sagledati u više pravaca. Ono što treba promijeniti, da se trenutno stanje popravi, nije tako lako predložiti i prihvatiti. Međutim, moramo shvatiti da i tu postoji redoslijed važnijih i manje važnih zadataka. Treba najprije poći od našeg shvatanja da je najvažnije u radu zdravlje i vaspitanje, a poslije ovoga su operativni programski zadaci. U iznijetom izlaganju dotakli smo se samo dijela problematike vezane za sprovođenje nastave fizičkog vaspitanja. Jedan od značajnih pravaca daljeg razvoja i preobražaja ove

aktivnosti je doslednije angažovanje svih onih koji vode brigu o vaspitanju i obrazovanju učenika, da preuzmu određene obaveze, odgovornost za stalnu brigu o nastavi fizičkog vaspitanja. Da nastoje poboljšati materijalne uslove kao i da organizuju sportske aktivnosti na način koji obezbeđuje neposrednije angažovanje učenika.

Programi fizičkog vaspitanja u školi, u budućnosti, sve više trebaju se zasnivati na stvarnim potrebama učenika, njegovih osobina, sposobnostima i interesima (Bjelica i Krivokapić, 2011a). Jedno od osnovnih pitanja u nastavi fizičkog vaspitanja, jeste pitanje cilja i zadataka. U tom pogledu fizičko vaspitanje mora se bazirati na sve izrađenijim potrebama i interesima učenika za obogaćivanjem i osmišljavanjem programa.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 12 May 2019 | **Accepted:** 29 July 2019 | **Published:** 11 October 2019

References

- Bakovljević, M. (1997). *Osnovi metodologije pedagoških istraživanja*. Beograd: Naučna knjiga.
- Bjelica, D. i Krivokapić, D. (2010a). *Teorijske osnove fizičke kulture*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. i Krivokapić, D. (2010b). Učenje kroz igru. In *Book of abstract 2nd International Scientific Conference "Anthropological aspects of sports, physical education and recreation"* (25). Banja Luka: Faculty of Physical Education and Sports.
- Bjelica, D. i Krivokapić, D. (2019). *Teorija sporta i tjelesnog vježbanja*. Nikšić: Fakultet za sport i fizičko vaspitanje, Podgorica: Crnogorska sportska akademija.
- Bjelica, D. i Krivokapić, D. (2011a). *Teorija igre*. Nikšić: Fakultet za sport i fizičko vaspitanje, Podgorica: Crnogorska sportska akademija.
- Bjelica, D. i Krivokapić, D. (2011b). Zdravstveno-preventivna uloga tjelesne aktivnosti omladine. U *Zborniku radova VI Međunarodne konferencije "Menadžment u sportu"* (144-149). Beograd: Alfa univerzitet, Fakultet za menadžment u sportu, Olimpijski komitet Srbije.
- Bjelica, D. i Petković, J. (2009). *Teorija fizičkog vaspitanja i osnove školskog sporta*. Podgorica: Crnogorska sportska akademija, Nikšić: Fakultet za sport i fizičko vaspitanje.
- Đorđić, V. (2005). *Školsko fizičko vaspitanje*. Novi Sad: Fakultet sporta i fizičkog vaspitanja.
- Dragutinović, K., & Mitrović, M. (2019a). Differences in the Attitudes of Fourth Grade Pupils of the Elementary School According to the Teaching of Physical Education in Terms of Gender. *Journal of Anthropology of Sport and Physical Education*, 3(1), 29-34.
- Dragutinović, K., & Mitrović, M. (2019b). Teachers attitudes of the teaching of Physical Education. In *Book of Abstract XVI Annual Scientific Conference "Sport, Physical Activity and Health: Contemporary Perspectives"* (100-101). Podgorica: Crnogorska sportska akademija, Nikšić: Fakultet za sport i fizičko vaspitanje.
- Džibrić, Dž., Bašinac, I., Biberović, A. i Tuzović, A. (2014). Razlike u stavovima srednjoškolaraca prema fizičkom vaspitanju. U *Zborniku radova X međunarodne konferencije "Izazovi savremenog menadžmenta u sportu"* (95-100). Beograd: Alfa univerzitet-Fakultet za menadžment u sportu, Sarajevo: Fakultet sporta i tjelesnog odgoja.
- Krulj, Kačapor, Kulić (2001): *Pedagogija*. Beograd: Svet knjige.
- Lazarević, D., Orlić, A., Lazarević, B. i Radisavljević-Jelić, S. (2015). Stavovi učenika ranog adolescentnog uzrasta prema fizičkom vaspitanju. *Fizička kultura*, 69(2), 88-98.
- Međedović, E., Murić, B. i Međedović, A. (2005). Stavovi učenika četvrtih razreda osnovnih škola prema nastavi fizičkog vaspitanja. U *Zbornik radova XI Nacionalnog naučnog skupa sa međunarodnim učešćem "Fis komunikacije 2005"* (92-99). Niš: Fakultet fizičke kulture.
- Radisavljević, S. i Višnjić, D. (2004). Stavovi učenika prema fizičkom vaspitanju. *Godišnjak*, 12, 141-153.
- Radovanović, Đ., Madić, D. i Nikolić, M. (2003). Mišljenje učenika petog razreda o nastavi fizičkog vaspitanja u mlađim razredima osnovne škole. U *Zbornik radova X Nacionalnog naučnog skupa sa međunarodnim učešćem "Fis komunikacije 2003"* (90-93). Niš: Fakultet fizičke kulture.
- Rot, N. (1983). *Osnove socijalne psihologije*. Beograd: Zavod za udžbenike i nastavna sredstva

Šimleša, P. i Potkonjak, N. (1989). *Pedagoška enciklopedija*. Beograd: ZZUINS.

Švraka, N. (2012). Stavovi mlađeg školskog uzrasta o pripadnosti grupi u nastavi fizičkog vaspitanja. *SVAROG*, 4, 294 – 302.

Zrnzević, N. i Zrnzević, J. (2015). Efekti nastave fizičkog vaspitanja na funkcionalne sposobnosti učenika mlađeg školskog uzrasta. *Glasnik Antropološkog društva Srbije*, 50, 1-9.

ORIGINAL SCIENTIFIC PAPER

Evaluation of Municipal Fitness Programs for Women with Low Back Pain

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Abstract

The present study examines the impact among several factors and the Lower Back Pain (LBP) in 131 women, participating in municipal physical activity programs who declared LBP. The purpose was to investigate the effect of municipal sports programs in female's participants LBP intensity. Data from 131 LBP sufferer women collected in the beginning and after a six months period. Municipal physical activity programs did not affect negatively LBP. Specific features participants (mothers) present greater risk in future to appear LBP. Municipal physical activity programs reduce the felling of LBP in women without children.

Key words: *Physical Activity, Pain, Movement Restriction, Health*

Introduction

In 2000, a European Community working group met with the phenomenon of acute undetected pain in the lumbar spine. Experts from almost all countries in the European zone took place in this meeting, to identify potential harmful factors, to establish a framework for prevention and to formulate guidelines for the treatment of LBP among the field's professionals. In the results, the working group made widely known the poor association of the LBP feeling in connection to the radiographic diagnosis, as it often did not coincide with the pain in the particular area of the patient. It also included guidelines for exercise as a mean of preventing and of reducing LBP during its chronic phase. Advices were also given on what kind and at which intensity of exercise is appropriate.

LBP defined as the pain followed by concomitant discomfort, located between the lateral and the lower gluteus folds (Haryono, Kawilarang, & Prastowo, 2019). Acute LBP defined as the one lasts less than 6 weeks, sub acute the one between 6-12 weeks, while for more than 12 weeks characterized as chronic. Pain divided in two categories according to the causal risk factors. Red-flag sign defined the group who has predisposing factors to experience LBP. This category includes ages under 25 and over 55 years, recent history of violent trauma,

steadily worsening non-mechanical pain (not receding with bedtime), chest pain, malignant tumor medical history, prolonged use cortisone, unexplained weight loss, and various other factors of minor importance (Waddell, Feder, McIntosh, Lewis, & Hutchinson, 1998). The yellow-flag sign refers to secondary risk factors for LBP. Participants according to their working group stated that, work satisfaction, emotional issues (stress, depression, etc.), pain management, and low levels of exercise are factors that cause chronic LBP (Kendall, Linton, & Main, 1997).

The cost of any form of LBP evaluated as GDP percentage. In Finland, the total cost reaches 0.8% (Heikki, 2002), in the Netherlands 1.7% (Van Tulder, Koes, & Bouter, 1995), in Sweden 1.7% (Nachemson, 1991), in the United States is 2.2% (Frymoyer & Cats-Baril, 1991) of the GDP's of each country. UK has the highest numbers of back pain related absenteeism in the EU, nearly 10 million workdays were lost to back pain in 2014, (Miller, 2014), where the cost also reaches 2% of GDP (Maniadakis & Gray, 2000), agreed with survey results presented shown that 75-85% of absenteeism were due to back pain (Andersson, 1999; Waddell et al., 1998). Artur (2014), underline that LBP is the leading cause of disability for most of countries in Europe, North Africa and a part of Latin America. The majority of researches have so far not addressed LBP issues

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in municipal sports programs, except one research which dealt with this kind of issues (Kosmas, Marmara, & Stergioulas, 2008).

The risk factors for back pain emerged through surveys are, age (Bailey, 2009), body mass index (Heliövaara, 1989), and childbirth (Bailey, 2009). There is a clash about smoking as a risk factor for LBP. Some studies have shown LBP is associated with cigarette smoking (Heliövaara, Mäkelä, Knekt, Impivaara, & Aromaa, 1991; Liira, Shannon, Chambers, & Haines, 1996), while others have not shown a relationship (Smedly, Egger, Cooper & Coggon, 1995; Leboeuf-Yde, Kyvik, & Bruun, 1998).

Four factors were used establishing four research hypotheses as the base of the research. A) Body Mass Index (BMI). The higher the BMI the highest the LBP intensity (Spyropoulos et al., 2007). B) Childbirth. LBP intensity in women with childbirth is higher than those who have not (Bailey, 2009). C) Smoking. There is a connection of Smoking factor and LBP and D) Metabolic Equivalent (MET). Too high or too low physical activity levels deteriorate LBP (Lallukka et al., 2017). The purpose of this study was to outline the profile of the women participating in municipality sports programs, suffering from LBP. Based upon the profile and in relation with the independent variables, new guidelines for renewed physical activity municipality programmes could be created concerning the special needs of the population, contributing in better health and fitness results of the participants, and acting as factors for avoiding and reducing the risk of LBP evolvement.

Methods

Since January 2009, female Municipality Sports Programs participants from the municipalities of the region of Attica, Greece, asked to fill in a pen and pencil questionnaire with three parts; first the demographic questionnaire, second the

LBP severity questionnaire and third the activity evaluation in METs questionnaire. There was a written consent for their participation. The selection of the Municipalities based on the separation of the Prefecture of Athens into four (4) regional areas: central, eastern, southern and western (Athens Prefecture, 2008). Four (4) Municipalities selected from each regional sector. In each area, the most populated municipality, the less populated and two municipalities with random sampling selected for the survey (Thomas & Nelson, 2003). From the 635 fully answered and returned questionnaires, only 131 (20.7%) declared their LBP suffering and only these selected as participants. To assess the intensity of pain and the effect of the variables, descriptive statistics, t tests for independent samples and ANOVA independent sample carried out.

The mean values of the responses of the participants (dependent variables) compared with the independent variables (age, smoking, educational level, etc.). Data were analyzed using SPSS version 17.0 statistical software for Windows (SPSS, Inc., Chicago, IL, USA). The results presented as means (M) and Standard Deviations (SD), or as percentages. For continuous variables, the significance of the differences between the groups analyzed using one-way analysis of variance followed by post hoc comparisons within dependent samples t-test, and for categorical variables using Tukey criterion.

In a sample of 635 women, 131 (20.7%) reported LBP. The average age of women participating was at $M = 51 \pm 12.12$ years (Table 1). This means that the average age of participation in the municipal programs is about fifty-one years ($M=51$), ($S.D. = 12.11$) with an age range from min = 17 to max = 67 years. Our research, however, focused on those with LBP. Only the 131 participants declared LBP to investigate the demographic and habitual independent variables.

Table 1. The somatometric and demographic characteristics (mean values - standard deviations) of the participants in the present study (n = 131)

	M (n)= 131	S.D.
Age	41	1.10
Weight(kg)	70.61	0.71
Height(m)	1.67	0.005
BMI (kg/m ²)	25.31	3.44

Body weight of participants (N = 131), was 70kg ($M = 70.61 \pm 0.71$ kg) (min = 47 - max = 97), while their height 1.67m ($M = 1.67 \pm 0.005$) (min = 1.50 - max = 1.84). A percentage of 81.7% of the values are concentrated in the area of 1.50-1.71m. These variables combined provide the Body Mass Index (BMI).

Since January 2009, female Municipality Sports Programs participants from the municipalities of the region of Attica, Greece, asked to fill in a pen and pencil questionnaire with three parts; first the demographic questionnaire, second the LBP severity questionnaire and third the activity evaluation in METs questionnaire. There was a written consent for their participation.

LBP severity evaluated with the 'Grading the severity of chronic pain' questionnaire (Von Korff, Ormel, Keefe, & Dworkin, 1992). The questionnaire consists of eight questions (Q). Q1 How would you rate six months before your pain in the middle on a scale from 0 to 10 where 0 = no pain and 10 = the worst possible pain, q2 How would you currently rate your pain in the middle on a scale from 0 to 10 where 0 = no pain and 10 = the

worst possible pain?, q3 In the last six months, how intense was the worst back pain you felt on a scale from 0 to 10 where 0 = no pain and 10 = the worst possible pain?, q4 In the last six months, and on average, how severe was the waist pain you felt on a scale from 0 to 10 where 0 = no pain and 10 = the worst possible pain, q5 For how many days, in the last six months, have you abstained from your usual activities (work, school, home, sports) because of back pain? Abstinence days, q6 In the last six months, how has the pain acted in the middle of your everyday activities on a scale from 0 to 10 where 0 = no effect and 10 = impossible to do any activity?, q7 In the last six months how has your ability to participate in creative / social / family activities changed due to pain in the middle on a scale from 0 to 10 where 0 = no change and 10 = tremendous change?, and q8 In the last six months, how much pain has in the middle of your ability to work (including homework) on a scale of 0 to 10 where 0 = no change and 10 = huge change?. Participants report some observations they made about pain. For each observation they put a circle in a number from 0 to 6 to indicate whether their

physical activities such as crouching, weight lifting, walking, driving, sports affect or could affect their LBP at this time (0: absolutely disagree, 6: absolutely agree). For the evaluation of the activity of each participant in METs we used the questionnaire of Kriska and Caspersen (1997), which has been translated in Greek language in 2011 (Marmara, Papacharalambous, Kouloulis, Maridaki, & Baltopoulos, 2011). Physical exercise indices are expressed as energy expenditure variables per day or per week, and are usually given as calories per day (kcal / day) or as metabolic equivalents per day (MET / day). In various studies, physical activity is often categorized according to its type and intensity. Physical exercise habits are measured in the form of frequency and duration of physical activity. The types of physical exercise used in most studies include physical activity during working, leisure time, home routine, family care, etc. The intensity of physical activity represents the metabolic cost required to carry out the effort. Often effort is measured in MET units. A MET is the metabolic cost corresponding to the basal metabolism and equates to 4,184 kJ / kg / h, the energy spent in the sitting position or else with 3.5 ml of oxygen consumed per kg of body mass per minute, which is about 1 kcal / kg / h.

Results

The BMI groups defined as < 25 for normal body weight, = 25-30 for overweight, and > 30 for obese (Mimic, Vuki-

cevic, & Vujacic, 2019). BMI's results show that most of the participants are within the limits of normal body weight. The average ($M = 25.2$) is slightly above the woman's normal body weight limit (min = 18 & max = 35) and 30% has exceeded the accepted value [(Q5) $F(2,128) = 3.1$, $p < 0.05$, $F(2,128) = 3.93$, $p < 0.05$, $F(2,128) = 5.87$, $p < 0.05$, $F(2,128) = 3.09$, $p < 0.05$ respectively]. For the rest of the questions no statistically significant differences observed ($p < 0.05$, see Table 2).

Tukey post hoc analyzes for statistically significant differences showed that for Question 12c "In the last six months, how intense the worst LBP that you have felt", 25-30 group ($M = 3.63 \pm 0.91$) had a statistically significant difference with those who had MM. 31 < ($M = 2.9 \pm 0.55$) on pain intensity in the last six months. For Question 12f "In the last six months, how much pain has affected the LBP in your day-to-day activities" $0.31 < (M = 3.64 \pm 1.53)$ had a statistically significant difference with those who had MM. 25-30 ($M = 4.81 \pm 1.63$) and 0-24 ($M = 4.65 \pm 1.40$) on the effect of pain in the last six months on daily activities. For Question 4 (Q4) "I should not do activities that could aggravate my pain", 0-24 ($M = 5.32 \pm 1.54$) there was a statistical difference with those having MM. 31 < ($M = 4 \pm 1.69$) for activities that could exacerbate pain.

While for Question 5 (Q5) "I cannot do activities that would make my pain stronger", there were no statistically significant differences between women of different BMIs.

Table 2. Descriptive table ANOVA BMI $p < 0.05$

Question		Sum Square	df	Mean Square	F	p
que12a	Groups	1.554	2	.777	.543	.582
	Error	183.179	128	1.431		
	Total	184.733	130			
que12b	Groups	4.164	2	2.082	1.257	.288
	Error	212.065	128	1.657		
	Total	216.229	130			
que12c	Groups	6.594	2	3.297	3.100	.048*
	Error	136.109	128	1.063		
	Total	142.702	130			
que12d	Groups	2,368	2	1,184	1,078	.343
	Error	140,548	128	1,098		
	Total	142,916	130			
que12e	Groups	2.882	2	1.441	.391	.677
	Error	472.035	128	3.688		
	Total	474.916	130			
que12f	Groups	17.709	2	8.855	3.933	.022*
	Error	288.199	128	2.252		
	Total	305.908	130			
que12g	Groups	5.196	2	2.598	1.858	.160
	Error	178.972	128	1.398		
	Total	184.168	130			
que12h	Groups	5.641	2	2.820	1.794	.170
	Error	201.260	128	1.572		
	Total	206.901	130			
Q1	Groups	13.579	2	6.789	2.512	.085
	Error	345.963	128	2.703		
	Total	359.542	130			
Q2	Groups	1.244	2	.622	.327	.721
	Error					
	Total					

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Question		Sum Square	df	Mean Square	F	p
Q3	Error	243.305	128	1.901		
	Total	244.550	130			
	Groups	.419	2	.210	.089	.915
Q4	Error	300.375	128	2.347		
	Total	300.794	130			
	Groups	27.555	2	13.778	5.875	.004*
Q5	Error	300.170	128	2.345		
	Total	327.725	130			
	Groups	19.468	2	9.734	3.093	.049*
Q6	Error	402.777	128	3.147		
	Total	422.244	130			
	Groups	10.556	2	5.278	2.487	.087
	Error	271.688	128	2.123		
	Total	282.244	130			

The percentage of the participants who had given childbirth was 75.6%, and of those without childbirth were 24.4% (Table 3). This is indicative of the participation of women with children in

Primary Sports Programs where they are highly opposed to those without children. For this categorical variable (as for smoking variable) t tests for independent samples performed.

Table 3. The categorical variables (child – smoking – if there is a midwife – if there is a back pain) in absolute numbers and percentages. (n = 131).

n=131	NO	%	YES	%
Children	32	24.4	99	75.6
Smoking	95	72.5	36	27.5
Operation	131	100	0	0

In the control t panel (Table 4) we can observe that in four questions (Question 12d, Question 4 (Q4), Question 5 (Q5), Question 6 (Q6)) there are statistically significant differences [t (129), $p < 0.05$], t (129) = -2.27, $p < 0.05$, t (129) = -3.97, $p < 0.05$, t (129) = -2.29, $p < 0.05$, respectively. For the remaining questions, no statistically significant differences ($p < 0.05$, see Table 4) observed. The above statistical check implies that the highest

average of LBP within the last six months it was intense for those who had children compared with those who did not have (Question 12d). At the same time, childless participants seem to be more determined to engage in activities that would exacerbate their pain (Question Q4, Question Q5), and they think that sport reduced LBP. (Question Q6).

Table 4. Mean values, standard deviations, control t and statistical significance in women with children (N1 = 131) and without children (N2 = 504) in questions $p < 0.05$

Question	Children	No Children	t test	Sign.
que12a	3.51±1.11	3.50±1.41	0.062	0.950
que12b	3.64±1.10	4.31±1.67	-2.595	0.011
que12c	3.55±1.00	3.40±1.18	0.699	0.486
que12d	3.79±.89	3.06±1.29	3.605	0.000*
que12e	3.68±1.87	3.40±2.02	0.721	0.472
que12f	4.66±1.45	4.31±1.74	1.137	0.258
que12g	2.79±1.17	2.34±1.18	1.895	0.60
que12h	3.95±1.27	3.75±1.21	0.816	0.416
Q1	4.79±1.57	4.21±1.86	1.726	0.087
Q2	3.66±1.28	3.87±1.62	-0.746	0.457
Q3	4.12±1.43	4.15±1.79	-0.113	0.910
Q4	4.77±1.51	5.50±1.70	-2.272	0.025*
Q5	3.80±1.68	5.18±1.78	-3.972	0.000*
Q6	3.97±1.52	4.65±1.18	-2.294	0.023*

The majority of the participants do not smoke (72.5%) (Table 5). This evidences the hypothesis that those who are doing sport in a regular basis usually do not smoke. In the control t panel (Table 5), no statistically significant differences found

between smokers and non-smokers. A finding that shows that smoking does not differentiate the responses of the participants in the PMS programs for the LBP.

Table 5. Mean values, standard deviations, control t and statistical significance in smokers (N1 = 36) and non-smokers (N2 = 95) in questions

Question	Smokers	Non Smokers	t test	Sign.
que12a	3.53 ± 1.44	3.51 ± 1.09	0.96	0.924
que12b	3.69 ± 1.19	3.85 ± 1.42	-0.625	0.533
que12c	3.50±1.23	3.52±0.97	-.0.128	0.898
que12d	3.52±1.10	3.65±1.02	-0.607	0.545
que12e	3.72±1.92	3.57±1.91	0.382	0.703
que12f	4.47±1.57	4.62±1.52	-0.494	0.622
que12g	2.80±1.28	2.64±1.15	0.7	0.485
que12h	4.11 ± 1.30	3.83 ± 1.24	1.133	0.259
Q1	4.50±1.78	4.71±1.62	-0.662	0.509
Q2	3.61±1.53	3.75±1.31	-0.545	0.586
Q3	3.86±1.65	4.23±1.46	-1.247	0.215
Q4	4.97±1.66	4.94±1.56	0.080	0.937
Q5	4.19±1.78	4.12±1.81	0.192	0.848
Q6	4.19±1.45	4.12±1.48	0.235	0.814

Metabolism defined as the metabolic cost corresponding to the basic metabolism and considered as the energy spent in a seated sedentary position. It equals 1.0 (4.184 kJ) / kgr / h, or else with 3.5ml / kg / min of oxygen consumed i.e. about 1kcal / kg / h (Kriska & Caspersen, 1997). Ainsworth et al., (2000) set

the intensity limits for physical activity classification as follows: light < 3 MET, moderate 3-6 METs, heavy > 6 MET. Based on these limits and by dividing the METs of the participants, we can highlight some important elements. Table 6, shows a high average of those who have shown LBP ($M = 49.62 \pm 0.912$).

Table 6. The MET of exercising women's pain (mean-type, n = 131)

	M	S.D.
MET PAIN	49.62	0.912

Finally the presented results show that women choose the specific programs regardless of their low or high weight.

Discussion

The aim of the present study focus on the municipal physical activity programs for female participants. The main goals of these programs are to improve fitness and to increase activity levels of female participants, minimizing at the same time the fear of LBP, or the LBP itself, for those suffering of it. This study evaluated the effect of municipality sports programs for women considering sociodemographic and lifestyle factors to the improvement or not of the LBP. Measures of the self-declared replies in the beginning and six months after implementation can also provide useful results of the effect of the municipal sports program to the intensity of the LBP (Le Borgne, Boudoukha, Petit, & Roquelaure, 2017).

Hypotheses concerning the factors connected to the LBP are supported by data in this research. Variables such as age, Body Mass Index (BMI), Childbirth, Smoking, and too high or too low physical activity by counting the Metabolic Equivalent (MET), demonstrate statistical significance with the appearance and the intensity of LBP. Analyses of the given answers shows a greater effect of pain on participants with BMI between 25-30 & 0-24 compared to participants with BMI > 31. As indicated by the participants belong to these two groups of BMI (25-30 & 0-24), pain effected more their daily activities, without this intensity exceeding 5 on a scale from 1-10. Women with BMI 25-30 show more severe pain than those who had higher BMI ($31 \leq$) during the last six months, while the intensity did not exceed 5 on the above-mentioned scale. Participants with normal BMI (0-24) seemed to be more afraid of exercise at the risk

of feeling more intense pain than participants who were obese ($31 <$). Similar surveys' results have shown that the highest the BMI, the more intensive the pain (Björck-van et al., 2008). LBP seems to be statistically significant higher ($p = 0.000$) in the female group who had been given birth than those who had not. Baily (2009) states that one of the reasons concerns hormonal factors. He considers that specificity of female nature, where changes in hormone levels are continuous, creates an additional risk factor, thus the percentage of women suffering from LBP is significantly higher than this of men. A percentage of 50-60% of women who gave birth seems to be more likely to suffer from LBP, while those who did not have a pregnancy suffer only at 15% (Ostgaard, Andersson, & Karlsson, 1991), confirming our research case.

Another notable finding is that there is no correlation between smoking and LBP. There was no statistical significance in any query but in any common feature of the two groups (smokers-non-smokers). Goldberg, Scott, & May, (2000), came to the same conclusion as the data obtained had a vague meaning. As reported by Goldberg et al. in the review, there were not enough articles to create a clear and strong correlation between smoking and LBP. Perhaps in our own study the sample was not able to give us the real dimension of the situation. Contrariwise, other studies' results correlate smoking to LBP (Shiri, Karppinen, Leino-Arjas, Solovieva, & Viikari-Juntura, 2010). The main conclusion of a resent review concerning the correlation between smoking and LBP seems to be positive, especially in adolescents but also in smokers, no matter if they gave up smoking or not (Shiri et al., 2010). Women ($N = 131$) who reported LBP had almost all a high MET score. Only 3% of the participants scored less than 25 METs. They respond they participate in a variety of physical activities, while several

declared that constant intensity prevailed in their daily routine. Mean value is $M = 49.61$ while the intermediate $MO = 45.12$. The difference in variance measures is that there is more than one prevailing price. More than half of those who diagnosed with LBP have between 50 and 64 METs, while about 40% has less than 50 METs.

An assumption concerning the participants declared LBP could be that the intensity of the activity creates high loads in their spine resulting in strained and manifested pain. Another possibility is that their daily routine activities add them an extra load. Eurobarometer in 2004 published a study carried out in 15 EU Member States (including Greece) in the framework of its actions on health and physical activity. Research aimed to assess the MET/week in each country through personal interviews by all participants. Greece had one of the highest scores in METs/week concerning females participants ($M = 35.12$). The results of the abovementioned research confirm the present study's results since high METs score is their common feature. Under the consideration that Eurobarometer conducted the research in a general population while the present study in physical active female participants, it is easily understandable that weekly loads for women in Greece are such that they cause LBP problems. Mortimer, Pernold, & Wiktorin (2006) report the direct association of high METs scores with LBP. Similar findings presented in other studies (Mortimer et al., 2001). The common feature of all of these findings is the correlation between METs scores and LBP. Researches' results show, that excesses in activities would have the same effects as inactivity for those who experienced LBP. These views match to Campello, Nordin, & Weiser (1996) theory of the inverted "U". These positions and views later agreed by other researchers (Heneweel, Vanhees, & Picavet, 2009).

Based upon the findings of the research, a good perspective is to include special training programs for pregnant women in their design as part of innovation that will create a more competitive organization in the sports product market (Spais, 2007). Informing young people, through organized workshops, newsletters distribution in schools etc., about the benefits of exercise could be the basis for preventing LBP in young age groups. The marketing department of the municipal sports organization should take initiatives to update the age target group's 17-26 programs, while there is another good opportunity for the organization to create specialized programs focused on women's age target groups. Future research should identify exercise's variables (intensity, type of exercise, etc.) of the program over a period of more than six months, which aim will be the substantially reduce the LBP. This will be in order to prevent the LBP annoying symptom; citizens should enjoy quality and effective services always for their own wellbeing and health.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 1 June 2019 | **Accepted:** 4 August 2019 | **Published:** 11 October 2019

References

Ainsworth, B.E., Haskell, W.L., Whitt, M.C., Irwin, M.L., Swartz, A.M., Strath, S.J., O'Brien, W.L., Bassett Jr, D.R., Schmitz, K.H., Emplaincourt, P.O., Jacobs Jr, D.R., & Leon, A.S. (2000). Compendium of Physical Activities: an Update of Activity Codes and MET Intensities. *Medicine & Science in Sports & Exercise*, 32(9), S498-S504. doi: 10.1097/00005768-200009001-00009

Andersson, G.B. (1999). Epidemiological features of chronic low-back pain. *The Lancet*, 354(9178), 581-585. doi: 10.1016/S0140-6736(99)01312-4

Arthur, M. (2014). Institute for Health Metrics and Evaluation. *Nursing Standard*, 28(42), 32-32. doi: 10.7748/ns.28.42.32.s33

Bailey, A. (2009). Risk factors for low back pain in women: still more questions to be answered. *Menopause*, 16(1), 3-4. doi: 10.1097/gme.0b013e31818e10a7

Campello, M., Nordin, M., & Weiser, S. (1996). Physical exercise and low back pain. *Scandinavian journal of medicine & science in sports*, 6(2), 63-72.

Frymoyer, J.W., & Cats-Baril, W.L. (1991). An overview of the incidences and costs of low back pain. *The orthopedic clinics of North America*, 22(2), 263-271.

Goldberg, M.S., Scott, S.C., & Mayo, N.E. (2000). A review of the association between cigarette smoking and the development of nonspecific back pain and related outcomes. *Spine*, 25(8), 995-1014.

Haryono, I. R., Kawilarang, M., & Prastowo, N. A. (2019). Leg length discrepancy in college students and its association with low back pain: a preliminary study. *Journal of Anthropology of Sport and Physical Education*, 3(2), 15-18. doi: 10.26773/jaspe.190403

Heliövaara, M. (1987). Body Height, Obesity, and Risk of Herniated Lumbar Intervertebral Disc. *Spine*, 12(5), 469-472. doi: 10.1097/00007632-198706000-00009

Heliövaara, M., Mäkelä, M., Knekt, P., Impivaara, O., & Aromaa, A. (1991). Determinants of Sciatica and Low-Back Pain. *Spine*, 16(6), 608-614. doi: 10.1097/00007632-199106000-00002

Heneweel, H., Vanhees, L., & Picavet, H.S.J. (2009). Physical activity and low back pain: a U-shaped relation? *Pain*, 143(1-2), 21-25. doi: 10.1016/j.pain.2008.12.033

Kendall, N.A.S., Linton, S.J., & Main, C.J. (1997). *Guide to assessing psychosocial yellow flags in acute low back pain: risk factors for long-term disability and work loss*. Wellington: Accident Rehabilitation & Compensation Insurance Corporation of New Zealand and the National Health Committee

Kosmas, I., Marmara, H., & Stergioulas, A. (2009). Organization of traumatic injuries of women trained in municipal sports programs. *Organization of sport*, 7, 30-38.

Kriska, A.M., & Caspersen, C.J. (1997). Introduction to a collection of physical activity questionnaires. *Medicine & Science in Sports & Exercise*, 29(6), 5-9. doi: 10.1097/00005768-199706001-00003

Lallukka, T., Viikari-Juntura, E., Viikari, J., Kähönen, M., Lehtimäki, T., Raitakari, O., & Solovieva, S. (2017). Early work-related physical exposures and low back pain in midlife: the Cardiovascular Risk in Young Finns Study. *Occupational and Environmental Medicine*, 74(3), 163-168. doi: 10.1136/oemed-2016-103727

Le Borgne, M., Boudoukha, A. H., Petit, A., & Roquelaure, Y. (2017). Chronic low back pain and the transdiagnostic process: How do cognitive and emotional dysregulations contribute to the intensity of risk factors and pain? *Scandinavian journal of pain*, 17(1), 309-315. doi: 10.1016/j.sjpain.2017.08.008

Leboeuf-Yde, C., Kyvik, K. O., & Bruun, N. H. (1998). Low Back Pain and Lifestyle: Part I: Smoking. Information From a Population-based Sample of 29,424 Twins. *Spine*, 23(20), 2207-2213.

Liira, J.P., Shannon, H.S., Chambers, L.W., & Haines, T.A. (1996). Long-term back problems and physical work exposures in the 1990 Ontario Health Survey. *American Journal of Public Health*, 86(3), 382-387. doi: 10.2105/AJPH.86.3.382

Maniadakis, N., & Gray, A. (2000). The economic burden of back pain in the UK. *Pain*, 84(1), 95-103. doi: 10.1016/S0304-3959(99)00187-6

Marmara, E.A., Papacharalambous, X.N., Kouloulas, V.E., Maridaki, D.M., & Baltopoulos, J.P. (2011). Physical activity and mammographic parenchymal patterns among Greek postmenopausal women. *Maturitas*, 69(1), 74-80. doi: 10.1016/j.maturitas.2011.02.003

Miller, J. (2014, February 25). *Is back, neck and muscle pain hurting the UK economy?* Retrieved from BBC website: <http://www.bbc.co.uk/news/business-26338889>

Mimic, J., Vukicevic, V., & Vujacic, N. (2019). Comparative analysis of certain morphological characteristics of top tennis players in different periods. *Journal of Anthropology of Sport and Physical Education*, 3(2), 35-41. doi: 10.26773/jaspe.190407

Mortimer, M., Pernold, G., & Wiktorin, C. (2006). Low back pain in a general population. Natural course and influence of physical exercise—a 5-year follow-up of the Musculoskeletal Intervention Center-Norråljä Study. *Spine*, 31(26), 3045-3051. doi: 10.1097/01.brs.0000250324.03210.3f

Mortimer, M., Wiktorin, C., Pernold, G., Svensson, H., Vingård, E., & Music-Norråljä study group. (2001). Sports activities, body weight and smoking in relation to low-back pain: a population-based case-referent study. *Scandinavian journal of medicine & science in sports*, 11(3), 178-184. doi: 10.1046/j.1524-4725.2001.110308.x

- Nachemson, A. (1991). *Back pain, etiology, diagnostics, and treatment*. Stockholm: Swedish Council on Technology Assessment in Health Care.
- Ostgaard, H.C., Andersson, G.B.J., & Karlsson, K. (1991). Prevalence of Back Pain in Pregnancy. *Spine*, 16(5), 549–552. doi: 10.1097/00007632-199105000-00011
- Shiri, R., Karppinen, J., Leino-Arjas, P., Solovieva, S., & Viikari-Juntura, E. (2010). The Association between Smoking and Low Back Pain: A Meta-analysis. *The American Journal of Medicine*, 123(1), 87.e7–87.e35. doi: 10.1016/j.amjmed.2009.05.028
- Smedley, J., Egger, P., Cooper, C., & Coggon, D. (1995). Manual handling activities and risk of low back pain in nurses. *Occupational and environmental medicine*, 52(3), 160–163. doi: 10.1136/oem.52.3.160
- Spais, G. (2007). *Introduction to the management of technological innovations*. Athens: Kritiki publications.
- Spyropoulos, P., Papathanasiou, G., Georgoudis, G., Chronopoulos, E., Koutis, H., & Koumoutsou, F. (2007). Prevalence of low back pain in Greek public office workers. *Pain physician*, 10(5), 651–660.
- Thomas, N., & Nelson, J. (2003). *Research Methods in Physical Activity* [in Greek]. Athens: Paschalidis.
- Van Tulder, M.W., Koes, B.W., & Bouter, L.M. (1995). A cost-of-illness study of back pain in The Netherlands. *Pain*, 62(2), 233–240. doi: 10.1016/0304-3959(94)00272-G
- Waddell, G., Feder, G., McIntosh, A., Lewis, M., & Hutchinson, A. (1998). (1996) Low Back Pain Evidence Review London: Royal College of General Practitioners. *Journal of Manual & Manipulative Therapy*, 6(3), 151–153. doi: 10.1179/jmt.1998.6.3.151

ORIGINAL SCIENTIFIC PAPER

Differences in Anthropometric Characteristics between Athletes of Different Orientation, Handball and Volleyball

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Abstract

The main goal of this research is to determine whether there are statistically significant differences in the level of anthropometric indicators among athletes of different sports directions (handball and volleyball players), that is, to determine the differences in the changes in the anthropometric indicators of athletes. The study was conducted on a sample of 50 subjects, a male gender divided into two subsample, 25 handball players and 25 volleyball players aged 13-15. In this study, 12 anthropometric variables were tested: body height, arm length, leg length, knee diameter, bicondylar range, bichromium range, body mass, median volume of the thorax, circumference, upper abdomen, skin abdomen and skin set of lower legs. For all applied variables, the central and dispersion parameters as well as measures of asymmetry and flattening were calculated, and the distribution normality was verified by Kolmogorov-Smirnov test. T-test was applied to verify that the system of applied variables has statistically quantitative differences between groups. On the basis of the obtained results, it can be concluded that there are statistically significant differences in anthropometric indicators in athletes of different sports orientations. This research can be used by trainers who realize training programs with these and other athletes in order to achieve better results.

Key words: Handball, Volleyball, Anthropometric Characteristics, Montenegro

Uvod

Sport je fizička aktivnost, izvedena u okviru javno određenih pravila i običaja, koja se najčešće bavi takmičenjem, ali služi i za druge namjene: razonodu, razvijanje karakteristika, očuvanje zdravlja ili kombinaciju ovih elemenata (Bjelica, 2002). Najšire gledano, pod sportom se podrazumijeva specifična ljudska aktivnost koja ima za cilj sportski rezultat (Bjelica, 2006a). Smatra se da je jedan od osnovnih motiva koji pokreće ljude da se bave sportom upravo težnja za usavršavanjem i mjerenjem svojih sposobnosti i karakteristika, koje se mogu razvijati pod uticajem trenažnog procesa (Bjelica, 2005). Zadovoljenje sportskih potreba i interesa značajno je sa aspekta zdravlja, svestranog razvoja ličnosti sportiste, podizanja radne sposobnosti do nivoa sportske forme, te sposobnosti za učešće u sistemu takmičenja (Bjelica, 2006b). Sport u savremenim uslovima života predstavlja složen sistem različitih ak-

tivnosti i oblika organizovanja kroz koje se ispoljavaju brojne i različite potrebe i interesi sportista (Bjelica, 2013; Popovic, 2017; Vukotic, Corluka, Vasiljevic, i Bubanja, 2018). Pošto je poznato da rezultat koji postižu sportisti u velikoj mjeri zavisi od morfoloških karakteristika (Masanovic, 2018; Masanovic, 2019; Vukasevic, Vukotic, i Masanovic, 2018; Vukotic i Georgiev, 2019a), bolje upoznavanje morfoloških karakteristika i konstitucije sportista ima za cilj: upravljanje ovim svojstvima i prilagođavanje sportskog treninga individualnim sposobnostima (Masanovic, Vukotic, Popovic, i Bjelica, 2018; Masanović, Vukotic, i Vukasevic, 2018; Vukotic i Georgiev, 2019b).

Rukomet i odbojka, kao sportske igre, zahtijevaju različite prethodno navedene dimenzije, a takođe i procese sportskog treninga i obučavanja (Vukotic, 2011; Masanovic, Milosevic, i Corluka, 2018; Masanovic, Corluka, i Milosevic, 2018; Vukotic, 2018). Jedan od neophodnih uslova za postizanje uspeha u

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**Montenegro
Sport**

rukometu i odbojci je i talenat, jer sportisti koji posjeduju visok potencijal lakše i brže napreduju nego sportisti sa manjim nivoom potencijala, te se nameće zaključak da je rana identifikacija i selekcija talenta veoma važna karika na putu do uspjeha (Bjelica, 2004). Ako se uzme u obzir da je potrebno barem pet godina napornog treninga, i da je relativno mali vremenski period kada sportista može očekivati vrhunski učinak, postaje jasno da je veoma važno da se sportski potencijal otkrije što ranije i kontinuirano prati kako bi omogućio dovođenje do vrhunskog rezultata (Bompa, 2000). Za sportske trenere je od posebnog značaja da poznaju zakonitosti rasta i razvoja antropometrijskih karakteristika, motoričkih i funkcionalnih sposobnosti da bi primjena različitih treninga bila što efikasnija (Bjelica, Popovic, i Gardasevic, 2016a; Bjelica, Popovic, i Gardasevic, 2016b; Bjelica, Popović, Kezunovic, Petkovic, Jurak, i Grasgruber, 2012; Masanovic, Vukotic, Bjelica, i Popovic, 2018; Popovic, Bjelica, Vukotic, i Masanovic, 2018).

Antropometrijske karakteristike su veoma značajne za uspjeh u sporskim igrama kao što su rukomet i odbojka. U prostoru morfoloških dimenzija rukometaša i odbojkaša definisani su posebni faktori označeni kao latentne morfološke dimenzije: longitudinalna dimenzionalnost skeleta, transverzalna dimenzionalnost skeleta, masa i volumen tela i potkožno masno tkivo (Bjelica i Fratric, 2011). Morfološki status vrhunskih sportista su relativno homogeni, u zavisnosti od sporta, i mogu biti definisani kao modeli sportskog postignuća (Misigoj-Durakovic, Matkovic, i Medved, 1995). Sve više rukometaši pokazuju snažnu i obimnu muskulaturu gornjeg dijela tijela, što rezultira jačem i bržem šutu. Visina daje prednost u rukometu, jer ona igračima omogućava da lakše dođu do dobre pozicije za šut u napadu, odnosno lakše blokiraju protivnički napad (Arifi, Bjelica, i Masanovic, 2019). Prema igračkim mjestima najviši su spoljni igrači i golman, a najniži su krilni igrači (Vukotic, 2010). Većina elemenata naročito onih sa loptom su veoma složeni i specifični. Za njihovo usvajanje i besprekornu primjenu u igri potreban je visok nivo razvijenosti cjelokupnog motornog aparata (Vukasevic, Vukotic, i Masanovic, 2018). Savremenu odbojku, karakteriše dinamička, vrlo brza igra, sa mnogo eksplozivnih pokreta, skokova, preciznog odbijanja lopte, snažnog i preciznog servisa, smeća u utakmicama koje mogu trajati vrlo dugo. Da bi jedna vrhunska odbojkaška ekipa mogla da odgovori svim ovim zahtjevima, mora da ima visok nivo razvijenosti navedenih sposobnosti i karakteristika u kojima osnovu čine određene motoričke i funkcionalne sposobnosti kao i antropološke karakteristike (Vukotic, 2010).

Osnovni cilj ovog istraživanja je da se utvrdi da li postoje statistički značajne razlike u nivou antropometrijskih karakteristika između sportista različitog sportskog usmjerenja (rukomet i odbojka), odnosno da se preciznije ukaže na to kolike su ove razlike i gdje su najviše.

Metod

Uzorak ovog istraživanja čini 50 ispitanika, muškog pola podijeljenih na dva subuzorka. Prvi subuzorak čini 25 dječaka članova Rukometnog kluba „Sutjeska“, iz Nikšića, prosječnog uzrasta 13.73 ± 6.18 , dok drugi subuzorak su čini 25 dječaka članova Odbojkaškog kluba „Volley star“, iz Nišića, prosječnog uzrasta 14.05 ± 7.15 godina.

Antropometrijsko istraživanje sprovedeno je uz poštovanje osnovnih pravila i principa vezanih za izbor mjernih instrumenata i tehnike mjerenja koji su standardizovani, prema uputstvima Internacionalnog Biološkog Programa. Odabrane varijable u ovom istraživanju hipotetski pokrivaju prostor antropometrijskih karakteristika (12) i to: visina tijela, dužina ruke, dužina noge, dijametar koljena, bikristalni raspon, biakromijalni raspon, masa tijela, srednji obim grudnog koša, obim natkoljenice, kožni nabor nadlaktice, kožni nabor trbuha i kožni nabor potkoljenice.

Podaci dobijeni ovim istraživanjem obrađeni su postupcima deskriptivne i komparativne statističke procedure. Za sve primijenjene varijable izračunati su centralni i disperzioni parametri kao i mjere asimetrije i spljoštenosti, a normalnost distribucije provjerena je Kolmogorov–Smirnovljevim testom. Razlike u antropometrijskim karakteristikama između dvije grupe ispitanika (rukometaša i odbojkaša) utvrđene su primjenom diskriminativne parametrijske procedure, t-testom za male nezavisne uzorke, sa statističkom značajnošću od $p < 0.05$.

Rezultati

U Tabelama 1 i 2 prikazani su osnovni deskriptivni statistički parametri antropometrijskih varijabli rukometaša i odbojkaša kadetskog uzrasta, gdje su izračunate vrijednosti mjera centralne i disperzione tendencije i to: aritmetička sredina, standardna devijacija, minimalne vrijednosti, maksimalne vrijednosti, koeficijent zakrivljenosti, koeficijent izduženosti i Kolmogorov–Smirnovljev test. Prvo su analizirani centralni i disperzioni parametri varijabli za procjenu antropometrijskih karakteristika rukometaša kadeta (Tabela 1).

Tabela 1. Centralni i disperzioni parametri varijabli za procjenu antropometrijskih karakteristika rukometaša kadetskog uzrasta (N=25)

	Mean	SD	Min	Max	Skew.	Kurt.	KS
Visina tijela	167.08	8.47	154.00	186.00	0.29	-0.69	0.82
Dužina ruke	70.76	4.99	61.00	80.00	-0.22	-0.64	0.87
Dužina noge	89.58	7.40	78.50	104.00	0.02	-0.93	0.88
Dijametar koljena	8.88	0.52	8.00	10.00	0.66	0.12	0.26
Bikristalni raspon	35.06	2.48	31.00	41.00	0.60	0.40	0.91
Biakromijalni raspon	37.39	2.73	33.00	46.50	1.49	4.16	0.57
Masa tijela	57.28	8.71	45.00	90.00	2.05	7.94	0.08
Srednji obim grudnog koša	80.42	5.40	71.00	95.50	0.67	1.13	0.91
Obim natkoljenice	41.23	3.93	34.00	50.50	0.54	0.26	0.60
Kožni nabor nadlaktice	11.36	2.01	6.67	15.30	-0.28	-0.14	0.55
Kožni nabor trbuha	10.51	2.84	5.67	15.67	0.04	-0.65	1.00
Kožni nabor potkoljenice	11.15	3.11	3.67	19.67	0.64	2.49	0.31

Legenda: Mean – aritmetička sredina; SD – standardna devijacija; Min – minimalni rezultat; Max – maksimalni rezultat; Skew. – koeficijent zakrivljenosti; Kurt. – koeficijent izduženosti i KS – Kolmogorov–Smirnov test

Inspekcijom tabele uočljivo je da su dobijeni rezultati normalno distribuirani (prema K-S testu). Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtozisa, može se konstatovati da su sve varijable u granicama normalne raspodjele. Vrijednosti skjunisa kod varijabli sa pozitivnim predznakom, govore da su vrijednosti većine ispitanika u zoni slabijih vrijednosti i one su normalno asimetrične ili umjereno asimetrične, dok u mjerama dužina ruke

i kožni nabor nadlaktice, koje su sa negativnim predznakom, govore da se vrijednosti većine ispitanika u zoni boljih rezultata. Negativne vrijednosti kurtozisa kod varijabli pokazuju veću spljoštenost (nagnutost) od normalnog oblika, tj. plati-kurtičnost. To ukazuje na heterogenost dobijenih rezultata, odnosno izraženiju diskriminativnost među ispitanicima. U mjeri masa tijela (Kurt=7.94) možemo konstatovati da su rukometaši najhomogeniji (leptokurtičnost).

Tabela 2. Centralni i disperzioni parametri varijabli za procjenu antropometrijskih karakteristika odbojkaša kadeta (N=25)

	Mean	SD	Min	Max	Skew.	Kurt.	KS
Visina tijela	178.58	8.13	161.00	190.00	0.22	-0.85	0.92
Dužina ruke	76.54	3.47	70.00	85.00	0.49	0.41	0.53
Dužina noge	98.16	5.17	88.00	109.00	0.04	-0.20	0.81
Dijametar koljena	8.85	0.60	7.80	10.00	0.26	-0.74	0.96
Bikristalni raspon	35.02	2.02	31.00	39.00	-0.44	0.08	0.20
Biakramijalni raspon	38.72	2.85	33.00	45.00	0.09	0.25	0.39
Masa tijela	66.28	10.60	35.00	80.00	-0.12	0.05	0.90
Srednji obim grudnog koša	80.20	5.44	70.00	94.00	0.21	0.71	0.98
Obim natkoljenice	42.84	3.68	34.00	52.00	-0.08	1.34	0.47
Kožni nabor nadlaktice	10.83	2.88	6.00	16.67	0.18	-0.69	0.98
Kožni nabor trbuha	8.48	3.44	4.00	18.30	1.29	1.55	0.28
Kožni nabor potkoljenice	7.85	2.51	2.67	11.70	-0.64	-0.23	0.32

Na osnovu dobijenih rezultata utvrđeno je da su rezultati normalno distribuirani (prema K-S testu). Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtozisa, može se konstatovati da su sve varijable u granicama normalne raspodjele. Vrijednosti skjunisa kod varijabli sa pozitivnim predznakom, govore da su vrijednosti većine ispitanika u zoni slabijih vrijednosti i one su normalno asimetrične ili umjereno asimetrične, dok su u mjeri kožni nabor trbuha sa odgovara-

jućim vrijednostima 1,29 izraženo asimetrični i u zoni slabijih rezultata. Negativne vrijednosti kurtozisa kod varijabli pokazuju veću spljoštenost (nagnutost) od normalnog oblika, tj. leptokurtičnost. To ukazuje na heterogenost dobijenih rezultata, odnosno izraženiju diskriminativnost među ispitanicima. Da bi se moglo utvrditi da li ima statistički značajnih razlika u analiziranim varijablama kod rukometaša i odbojkaša kadetskog uzrasta, primijenjena je statistička procedura t-test (Tabela 3).

Tabela 3. Razlika između aritmetičkih sredina varijabli za procjenu antropometrijskih karakteristika kadeta rukometaša (N=25) i kadeta odbojkaša (N=25)

Varijabla	Grupe	Mean	SD	t	Sig
Visina tijela	Rukometaši	167.08	8.47	2.34	.030*
	Odbojkaši	178.58	8.13		
Dužina ruke	Rukometaši	70.76	4.99	3.11	.000*
	Odbojkaši	76.54	3.47		
Dužina noge	Rukometaši	89.58	7.40	2.51	.023*
	Odbojkaši	98.16	5.17		
Dijametr koljena	Rukometaši	8.88	0.52	-1.05	.070
	Odbojkaši	8.85	0.60		
Bikristalni raspon	Rukometaši	35.06	2.48	3.69	.105
	Odbojkaši	35.02	2.02		
Biakramijalni raspon	Rukometaši	37.39	3.15	1.98	.126
	Odbojkaši	38.72	2.73		
Masa tijela	Rukometaši	57.28	10.34	2.99	.012*
	Odbojkaši	56.28	10.60		
Srednji obim grudnog koša	Rukometaši	80.42	5.40	3.19	.184
	Odbojkaši	80.20	5.44		
Obim natkoljenice	Rukometaši	41.23	3.39	3.15	.013*
	Odbojkaši	42.84	3.68		
Kožni nabor nadlaktice	Rukometaši	11.36	2.18	1.50	.177
	Odbojkaši	10.83	2.88		
Kožni nabor trbuha	Rukometaši	10.51	2.84	3.18	.014*
	Odbojkaši	8.48	3.44		
Kožni nabor potkoljenice	Rukometaši	11.15	3.11	3.42	.003*
	Odbojkaši	7.85	2.51		

Legenda: Mean – aritmetička sredina; SD – standardna devijacija; t - vrijednosti t testa i Sig. – značajnost razlike; * - značajna razlika između grupa

Na osnovu rezultata se može vidjeti da razlike postoje, što nam potvrđuju i vrijednosti T- testa sa statističkom značajnošću na nivou $p < 0.05$. Prilikom utvrđivanja razlika kod antropometrijskih parametara, očigledna je statistička značajnost kod sljedećih varijabli: visine tijela, dužina ruke, dužina noge, mase tijela, obim natkoljenice, kožni nabor trbuha i kožni nabor potkoljenice, dok statistička razlika ne postoji kod varijabli: dijametar koljena, bikristalni raspon, biakromijalni raspon, srednji obim grudnog koša i kožni nabor nadlaktice.

Diskusija

Na osnovu dobijenih deskriptivnih parametara ovog istraživanja prosječna tjelesna visina rukometaša je 167.08 cm, a masa tijela rukometaša je 57.28 kg, što pokazuje da oni imaju veoma slične vrijednosti sa igračima iz Hrvatske koji su istog uzrasta (Ostroski, Milanovic, i Metikos, 2014) gdje je prosječna tjelesna visina rukometaša bila 167.00 cm a masa tijela je iznosila 62.70 kg. Opšti zaključak je da postoji sličnost naših i hrvatskih igrača i to ohrabruje, jer hrvatski rukometaši su godinama u svjetskom vrhu, što može da znači da smo na dobrom putu da se i mi tamo nađemo. Posljednjih godina se u rukomet puno ulaže o čemu govore i rezultati muške kadetske reprezentacije Crne Gore koja je osvojila srebrnu medalju na Mediteranskom prvenstvu u Egiptu 2019. godine. Seniorska reprezentacija uspjela je da se plasirala na evropsko prvenstvo, jer su u kvalifikacijama pobjeđivali najjače evropske ekipe među kojima i aktuelnog osvajača zlatne olimpijske medalje i aktuelnog svjetskog prvaka Dansku.

Na osnovu dobijenih deskriptivnih parametara ovog istraživanja prosječna tjelesna visina odbojkaša je 178.58 cm, a masa tijela odbojkaša je 66.28 kg, pa vidimo da oni imaju nešto niže vrijednosti u odnosu na reprezentativce Srbije istog uzrasta (Bogdanovic et al., 2014) gdje je prosječna tjelesna visina bila 181.87 cm a masa tijela je iznosila 70.53 kg. Možda je to osnovni razlog zašto crnogorski odbojkaši nijesu uspješni na međunarodnim takmičenjima kao srbijanski odbojkaši sa kojima je vršeno poređenje. Na osnovu navedenog odbojkaški stručnjaci bi trebalo da više vode računa o selekciji jer imaju populaciju koja je veoma visoka (Bjelica, i sar. 2016a; Masanovic, 2017; Vukotic, 2018; Masanovic, 2018b).

Može se zaključiti da kod primijenjenog sistema antropometrijskih mjera postoje statistički značajne razlike između rukometaša i odbojkaša, što se vidi na osnovu rezultata T-testa za male nezavisne uzorke. Dobijeni rezultati dijelom potvrđuju važnost longitudinalne dimenzionalnosti skeleta u razlikovanju selekcionisanih rukometaša i odbojkaša kadetskog uzrasta (Cselko, Laszlo, Tekus, i Wilhelm, 2013; Katic, Grgantov, i Jurko, 2006), jer razlike ukazuju da su za uspješnost u igri kod mladih rukometaša i odbojkaša potrebne različite morfološke karakteristike (Vukotic, 2010). U sportskoj praksi dobro je poznat problem prerane specijalizacije kao i neuvažavanja fiziološke zrelosti u procjeni krajnjih mogućnosti mladih sportista (Masanovic, Popovic, i Bjelica, 2018) što može dovesti na krivi put. Korisnije je da se sve u cilju postizanja vrhunskih rezultata, pokuša pronaći adekvatan somatotip za određene sportove, nivo takmičenja i igračke pozicije (Massuca, i Frago, 2011).

Rezultati koji su dobijeni ovim istraživanjem mogu poslužiti kao modelni parametri u procjenjivanju varijabla za sve ostale igrače istog ranga takmičenja u Crnoj Gori, kao i za upoređivanje sa rezultatima dobijenim u drugim, sličnim istraživanjima. Treba napomenuti da igrači koje žele uspješno nastupati u ligama gdje se igra najkvalitetniji rukomet i odbojka,

moraju imati karakteristike utvrđene standardima za taj sport, a to se prije svega ogleda u antropometrijskim karakteristikama. Jer prilikom selekcije bitno je dati prednost onim igračima koji svojim antropometrijskim karakteristikama zadovoljavaju zahtjeve prema igračkim pozicijama u navedenim sportskim igrama (Gabbett i Georgieff, 2007), a ostale bi trebalo usmjeriti ka sportovima gdje prema svojim morfološkim karakteristikama mogu imati više uspjeha jer veliki je broj vrhunskih sportista koji su čak i u ovim godinama promijenili sport pa vrhunske uspjehe postigli u sportu koji nije bio njihov prvi izbor.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 10 June 2019 | Accepted: 29 July 2019 | Published: 11 October 2019

References

- Arifi, F., Bjelica, D., & Masanovic, B. (2019). Differences in anthropometric characteristics among junior soccer and handball players. *Sport Mont*, 17(1), 45-49. doi: 10.26773/smj.190208
- Bjelica, D. (2002). *Opšti pojmovi sportskog treninga: (skraćena verzija)*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2004). *Utjecaj sportskog treninga na antropomotoričke sposobnosti: (fudbalskih kadeta Crne Gore)*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2005). *Sistematizacija sportskih disciplina i sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2006a). *Sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2006b). *Teorijske osnove tjelesnog i zdrastvenog obrazovanja*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2013). *Teorija sportskog treninga*. Podgorica: Univerzitet Crne Gore.
- Bjelica, D., & Fratric, F. (2011). *Sportski trening: teorija, metodika i dijagnostika*. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Bjelica, D., Popović, S., i Gardašević, J. (2016a). Modeli fizičke pripreme vrhunskih sportaša i doziranje opterećenja. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (185-189), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bjelica, D., Popović, S., i Gardašević, J. (2016b). Opći principi planiranja i programiranja fizičkih priprema sportaša. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (190-192), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bogdanovic, Z., Smajic, M., Jaksic, D., Obradovic, M., Gogic, A., Vidakovic, H.M., Ljubicavljivic, M., Draskovic, V., Visnjic, S., Mekic, H., Stankovic, R., Ivancic, G., & Popovic, S. (2014). Lumbar and abdominal muscles isometric potential in volleyball cadets. *International Journal of Morphology*, 32(3), 1036-1042.
- Bompa T. (2000). *Periodizacija: Teorija i metodika treninga*. Zagreb: Hrvatski košarkaški savez.
- Cselko, A., Laszlo, Z., Tekus, E., & Wilhelm, M. (2013). Morphological, motor and situation-motor characteristics of elite female handball players according to playing performance and position. *Collegium Anthropologicum*, 34(4), 1355-61.
- Gabbett, T. & Georgieff, B. (2007). Physiological and anthropometric characteristics of Australian junior national, state, and novice volleyball players. *Journal of Strength and Conditioning Research*, 21(3), 902-8.
- Gaurav, V., Singh, M., & Singh, S. (2010). Anthropometric characteristics, somatotyping and body composition of volleyball and basketball players. *Journal of Physical Education and Sports Management*, 1(3), 28-32.
- Masanovic, B. (2018). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from Serbian national league. *Sport Mont*, 16(3), 19-24. doi: 10.26773/smj.181004
- Masanovic, B. (2019). Comparative Study of Morphological Characteristics and Body Composition between Different Team Players from Serbian Junior National League: Soccer, Handball, Basketball and Volleyball. *International Journal of Morphology*, 37(2), 612-619.
- Masanovic, B., Corluca, M., & Milosevic, Z. (2018). Comparative Study of Anthropometric Measurement and Body Composition of Junior Soccer and Handball Players from the Serbian National League. *Kinesiologia Slovenica*, 24(3), 37-46.
- Masanovic, B., Milosevic, Z., & Corluca, M. (2018). Comparative Study of Anthropometric Measurement and Body Composition between Junior Handball and Volleyball Players from Serbian National League. *International Journal of Applied Exercise Physiology*, 7(4), 1-6. <https://doi.org/10.30472/ijaep.v7i4.313>

- Masanovic, B., Vukotic, M., & Vukasevic, V. (2018). Comparative study of morphological characteristics and body composition between elite basketball players from different regions. *Journal of Anthropology of Sport and Physical Education*, 2(4), 103-107. doi: 10.26773/jaspe.181019
- Masanovic, B., Vukotic, M., Popovic, S., & Bjelica, D. (2018). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from Serbian national league. *World Congress of Performance Analysis of Sport XII* (340). Croatia: University of Zagreb.
- Massuca, L., & Fragoso, I. (2011). Study of Portuguese handball players of different playing status. A morphological and biosocial perspective. *Biology of Sport*, 28(1), 37-44.
- Misigoj-Durakovic, M., Matkovic, B., & Medved, R. (1995). *Morfološka antropometrija u športu*. Morphological anthropometry in sports. Zagreb, Croatia: Fakultet za fizičku kulturu.
- Ostroski, S., Milanovic, M., i Metikos, B. (2014). Razlike nogometaša i rukometaša dobi 13 do 14 godina uključenih u školski sportski klub u nekim kondicionim obilježima. U *zborniku radova 23 Ljetnje škole ki-neziologa Republike Hrvatske* (231-236), Zagreb: Hrvatski kineziološki savez.
- Popovic, S. (2017). Local Geographical Differences in Adult Body Height in Montenegro. *Montenegrin Journal of Sports Science and Medicine*, 6(1), 81-87.
- Popović, S., Bjelica, D., Vukotic, M., & Masanovic, B. (2018). Describing Physical Activity Profile of Older Montenegrin Females Using the International Physical Activity Questionnaire (IPAQ). In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (60-61), Podgorica: Montenegrin Sports Academy.
- Vukasevic, V., Vukotic, M., & Masanovic, B. (2018). Comparative study of morphological characteristics and body composition between basketball players from second leagues in Montenegro and Serbia. *Journal of Anthropology of Sport and Physical Education*, 2(3), 21-25. doi: 10.26773/jaspe.180704
- Vukotic, M. (2010). *Nivo morfoloških karakteristika, motoričkih i funkcionalnih sposobnosti sportista različitog sportskog usmjerenja*. Neobjavljena magistarska teza. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Vukotic, M. (2011). Differences of anthropometric characteristic and motor abilities of different sport orientation. *Sport Mont*, 9(28-29-30), 112-8.
- Vukotic, M. (2018). Comparative analysis of antropmtric indicators of sportisis of different soprts guidance. In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (75-76), Podgorica: Montenegrin Sports Academy.
- Vukotic, M., & Georgiev, G. (2019a). Comparative analysis of anthropometric characteristics between athlets of different orientation, basketball and handball. *Journal of Anthropology of Sport and Physical Education*, 3(2), 19-23. doi: 10.26773/jaspe.190404
- Vukotic, M., & Georgiev, G. (2019b). Comparative analysis of anthropometric characteristics between athlets of different orientation. *Journal of Anthropology of Sport and Physical Education*, 3(1), 41-45. doi: 10.26773/jaspe.190108
- Vukotic, M., Corluka, M., Vasiljević, I., & Bujanja, M. (2018). Differences in the Morphological Characteristics and Body Composition of Handball Players WHC Levalca in Montenegro and WHC Grude in Bosnia and Herzegovina. *Journal of Anthropology of Sport and Physical Education*, 2(2), 49-53. doi: 10.26773/jaspe.180409

REVIEW PAPER

Scientific Artical of the Lecturers from the Faculty for Sport and Physical Education in Niksic Published in the Journal Sport Mont between 2016 and 2018: A Content Analysis

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Abstract

Sport Mont is a scientific journal which has been existing since 2003 and it has published a great number of works of renowned scientists in the field of sport sciences and medicine. The journal has been indexed in more international database, it is released three times a year, and the published works are mostly presented on the scientific conference of the Montenegrin Sport Academy. Sport Mont, in addition to the original scientific works, publishes reviews, editorials, short reports, as well as the calls and awards for works in the field of sport sciences and medicine. This essay deals with the contents analysis of the scientific research works where the authors and co-authors are the lecturers of the Faculty for Sport and Physical Education in Nikšić, published in the journal Sport Mont between 2016 and 2018. They have been sorted according to the scientific fields they belong to, and through discussion, a short overview of the obtained results has been given. This way makes possible an easier search of the mentioned articles for the authors who express their interest for them.

Key words: Sport Science, Sport Mont, Medicine

Uvod

Crnogorska sportska akademija je osnovana 2003. godine, kada je pokrenut i naučni časopis Sport Mont. Ovaj naučni časopis ima dugogodišnju tradiciju, koja baštini visoke standarde i norme stručnog, naučnog, moralnog i etičkog digniteta. Prepoznatljiv u okruženju, regionalno i mnogo šire, Sport Mont predstavlja značajnu naučnu bazu, u kojoj se mogu naći radovi eminentnih stručnjaka iz oblasti sportskih nauka, medicine i fizičkog vaspitanja. Zapravo, on pokriva veliki broj naučnih disciplina čiji su izvori saznanja od posebnog značaja za razumijevanje brojnih kinezioloških fenomena. To se prije svega odnosi na biomehaniku, fiziologiju, kineziterapiju, fizikalnu terapiju, traumatologiju sportskih povreda, trenažnu tehnologiju, antropologiju sporta, pedagogiju-sociologiju i psihologiju sporta, sportski menadžment i sve ostale aspekte vježbanja,

zdravlja i sporta. Časopis je indeksiran u 23 međunarodne naučne baze podataka, od kojih je Scopus najprestižnija. Izlazi tri puta godišnje i radovi koji su objavljeni u njemu su većinom predstavljeni na naučnoj konferenciji Crnogorske sportske akademije, koja se tradicionalno održava svake godine.

Pored originalnih naučnih radova, Sport Mont objavljuje pregledne radove, uvodnike, kratke izvještaje, nagrade radova u oblasti sportskih nauka i medicine i funkcionise kao otvoreni forum za diskusiju o značajnim aktuelnim pitanjima. Ovaj časopis obezbjeđuje transparentnost i dostupnost naučnim radnicima širom svjetskih meridijana. Stoga, on pruža online otvoren pristup, brzo objavljivanje, mišljenje stručnjaka i istraživača, post publikacijske alate za pokazivanje kvaliteta i učinka, zajednicu zasnovanu na zajedničkom jeziku članaka i svjetsku medijsku pokrivenost.

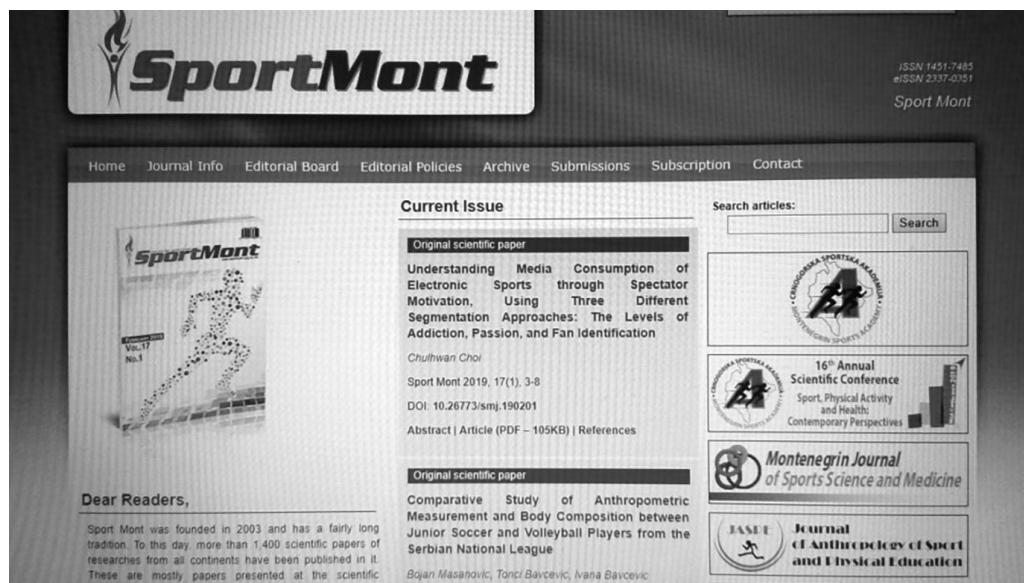
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**Montenegro
Sport**



Slika 1. Internet strana časopisa

Sport Mont je, doživljavajući brojne transformacije, prelazio evolutivni put napretka, čije ga tekovine danas postavljaju u ravan najpriznatijih časopisa istog stvaralačkog područja u cijelom svijetu. Egzistiranje časopisa u današnjem obliku je rezultat stručnog, posvećenog i pregalačkog rada glavnih urednika časopisa i članova uredničkog odbora, koji su uspjeli na ovaj način značaj-

no doprinijeti razvoju kineziološke nauke na našim prostorima. Glavni i odgovorni urednici časopisa su prof. dr Duško Bjelica sa Univerziteta Crne Gore i prof. dr Zoran Milošević sa Univerziteta u Novom Sadu. Izvršni urednik je mr Jovan Gardašević sa Univerziteta Crne Gore. Uređivački odbor čini 15 profesora sa Univerziteta Crne Gore, regiona, Evrope i Sjeverne Amerike.



Slika 2. Prof. dr Duško Bjelica

Do sada su izvršene brojne analize sadržaja iz časopisa Sport Mont, a cilj ovog rada je selektovanje članaka objavljenih od strane predavača sa Fakulteta za sport i fizičko vaspitanje-Nikšić, u periodu između 2016. i 2018. godine, po odgovarajućim naučnim oblastima kojima pripadaju. Na ovaj način će biti omogućen lakši pristup i pretraga onim autorima koji se budu služili radovima objavljenim u pomenutom vremenskom okviru.

Metod rada

U ovom radu biće korišćena metoda analize sadržaja, koja se efikasno primjenjuje u raznovrsnim područjima ljudskog rada i stvaralaštva. Ona podrazumijeva korišćenje pisanih izvora kao osnovne jedinice izvora informacija. Ovdje su to svi naučni član-

ci pod autorstvom ili koautorstvom predavača Fakulteta za sport i fizičko vaspitanje-Nikšić, objavljeni u časopisu Sport Mont između 2016. i 2018. godine. Ukupno je objavljeno 35 naučnih radova. Klasifikovani su prema sljedećim naučnim oblastima: sportski trening i metodika treninga; ishrana i suplementacija; fizičko vaspitanje djece sa posebnim potrebama; antropologija sporta; sociologija sporta i naučno izdavaštvo; korektivna gimnastika; istorija sporta; menadžment i marketing u sportu.

Rezultati

Nakon sprovedene kompletne analize sadržaja iz Sport Mont-a, izvršena je klasifikacija radova u osam oblasti, kako je prikazano u tabeli 1.

Tabela 1. Podjela radova prema naučnim oblastima

Naučna oblast	Broj radova
Sportski trening i metodika treninga	7
Ishrana i suplementacija	2
Fizičko vasp. djece sa posebnim potrebama	1
Antropologija sporta	16
Sociologija sporta i naučno izdavaštvo	3
Korektivna gimnastika	1
Istorija sporta	2
Menadžment i marketing u sportu	3

Sportski trening i metodika treninga kao sastavni dio trenažne tehnologije, omogućavaju transformacione procese pojedinih segmenata antropološkog statusa sportiste i usavršavanje tehničko-taktičkih elemenata, što u krajnjem vodi ka postizanju vrhunskih sportskih dostignuća. Iz ovog veoma važnog segmenta su prikazani sljedeći radovi: "Repulsion of the futsal ball depending on the pressure in it" (Bjelica, Milošević, Talović i Bajramović, 2018), "Six-week preparation period and its effects on transformation movement speed" (Gardašević, Bjelica i Vasiljević, 2016), "The effects of the training in the preparation period on the repetitive strength transformation with cadet level football players" (Gardašević, Bjelica, Vasiljević i Milašinović, 2016), "The strength of kicking the ball after preparation period with U15 football players" (Gardašević, Bjelica i Vasiljević, 2017), "The impact of the preparation period on endurance at football players U16" (Gardašević, Bjelica i Ćorluka, 2018), "Methods for evaluation of some psychomotor abilities" (Krivokapić i Tanase, 2016), "Theories of tiredness in sport" (Krivokapić, 2017).

Ishrana i suplementacija je naučna oblast koja se ubrzano razvija. Sve više istraživača se usmjerava na nju, a sve veći broj naučnih članaka ukazuje na vezu između adekvatne ishrane i sportskih postignuća. Radovi iz ove oblasti su: "Knowledge of physical education students on sports nutrition" (Bjelica, Kezunović i Gardašević, 2016), "Waist circumference as an indicator abdominal obesity in middle age" (Vasiljević, Gardašević, Kezunović i Bojanić, 2017).

Fizičko vaspitanje djece sa posebnim potrebama je zasebna oblast u okviru kineziologije, koja u posljednje vrijeme, shodno rezultatima brojnih studija iz ovog domena, doprinosi rasvjetljavanju brojnih okolnosti prilikom primjene raznovrsnih formi i oblika fizičkog vježbanja kod djece sa posebnim potrebama. Ova oblast u Sport Mont časopisu je bila zastupljena radom: "The effect of regular physical education in the transformation motor development of children with special needs" (Bojanić, Ljubojević i Bubanja, 2016).

Antropologija je naučna disciplina koja se bavi konstrukcijom antropoloških modela, utvrđivanjem strukture i relacija antropoloških karakteristika, kao i utvrđivanjem relacija u odnosu na odgovarajuće kineziološke fenomene. Zastupljeni radovi iz ove oblasti su sljedeći: "Relationship between foot length measurements and body height: A prospective regional study among adolescents in central region of Kosovo"

(Arifi, Gardašević i Mašanović, 2018), "Effect of morphological characteristics and motor abilities on the execution of technical elements in alpine skiing" (Bubanja, Milašinović i Bojanić, 2016), "Differences in the morphological characteristics and body composition of football players of HSC Zrinjski Mostar and FC Siroki Brijeg in Bosnia and Herzegovina" (Ćorluka, Bjelica, Vasiljević, Bubanja, Georgiev i Zeljko, 2018), "Relationship between sitting height measurements and standing height: A prospective regional study among adolescents in eastern region of Kosovo" (Gardašević, 2018), "Relationship between tibia length measurements and standing height: A prospective regional study among adolescents in southern region of Kosovo" (Gardašević, Mašanović i Arifi, 2018), "Sport-specific morphology profile: differences in anthropometric characteristics among elite soccer and handball players" (Gušić, Popović, Molnar, Mašanović i Radaković, 2017), "Effects of various physical education curriculum on motor skills in students of final grades in primary school" (Ljubojević, Muratović i Bubanja, 2016), "Structure of cognitive abilities and skills of lifeguards" (Ljubojević, Muratović i Bubanja, 2017), "Relationship between foot length measurements and body height: A prospective regional study among adolescents in eastern region of Kosovo" (Mašanović, Gardašević i Arifi, 2018), "Relationship between foot length measurements and standing height: A prospective regional study among adolescents in southern region of Kosovo" (Mašanović, Gardašević i Arifi, 2018), "Standing height and its estimation utilizing arm span and foot length measurements in Dinaric Alps population: A systematic review" (Mašanović, 2018), "Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from the Serbian national league" (Mašanović, 2018), "Body height and its estimation utilizing arm span measurements in male adolescents from southern region in Montenegro" (Milašinović, Popović, Matić, Gardašević i Bjelica, 2016), "Body height and its estimation utilizing arm span measurements in female adolescents from southern region in Montenegro" (Milašinović, Popović, Jakšić, Vasiljević i Bjelica, 2016), "Ranging the results achieved in biomotoric and specific-motoric skills in handball players and non-athletes cadet" (Muratović i Bojanić, 2016), "Standing height and its estimation utilizing foot measurements in adolescents from western region in Kosovo" (Popović, Gardašević, Mašanović, Arifi i Bjelica, 2017).



Slika 3. Naslovna strana časopisa (2017)

Iz oblasti sociologije sporta i naučnog izdavaštva, koje objašnjavaju odnose pojedinca prema grupi ili široj društvenoj zajednici, zakonitosti unutar složenih pedagoških okvira, kao i trendove kretanja i razvoja stručnog i naučnog izdavaštva u sportskim naukama, zastupljeni su sljedeći radovi: "Youth and sport in Montenegro" (Maroš, 2018), "A content analysis of published articles in Montenegrin journal of sportsojscience and medicine from 2012 to 2018" (Maroš, 2018), "Research and writing development in the area of sport science publishing in Montenegro" (Popović, 2018).

Korektivna gimnastika predstavlja strukturalni segment kineziterapije i primjenjuje se u prevenciji i korekciji funkcionalnih poremećaja i promjena na lokomotornom aparatu. Iz ove oblasti je objavljen naučni rad: "Frequencu of foot deformity among students of Faculty for sport and physical education" (Muratović, Vasiljević, Bojanić i Ljubojević, 2016).

U oblasti istorija sporta, koja se bavi izučavanjem istorijskih fakata i okolnosti bitnih za razumijevanje evolutivnih tokova kompleksnog razvoja sporta, dati su sljedeći radovi: "Historical development of skiing with special retrospection in Montenegro" (Milašinović i Popović, 2016), "Historical development of skiing: Case study in Durmitor area" (Milašinović i Bjelica, 2017).

Menadžment i marketing u sportu, u uslovima specifičnih socio-ekonomskih prilika današnjice, predstavljaju veoma bitne karike u složenom sistemu društvenih odnosa u sportu. Iz ovih oblasti su izdvojeni radovi: "Ethical dilemmas of sport advertising" (Bjelica, Gardašević, Vasiljević i Popović, 2016), "Attitudes of consumers from Autonomous province of Vojvodina toward advertising through sport in relation with the frequencu of watching sports events" (Mašanović, 2018), "Model of advertising communication in sport" (Popović i Milašinović, 2016).

Diskusija

U ovom radu izvršena je klasifikacija naučnih članaka predavača sa Fakulteta za sport i fizičko vaspitanje u Nikšiću, koji su objavljeni u časopisu Sport Mont između 2016. i 2018. godine. Metodom analize sadržaja utvrđeno je da su isti iz sljedećih oblasti: sportski trening i metodika treninga (7), ishrana i suplementacija (2), fizičko vaspitanje djece sa posebnim potreba-

ma (1), antropologija sporta (16), sociologija sporta i naučno izdavaštvo (3), korektivna gimnastika (1), istorija sporta (2) i menadžment i marketing u sportu (3).

Najveći broj istraživanja je iz antropologije sporta, što i ne čudi obzirom na veliki opseg ove naučne discipline i sami značaj problematike koju tretira. U ovim radovima pažnja je bila usmjerena na utvrđivanju raznovrsnih uticaja i relacija između pojedinih segmenata antropološkog statusa, kao i njihov prediktivni aspekt na raznovrsne kineziološke manifestacije. Ipak, tematika se u najvećem broju studija odnosila na ispitivanje relacija pojedinih antropometrijskih pokazatelja kod adolescenata sa teritorija Crne Gore i Kosova. Dobijeni rezultati ovih istraživanja, daju značajan doprinos u objašnjavanju etnografskih razlika i utvrđivanju sličnosti i odstupanja od prethodno utvrđenih standarda u ispitivanim odnosima fokusiranih morfoloških mjera. Ostala prikazana istraživanja iz ove kompleksne oblasti rasvjetljavaju brojne nedoumice vezane za kompleksno funkcionisanje "sistema čovjek" u najraznovrsnijim kretnim postavkama. Poslije antropologije sporta, najviše radova je bilo iz oblasti sportskog treninga i metodike treninga. U njima je težište stavljeno na utvrđivanju efekata dejstva raznovrsnih trenajnih tretmana na unaprijeđivanje kondicionih potencijala u pojedinim sportovima. Takođe, u kontekstu efikasnosti primjene djelotvornih trenajnih transformacionih procesa, u pojedinim radovima su se razmatrale i metode za procjenu pojedinih psihomotoričkih sposobnosti, kao i teorije zamora u sportu. U krajnjem, rezultanta svih studija iz ove oblasti je razotkrivanje načina efikasnog djelovanja na one antropološke operatore koji dovode do vrhunskog sportskog postignuća. Ovdje je takođe naglašen broj radova iz menadžmenta i marketinga u sportu. Ove teme su doprinijele shvatanju brojnih dešavanja u sportu i tržišnoj ekonomiji u kontekstu savremenog poimanja socioekonomskih fenomena. Zapravo, govorilo se o kreiranju konceptualnog modela reklamne komunikacije, etičkim dilemama sportskog oglašavanja i markentiškim djelatnostima u sportu postavljenim u međudnos sa potrebama i stavovima potrošača, odnosno kupaca sportskih proizvoda. Radovi autora iz oblasti sociologije sporta i naučnog izdavaštva, takođe su dali značajne činjenične postavke bitne za poimanje složenih socioloških fenomena u uslovima ubrzanog razvoja sporta i njegovih pratećih

djelatnih kategorija, kao i tokova savremenog stručnog i naučnog izdavaštva u sportskim naukama. Po svojoj zastupljenosti slijede radovi koji se tiču ishrane i suplementacije, a koji su, umnogome doveli do razumijevanju međuzavisnosti adekvatnog nutritivnog statusa i pojedinih pokazatelja koji se direktno ili indirektno povezuju sa zdravljem čovjeka. Iako su autori iz istorije sporta, fizičkog vaspitanja djece sa posebnim potrebama i korektivne gimnastike zastupljeni sa po jednim radom, oni su obzirom na značaj predmeta proučavanja ipak ostavili značajan trag u izdanjima časopisa u navedenom vremenskom periodu. Ustanovljena su brojna stanovišta na kojima počivaju postulati pravilnog shvatanja evolutivnog razvoja sporta, adekvatne primjene sredstava i metoda fizičkog vježbanja u specifičnim uslovima kod djece sa posebnim potrebama i značaju statike i dinamike stopala kao segmenta lokomotornog aparata čovjeka. Sumirajući sve naprijed rečeno, konstatuje se izuzetna stručna, naučna i stvaralačka djelatnost predavača sa Fakulteta za sport i fizičko vaspitanje u Nikšiću, koja je u krajnjem dovela do značajnih zakonomjernih naučnih odrednica, koje će, sada i ubuduće, biti vodilja u pravcima daljeg istraživačkog aktiviteta i unaprijeđivanja složenih procesa u fizičkom vaspitanju i sportu.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 13 July 2019 | **Accepted:** 20 August 2019 | **Published:** 11 October 2019

References

- Arifi, F., Gardašević, J., & Mašanović, B. (2018). Relationship between foot length measurements and body height: A prospective regional study among adolescents in central region of Kosovo. *Sport Mont*, 16(3), 69-74.
- Bjelica, D., Gardašević, J., Vasiljević, I., & Popović, S. (2016). Ethical dilemmas of sport advertising. *Sport Mont*, 14(3), 41-43.
- Bjelica, D., Milošević, Z., Talović, M., & Bajramović, I. (2018). Repulsion of the futsal ball depending on the pressure in it. *Sport Mont*, 16(2), 61-67.
- Bojanić, D., Ljubojević, M., & Bubanja, M. (2016). The effect of regular physical education in the transformation motor development of children with special needs. *Sport Mont*, 14(1), 21-23.
- Bubanja, M., Milašinović, R., & Bojanić, D. (2016). Effect of morphological characteristics and motor abilities on the execution of technical elements in alpine skiing. *Sport Mont*, 14(3), 11-14.
- Čorluka, M., Bjelica, D., Vasiljević, I., Bubanja, M., Georgiev, G., & Zeljko, I. (2018). Differences in the morphological characteristics and body composition of football players of HSC Zrinjski Mostar and FC Siroki Brijeg in Bosnia and Herzegovina. *Sport Mont*, 16(2), 77-81.
- Gardašević, J., Bjelica, D., & Vasiljević, I. (2016). Six-week preparation period and its effects on transformation movement speed. *Sport Mont*, 14(1), 13-16.
- Gardašević, J., Bjelica, D., Vasiljević, I., & Milašinović, R. (2016). The effects of the training in the preparation period on the repetitive strength transformation with cadet level football players. *Sport Mont*, 14(2), 31-33.
- Gardašević, J., Bjelica, D., & Vasiljević, I. (2017). The strength of kicking the ball after preparation period with U15 football players. *Sport Mont*, 15(2), 39-42.
- Gardašević, J., Bjelica, D., & Čorluka, M. (2018). The impact of the preparation period on endurance at football players U16. *Sport Mont*, 16(1), 21-24.
- Gardašević, J. (2018). Relationship between sitting height measurements and standing height: A prospective regional study among adolescents in eastern region of Kosovo. *Sport Mont*, 16(2), 15-19.
- Gardašević, J., Mašanović, B., & Arifi, F. (2018). Relationship between tibia length measurements and standing height: A prospective regional study among adolescents in southern region of Kosovo. *Sport Mont*, 16(3), 51-55.
- Gušić, M., Popović, S., Molnar, S., Mašanović, B., & Radaković, M. (2017). Sport-specific morphology profile: differences in anthropometric characteristics among elite soccer and handball players. *Sport Mont*, 15(1), 3-6.
- Krivokapić, D., & Tanase, G. (2016). Methods for evaluation of some psychomotor abilities. *Sport Mont*, 14(2), 17-19.
- Krivokapić, D. (2017). Theories of tiredness in sport. *Sport Mont*, 15(1), 35-37.
- Ljubojević, M., Muratović, A., & Bubanja, M. (2016). Effects of various physical education curriculum on motor skills in students of final grades in primary school. *Sport Mont*, 14(1), 25-28.
- Ljubojević, M., Muratović, A., & Bubanja, M. (2017). Structure of cognitive abilities and skills of lifeguards. *Sport Mont*, 15(1), 27-29.
- Maroš, M. (2018). Youth and sport in Montenegro. *Sport Mont*, 16(2), 97-100.
- Maroš, M. (2018). A content analysis of published articles in Montenegrin journal of sports science and medicine from 2012 to 2018. *Sport Mont*, 16(3), 97-102.
- Mašanović, B., Gardašević, J., & Arifi, F. (2018). Relationship between foot length measurements and body height: A prospective regional study among adolescents in eastern region of Kosovo. *Sport Mont*, 16(1), 9-13.
- Mašanović, B., Gardašević, J., & Arifi, F. (2018). Relationship between foot length measurements and standing height: A prospective regional study among adolescents in southern region of Kosovo. *Sport Mont*, 16(2), 27-31.
- Mašanović, B. (2018). Standing height and its estimation utilizing arm span and foot length measurements in Dinaric Alps population: A systematic review. *Sport Mont*, 16(2), 101-106.
- Mašanović, B. (2018). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from the Serbian national league. *Sport Mont*, 16(3), 19-24.
- Milašinović, B. (2018). Attitudes of consumers from Autonomous province of Vojvodina toward advertising through sport in relation with the frequency of watching sports events. *Sport Mont*, 16(3), 91-96.
- Milašinović, R., & Popović, S. (2016). Historical development of skiing with special retrospection in Montenegro. *Sport Mont*, 14(1), 39-41.
- Milašinović, R., Popović, S., Matić, R., Gardašević, J., & Bjelica, D. (2016). Body height and its estimation utilizing arm span measurements in male adolescents from southern region in Montenegro. *Sport Mont*, 14(2), 21-23.
- Milašinović, R., Popović, S., Jakšić, D., Vasiljević, I., & Bjelica, D. (2016). Body height and its estimation utilizing arm span measurements in female adolescents from southern region in Montenegro. *Sport Mont*, 14(23), 15-18.
- Milašinović, R., & Bjelica, D. (2017). Historical development of skiing: Case study in Durmitor area. *Sport Mont*, 15(1), 39-41.
- Muratović, A., & Bojanić, D. (2016). Ranging the results achieved in biomotoric and specific-motoric skills in handball players and non-athletes cadet. *Sport Mont*, 14(1), 29-31.
- Muratović, A., Vasiljević, I., Bojanić, D., & Ljubojević, M. (2016). Frequency of foot deformity among students of Faculty for sport and physical education. *Sport Mont*, 14(3), 19-20.
- Popović, S., & Milašinović, R. (2016). Model of advertising communication in sport. *Sport Mont*, 14(1), 31-38.
- Popović, S., Gardašević, J., Mašanović, B., Arifi, F., & Bjelica, D. (2017). Standing height and its estimation utilizing foot measurements in adolescents from western region in Kosovo. *Sport Mont*, 15(3), 3-7.
- Popović, S. (2018). Research and writing development in the area of sport science publishing in Montenegro. *Sport Mont*, 16(3), 31-36.
- Vasiljević, I., Bjelica, D., Kezunović, M., & Gardašević, J. (2016). Knowledge of physical education students on sports nutrition. *Sport Mont*, 14(1), 17-19.
- Vasiljević, I., Gardašević, J., Kezunović, M., & Bojanić, D. (2017). Waist circumference as an indicator abdominal obesity in middle age. *Sport Mont*, 15(1), 21-22.

REVIEW PAPER

Exercise Core Body Temperature is Adequately Regulated Following Spinal Cord Injury: A Meta-Analysis

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Abstract

Damage to the spinal cord results in malfunction of sympathetic pathways, which consequently influences thermoregulation during exercise. A consensus view is that athletes with spinal cord injury (SCI) are at a heightened thermal strain than the able-bodied athletes. However, a number of studies have reported similar increases in core body temperature in both populations. This study presented an up-to-date review of core body temperature response to exercise, from a meta-analytic point of view. Inclusion criteria were persons with SCI and control population (either healthy persons or wheelchair athletes without SCI) completed an exercise trial and their core body temperature responses were recorded under the same environmental conditions. Effect of SCI on thermoregulatory capabilities was quantified as raw mean core body temperature difference. Twelve studies examining 100 persons with SCI (Cervical (C2) to Lumbar (L5)) and 97 persons without SCI were meta-analyzed. The exercise interventions were submaximal exercise and the unweighted means $\pm$ standard deviations heat index and exercise duration were $31.5 \pm 11.9^\circ\text{C}$ and 57 ± 20 min, respectively. Regardless of injury, core body temperature was not different: raw mean difference, 0.048°C ; 95% confidence interval, -0.12°C to 0.22°C . In conclusion, under SCI sport-specific exercise and environmental conditions, SCI does not produce outsized thermoregulatory impact, though the influence could be variable as a result of different lesion levels.

Key words: *Paraplegia, Tetraplegia, Rectal Temperature, Heat Stress*

Introduction

Prolonged exercise in the heat elevates core body temperature, which impairs endurance performance and poses increased risks of heat illness (American College of Sports Medicine et al., 2007). Like the able-bodied athletes, athletes with spinal cord injury (SCI) not only face similar heat strain when exercise training and competition occur in hot and humid environments, but also their thermoregulatory capabilities are uniquely challenged. Traumatic damage to the spinal cord, especially with resultant tetraplegia or high paraplegia is associated with a significant malfunction of the sympathetic pathways (Walter & Krassioukov, 2018). Following SCI, the afferent pathways from the periphery to the thermoregulatory effectors in the hypothalamus are disrupted, accounting for the abnormal physiological control during physical activities and exercise (Walter & Krassioukov, 2018). During continuous submaximal exercise in temperate and warm conditions ($20\text{--}30^\circ\text{C}$),

athletes with SCI show elevated core body temperature, and this increase in core body temperature is more evident in athletes with high level lesion when traumatic damage occurs above T6 (Price, 2016; Price & Trbovich, 2018). This alteration of sympathetic nervous system activity below the lesion level also impairs sweating, increasing susceptibility of heat illness (Price, 2016). Athletes with SCI are therefore considered to be under a greater risk of hyperthermia when compared to the able-bodied athletes (Lepretre, Goosey-Tolfrey, Janssen, & Perret, 2016).

A growing number of studies however reported that thermoregulation in persons with SCI during exercise was more dynamic than traditionally believed. Evidence for this possibility has been revisited by Price and Trbovich (2018). Briefly, persons with paraplegia appear to show similar exercise core body temperature responses compared to the able-bodied in temperate and warm environments, while persons with tetraplegia appear to show gre-

ater heat imbalance due to greater loss of sympathetic nervous system control (Price & Trbovich, 2018). Despite our growing knowledge of thermoregulatory capabilities in various populations, there remains a paucity of information related to athletes with SCI. Therefore, the aim of the present review was to examine the thermoregulatory capabilities following SCI. Thermoregulation is usually described as a mechanism to maintain body temperature homeostasis in the cold/heat and characterized by a core internal temperature point. In this meta-analysis, the thermoregulatory capabilities are quantified by auditory canal, esophageal, gastrointestinal, oral, or rectal temperature as proxy measures of core body temperature response to exercise.

Methods

Literature search

An electronic literature search (EndNote, ver. X9, Clarivate Analytics, USA) was performed in the PubMed and Web of Science Core Collection databases from their earliest available date up to January 2019. A variety of search terms were explored to capture maximal relevant literature. The combination of final search terms and Boolean connectors used was: (spinal cord injur* OR paralympic OR wheelchair athlete* OR wheelchair player* OR paraplegia OR tetraplegia) AND (thermoregulation OR heat stress OR core body temperature (aural canal temperature, esophageal temperature, rectal temperature) OR skin temperature OR sweat OR sweating).

Two steps of records filtering were applied. The first step was based on screening of titles and abstracts, and those that did not meet the general inclusion criteria were eliminated based on this alone: English-language, full-length, experimental studies of persons with SCI that involved with an exercise protocol. The second step focused on the specific inclusion criteria: experimental protocol must have included a control group without SCI and the outcomes must have included measures of core body temperature. Following these records filtering, a forward citation map based on the references of eligible studies was yielded to identify any other relevant literature that potentially missed from the key words search. After these screening processes, thirteen studies met the inclusion criteria but one study was later excluded due to insufficient duration and environment to test the thermoregulatory capabilities. As a result, a total of twelve studies were included in this meta-analysis (Boot, Binkhorst, & Hopman, 2006; Castellani et al., 2001; Dawson, Bridle, & Lockwood, 1994; Fitzgerald, Sedlock, & Knowlton, 1990; Goosey-Tolfrey, Swainson, Boyd, Atkinson, & Tolfrey, 2008; Griggs, Havenith, Price, Mason, & Goosey-Tolfrey, 2017; Price & Campbell, 1997, 1999; Theisen, Vanlandewijck, Sturbois, & Francaux, 2001; Trbovich, Ortega, Schroeder, & Fredrickson, 2014; Veltmeijer, Pluim, Thijssen, Hopman, & Eijssvogels, 2014; Zacharakis, Kounalakis, Nassis, & Geladas, 2013). A flow diagram of the screening process is presented in Figure 1.

Data extraction

Participants' demographics, exercise protocols, and core body temperature profiles during exercise were extracted. Studies (Boot et al., 2006; Theisen et al., 2001; Trbovich et al., 2014) that included different lesion levels and reported the outcomes separately were extracted as independent data in the meta-analysis; otherwise, data (Goosey-Tolfrey et al., 2008) were extracted as a single composite score (Higgins & Green, 2008). As a result, a total of 16 independent data sets from the 12 included studies

were meta-analyzed. Core body temperatures originally reported in the graphical form were digitally reconstructed to numeric values (Photoshop, ver. CC2018, Adobe, USA). When standard deviations of core body temperatures were not reported directly, these values were calculated from standard errors and sample sizes presented in studies. The risk of bias derived from each study was evaluated using established procedures, with scores from 0 to 11 possible (Physiotherapy Evidence-Based Database Scale (Mooney, Herbert, Sherrington, & Maher, 2002).

Meta-analysis

Effect size in this meta-analysis was calculated as the raw mean difference of core body temperature during exercise between persons with and without SCI. Among the included studies, two studies reported final measurement (cessation of exercise) core body temperature (Boot et al., 2006; Castellani et al., 2001); four studies reported change-from-baseline core body temperature (Dawson et al., 1994; Fitzgerald et al., 1990; Theisen et al., 2001; Zacharakis et al., 2013); three studies reported both final measurement and change-from-baseline core body temperature (Goosey-Tolfrey et al., 2008; Griggs et al., 2017; Trbovich et al., 2014); and three studies reported change-from-baseline core body temperature directly and presented figures of core body temperature responses, allowing indirect data extraction of final measurement core body temperature and correlation coefficients of change-from-baseline core body temperature to be calculated (see below) (Price & Campbell, 1997, 1999; Veltmeijer et al., 2014). Considering baseline core body temperatures showed difference between experimental groups, the calculation of raw mean difference in this meta-analysis was based on change-from-baseline, when both final measurement and change-from-baseline data are available.

Correlation coefficients of change-from-baseline standard deviations were computed (Fu et al., 2013) for included studies in the meta-analysis and the results are inconsistent: mean 0.2978, 95% confidence interval (CI) 0.0384 to 0.5572 (range: -1 to 0.98). Correlations of three studies (Fitzgerald et al., 1990; Theisen et al., 2001; Zacharakis et al., 2013) cannot be calculated by retrospective method and therefore these values were imputed based on the mean correlation (0.2978) in the meta-analysis.

Two sets of sensitivity analysis were performed. First, the imputed correlation coefficient was replaced by 0.0384 and 0.5572 to evaluate the impact of data imputation. Second, although the difference in core body temperature at the commencement of exercise favored reporting change-from-baseline core body temperature in several studies, the correlation coefficients are inconsistent among the included studies and the overall mean is less than 0.5, suggesting final measurement may offer more precision. Therefore, studies reporting both final measurement and change-from-baseline core body temperature or allowing indirect data extraction of final measurement core body temperature were meta-analyzed, utilizing final measurement core body temperature instead this time.

All statistical analyses were carried out using the Comprehensive Meta-Analysis (ver. 3.3.070, Biostat, USA). The meta-analysis was based on a random-effects model, accounting for the differences in the participants and exercise protocols among studies. Non-overlapping 95% CI was considered a statistical significance. Heterogeneity was assessed using Q statistic but no subgroup or meta-regression analysis could be performed. The publication bias of included studies was assessed using the Egger's test (Egger, Davey Smith, Schneider, & Minder, 1997).

Table 1. Subject Characteristics from 12 Included Studies

Citation	n, sex*		Age (yr)		Body mass (kg)		Height (cm)		SCI level	SCI training background
	SCI	CON	SCI	CON	SCI	CON	SCI	CON		
Goosey-Tolfrey et al., 2008	7M/1F	7M	27.2±6.9	25.4±4.2	68.3±17.9	75.3±10.0	N/A	N/A	C2-L2a	Trained, tennis
Griggs et al., 2017	10	7	30.0±5.0	23.0±5.0	68.4±10.5	65.3±14.8	N/A	N/A	C5/6-C7	National level, rugby
Trbovich et al., 2014	6M	19M	34.5	30.8	N/A	N/A	N/A	N/A	C5-C7	Trained, basketball/rugby
Zacharakis et al., 2013	8M	9M	31.4±8.4	20.1±0.8	72.1±11.5	73.9±5.5	180.0±7.0	179.0±5.0	C7-T6	National level, basketball
Boot et al., 2006	4M	10M	31.0±4.0	33.0±5.0	77.0±7.0	80.0±14.0	183.0±9.0	180.0±8.0	T1-T6	N/A
Trbovich et al., 2014	5M	19M	27.6	30.8	N/A	N/A	N/A	N/A	T3-T5	Trained, basketball/rugby
Price & Campbell, 1999	9	11	28.5±4.5	30.3±7.4	67.9±14.0	78.3±7.2	N/A	N/A	T3/4-L1	National level, various sports
Price & Campbell, 1997	10	9	30.1±6.4	30.6±7.9	62.9±12.7	77.7±9.5	N/A	N/A	T3/4-L4	National level
Theisen et al., 2001	6	6	32.2±7.0	28.3±5.6	82.0±10.7	78.7±10.1	181.0±7.0	180.0±8.0	T5-T9	Recreational level
Veltmeijer et al., 2014	2	4	N/A	N/A	N/A	N/A	N/A	N/A	T6/L2b	International level, tennis
Castellani et al., 2001	4M	5M	36.0±14.0	33.0±9.0	66.9±11.8	72.5±13.2	171.0±8.0	171.0±4.0	T7/8-T11c	Highly trained, track/road race
Boot et al., 2006	6M	10M	29.0±6.0	33.0±5.0	68.0±13.0	80.0±14.0	181.0±6.0	180.0±8.0	T7-T12	N/A
Trbovich et al., 2014	6M	19M	37.3	30.8	N/A	N/A	N/A	N/A	T7-T12	Trained, basketball/rugby
Fitzgerald et al., 1990	5F	5F	27.0±3.8	24.6±2.7	59.0±6.9	64.6±7.2	N/A	N/A	T9-L5	No training history
Theisen et al., 2001	6	6	31.3±8.3	28.3±5.6	60.2±11.2	78.7±10.1	174.0±10.0	180.0±8.0	T10-T12	Recreational level
Dawson et al., 1994	5M	5M	24.8±3.7	26.0±3.2	71.9±7.5	86.8±5.8	179.0±4.7	183.3±5.4	T12-L3	Recreational level
Unweighted mean	6.3	8.1	30.5	27.7	68.7	75.3	178.4	178.7		

Note: Sixteen independent data set from a total of 12 included studies (see also methods). Data are expressed as mean ± standard deviation. CON, controls; F, female; M, male; N/A, data not available; SCI, spinal cord injury. *when the reporting of sex of the subjects is not available, the overall sample size is included. a included brittle bones (n=1), spina bifida (n=1), nerve damage (n=1). b included brain stem infarction (n=1). c including post-polio individual (n=1).

Table 2. Exercise Characteristics from 12 Included Studies

Citation (lesion level)	Ambient environment (Temperature, rH)	Heat index (°C)	CBT	Exercise regime	Modality	Exercise loading	
						SCI	CON
Goosey-Tolfrey et al., 2008	30.8°C, 60.6%	35	TAC	60 min, 50% PO _{peak}	WERG/CYCLE	43±21 W	129±25 W
Griggs et al., 2017	18.4-20.9°C, 31.1-45.1%	17-20	TGI	32 min, game play	WCR	158±44 W	200±74 W
Trbovich et al., 2014	21.1-23.9°C	N/A	TGI	60 min, ISP	WCB, WCR	N/A	N/A
Zacharakis et al., 2013	23°C, 40-55%	22-23	TOR	60 min, 55% VO _{2max}	WCR	37.1±2.6 W	48.6±1.1 W
Boot et al., 2006 (T1-T6)	35°C, 70%	50	TRE	45 min, 40% PO _{peak}	ACE	38±3.5 W	61±2.5 W
Price & Campbell, 1999	21.5°C, 47%	21	TAC	60 min, 60% VO _{2max}	ACE	1.22 L·min ⁻¹	2.07 L·min ⁻¹
Price & Campbell, 1997	21.5°C, 47%	21	TAC	90 min, 80% HR _{peak}	ACE	1.34 L·min ⁻¹	1.92 L·min ⁻¹
Theisen et al., 2001 (T5-T9)	23°C, 53%	23	TES	60 min, 50% PO _{peak}	ACE	50.0±11.0 W	56.7±2.6 W
Veltmeijer et al., 2014	21.2-24.8°C, 51.8-61.4%	21-25	TGI	45 min, game play (singles)	WCT	N/A	N/A
Castellani et al., 2001	25°C, 58.9% & 32.2°C, 66%	25, 40	TRE	20 min, 0.03 kg resistance·kg ⁻¹ BM	WCR	N/A	N/A
Boot et al., 2006 (T7-T12)	35°C, 70%	50	TRE	45 min, 40% PO _{peak}	ACE	53±2.4 W	61±2.5 W
Fitzgerald et al., 1990	24-25°C, 38-52%	23-25	TOR	90 min, 50-55% VO _{2max}	WCR	221±44.9 W	255±20.3 W
Theisen et al., 2001 (T10-T12)	23°C, 53%	23	TES	60 min, 50% PO _{peak}	ACE	55.8±11.1 W	56.7±2.6 W
Dawson et al., 1994	37.4°C, 33%	39	TRE	60 min, 55-60% VO _{2max}	ACE	N/A	N/A

Note: ACE, arm-crank exercise; BM, body mass; CBT, core body temperature; CYCLE, cycle ergometer; CON, controls; ISP, intermittent sprint exercise; N/A, data not available; PO_{peak}, peak power output; rH, relative humidity; SCI, spinal cord injury; TAC, auditory canal temperature; TES, esophageal temperature; TGI, gastrointestinal temperature; TOR, oral temperature; TRE, rectal temperature; W, watts; WCB, wheelchair basketball; WCR, wheelchair rugby; WCT, wheelchair tennis; WERG, wheelchair ergometer.

Results

Table 1 presents the participants' characteristics. A total of 100 persons with SCI and 97 persons without SCI were included in the meta-analysis. Three studies reported wheelchair athletes without SCI as control (Boot et al., 2006; Griggs et al., 2017; Veltmeijer et al., 2014), and the rest reported healthy persons as control. Eight studies recruited highly trained athletes with SCI. Table 2 summarizes the exercise protocol. Most experiments were conducted in thermoneutral and warm

environments with the unweighted means $\pm$ standard deviations ambient temperature and heat index being $26.1 \pm 5.7^\circ\text{C}$ and $31.5 \pm 11.9^\circ\text{C}$, respectively. Seven studies utilized wheelchair specific submaximal exercise and five studies utilized upper-body arm ergometer to simulate wheelchair exercise. The exercise intensity was prescribed based on relative maximal exercise capacity or wheelchair gameplay. The unweighted exercise duration was 57 ± 20 min. The quality scores for these studies ranged from 9 to 11, qualifying as high quality of design.

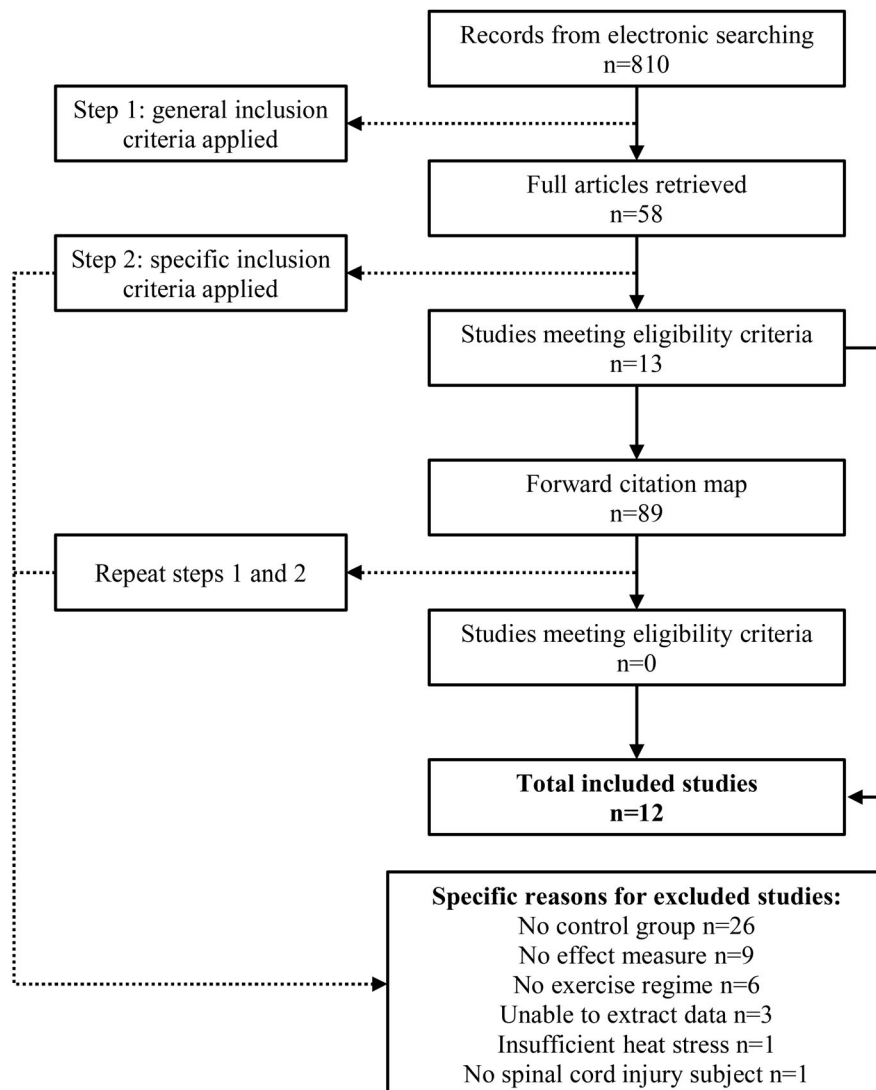


Figure 1. Flow diagram of the study selection process

Overall, there was no statistical or biological meaningful difference in the core body temperature response to exercise between persons with SCI and without SCI: raw mean difference 0.048°C (95% CI: -0.12°C to 0.22°C) (Figure 2). A sensitivity test based on final measurement core body temperature shifted the overall mean difference to 0.002°C (95% CI: -0.18°C to 0.19°C) (Figure 2), thus ruled out the possibility of selective reporting and analysis on the effect size. By assuming two

different correlation coefficients, the overall meta-analysis did not shift, confirming minimal impact of data imputation on the result. Significant heterogeneity was expected as a result of variable sample and experimental design. Due to the variance in sample population (i.e., mixed lesion levels within and between studies), no subgroup analysis however can be performed. The Egger's test, $P = 0.42$ (1-tailed), suggested low risk of publication bias.

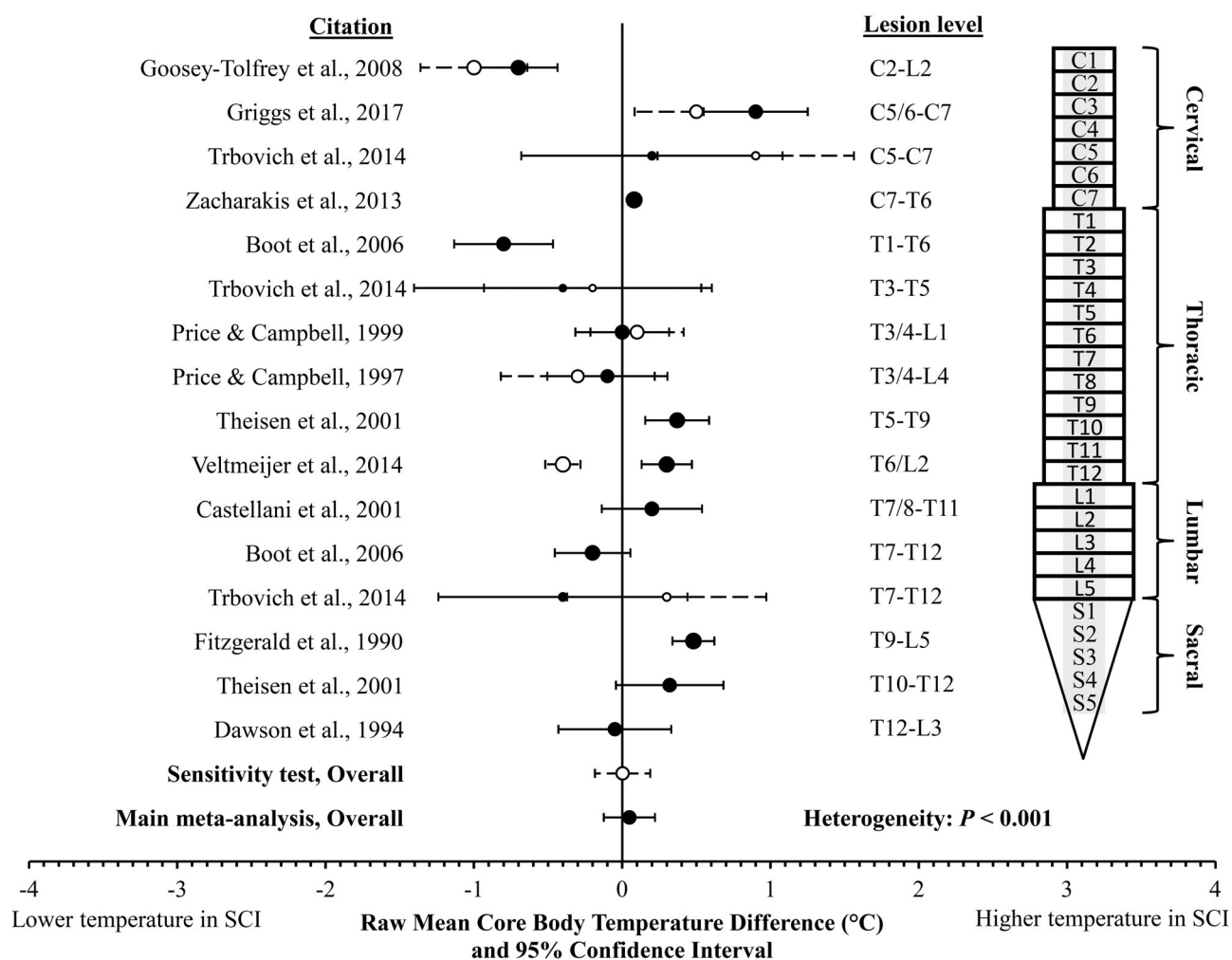


Figure2. Forest plot of core body temperature response to exercise between persons with spinal cord injury and without injury. Filled circle represents the main meta-analysis and open circle represents sensitivity analysis utilizing final measurement core body temperature when available (see also Table 1). The solid horizontal line depicts the 95% confidence interval of the main meta-analysis and the dashed horizontal line depicts the 95% confidence interval of the sensitivity analysis. The circle size was proportional to the weight of each study in the main random-effects meta-analysis.

Discussion

This is the first meta-analysis investigating the effect of SCI on thermoregulatory capabilities, quantified by the core body temperature responses during exercise. Notably, contrary to the common belief that impaired thermoregulatory mechanisms due to disrupted neural thermoregulatory pathways would increase risks of hyperthermia for athletes with SCI, current meta-analysis of core body temperature responses found that the thermoregulatory capabilities during sport-specific experiments do not show synchronized decline. Therefore, this population is not at higher risk of experiencing heat illness simply on the basis of partial disruption of thermoregulatory capabilities from SCI.

Heat illness occurrences, or life-threatening exertional heat stroke in sports, can be attributed to exercise duration and climatic conditions. In able-bodied athletes, heat illness occurs most often in marathon and ultra-distance races held in hot and humid conditions (American College of Sports Medicine et al., 2007). Considering these factors, athletes with SCI participating in marathon (T54 classification) or open field sports

such as tennis lasting over 60 min in hot weather, are more likely to experience heat illness. To the best of knowledge to date, however, there is an apparent rarity of occurrence of heat illness in this population. A recent study reporting illness at the 2015 Para Athletic World Championships (venue ambient temperature 24.6–36.0°C) found only one incident of heat illness as a direct result of pathological elevation of the core body temperature (T52 classification, 39.5°C at the cessation of competition) (Grobler, Derman, Racinais, Ngai, & van de Vliet, 2019). In a field study exploring the physiological responses of elite wheelchair athletes, it has been shown that core body temperature following a 25-km race was 38.7°C (Edwards et al., 2018), which is well below the threshold temperature of 40.5°C that could trigger thermoregulatory collapse (Casa et al., 2015). In view of the available epidemiologic profiles, claims that athletes with SCI are more susceptible to heat stress than the able-bodied athletes are unwarranted.

A traditional concept regarding the thermoregulatory pathways following SCI is based on disrupted ascending peripheral afferent input to the hypothalamus, which could be activated

by the rising temperature of viscera and skin (Price & Trbovich, 2018; Walter & Krassioukov, 2018). When the peripheral temperature sensing below the lesion level is malfunctioning, this could cause delayed activation of thermoregulatory center and consequently leads to reduced heat loss response when overheated (Price & Trbovich, 2018). However, the organization of the temperature sensing underlying afferent thermal pathways involves a complex interaction of multiple sites from the peripheral and central origins. The functional contribution of central temperature sensing to the thermoregulatory mechanisms has been reported in a variety of animal species. Classic experiments have identified that artificial heating of the hypothalamus could directly elicit cutaneous vasodilation (Folkow, Strom, & Uvnas, 1949), and artificial increase in hypothalamic temperature triggered comparable heat loss response to spinal cord temperature (Jessen & Ludwig, 1971). More recent studies provided additional evidences, showing that the brain temperature could trigger signaling of neural sensors in the hypothalamus, which strongly influences the core body temperature (Conti et al., 2006; Song et al., 2016). During exercise, the brain temperature mimics temperature response of the core in animals and humans (Fuller, Carter, & Mitchell, 1998; Nybo, Secher, & Nielsen, 2002; Walters, Ryan, Tate, & Mason, 2000). Considering hypothalamic neurons are sensitive to brain temperature and the set-point thermoregulation is around 37°C (Boulant, 2006), thermal sensors residing in this region provide direct and efficient input regarding the state of body temperature homeostasis to the thermoregulatory center. It is clear, from the animal model at least, central temperature sensing plays a significant role in initiating thermoregulatory mechanisms. It has been suggested that central temperature sensing could even override lower-priority peripheral temperature sensing from the skin (Fusco, Hardy, & Hammel, 1961; Shafton, Kitchen, McKinley, & McAllen, 2014). It is possible that, despite of malfunctioning of peripheral temperature sensing from the spinal cord, viscera, and skin following SCI, the afferent thermal pathways to the hypothalamus could still function properly during exercise. The complex organization and coordination of the ascending thermoregulatory pathways following SCI await further clarification.

Rising body temperature constitutes an immediate stimulus for the activation of the heat loss mechanism through dilation of blood vessels (cutaneous vasodilation) and water evaporation (sweating) in humans. Human lesion studies suggest impaired reflex cutaneous vasodilation responses to temperature rise (Theisen et al., 2001; Van Duijnhoven et al., 2009). However, current result suggests the functional consequences to global thermoregulation during exercise seems less than expected. It remains to be determined the threshold exercise duration and environmental conditions that may lead to excessive body temperature rise (above 40°C) following partial or complete loss of spinal reflex below the lesion level.

A limitation to the present meta-analysis should be noted. Most of the included studies prescribed sport-specific exercise intensity; as a result, the absolute exercise loading eliciting heat production was not accounted for the consequent change in core body temperature. However, it should be noted that these experimental protocols attempted to mimic SCI sport-specific physical demands hence their results should be considered to be ecologically valid.

In conclusion, the present analysis of core body temperature responses under SCI sport-specific exercise and environmental

conditions identified that thermoregulatory mechanisms can be adequately modulated in the presence of SCI, although special attention is still needed when adaptive endurance competitions are scheduled in hot and humid environments. Advancing our understanding of the precise physiological and neural mechanisms of thermoregulation has broad implications for this population to participate in physical activity and exercise, which could have enormous health and social benefits.

Acknowledgements

There are no acknowledgements.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Received: 20 July 2019 | **Accepted:** 1 September 2019 | **Published:** 11 October 2019

References

- American College of Sports Medicine, Armstrong, L.E., Casa, D.J., Millard-Stafford, M., Moran, D.S., Pyne, S.W., & Roberts, W.O. (2007). American College of Sports Medicine position stand. Exertional heat illness during training and competition. *Medicine and Science in Sports and Exercise*, 39(3), 556-572.
- Boot, C.R., Binkhorst, R.A., & Hopman, M.T. (2006). Body temperature responses in spinal cord injured individuals during exercise in the cold and heat. *International Journal of Sports Medicine*, 27(8), 599-604.
- Boulant, J.A. (2006). Neuronal basis of Hammel's model for set-point thermoregulation. *Journal of Applied Physiology*, 100(4), 1347-1354.
- Casa, D.J., DeMartini, J.K., Bergeron, M.F., Csillan, D., Eichner, E.R., Lopez, R.M., Ferrara, M.S., Miller, K.C., O'Connor, F., Sawka, M.N., & Yeargin, S.W. (2015). National Athletic Trainers' Association Position Statement: Exertional Heat Illnesses. *Journal of Athletic Training*, 50(9), 986-1000.
- Castellani, J.W., Armstrong, L.E., Kenefick, R.W., Pasqualicchio, A.A., Riebe, D., Gabaree, C.L., & Maresh, C.M. (2001). Cortisol and testosterone concentrations in wheelchair athletes during submaximal wheelchair ergometry. *European Journal of Applied Physiology*, 84(1-2), 42-47.
- Conti, B., Sanchez-Alavez, M., Winsky-Sommerer, R., Morale, M., Lucero, J., Brownell, S., Fabre, V., Huitron-Resendiz, S., Henriksen, S., Zorrilla, E.P., de Lecea, L., & Bartfai, T. (2006). Transgenic mice with a reduced core body temperature have an increased life span. *Science*, 314(5800), 825-828.
- Dawson, B., Bridle, J., & Lockwood, R. J. (1994). Thermoregulation of paraplegic and able bodied men during prolonged exercise in hot and cool climates. *Paraplegia*, 32(12), 860-870.
- Edwards, T., Barfield, J.P., Niemi, G.M., Beals, J.W., Broad, E.M., Motl, R.W., De Lisio, M., Burd, N.A., & Pilutti, L.A. (2018). Physiological responses during a 25-km time trial in elite wheelchair racing athletes. *Spinal Cord Series and Cases*, 4, 77.
- Egger, M., Davey Smith, G., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *British Medical Journal*, 315(7109), 629-634.
- Fitzgerald, P.I., Sedlock, D.A., & Knowlton, R.G. (1990). Circulatory and thermal adjustments to prolonged exercise in paraplegic women. *Medicine and Science in Sports and Exercise*, 22(5), 629-635.
- Folkow, B., Strom, G., & Uvnas, B. (1949). Cutaneous vasodilatation elicited by local heating of the anterior hypothalamus in cats and dogs. *Acta physiologica Scandinavica*, 17(4), 317-326.
- Fu, R., Vandermeer, B.W., Shamliyan, T.A., O'Neil, M.E., Yazdi, F., Fox, S.H., & Morton, S.C. (2013). *Handling continuous outcomes in quantitative synthesis*. Rockville, MD: Agency for Healthcare Research and Quality.
- Fuller, A., Carter, R.N., & Mitchell, D. (1998). Brain and abdominal temperatures at fatigue in rats exercising in the heat. *Journal of Applied Physiology*, 84(3), 877-883.
- Fusco, M.M., Hardy, J.D., & Hammel, H.T. (1961). Interaction of central and peripheral factors in physiological temperature regulation. *The American Journal of Physiology*, 200(3), 572-580.
- Goosey-Tolfrey, V., Swainson, M., Boyd, C., Atkinson, G., & Tolfrey, K. (2008). The effectiveness of hand cooling at reducing exercise-induced hyperthermia and improving distance-race performance in wheelchair and able-bodied athletes. *Journal of Applied Physiology*, 105(1), 37-43.
- Griggs, K.E., Havenith, G., Price, M.J., Mason, B.S., & Goosey-Tolfrey, V.L. (2017). Thermoregulatory responses during competitive wheelchair rugby match play. *International Journal of Sports Medicine*, 38(3), 177-183.
- Grobler, L., Derman, W., Racinais, S., Ngai, A.S.H., & van de Vliet, P. (2019). Illness at a Para Athletics Track and Field World Championships under hot and humid ambient conditions. *PM & R*. doi: 10.1002/pmrj.12086

- Higgins, J.P., & Green, S. (2008). Cochrane Handbook for Systematic Reviews of Interventions. In J.P. Higgins & S. Green (Eds.), *Cochrane Handbook for Systematic Reviews of Interventions* (pp. 177). Chichester: Wiley-Blackwell.
- Jessen, C., & Ludwig, O. (1971). Spinal cord and hypothalamus as core sensors of temperature in the conscious dog. II. Addition of signals. *Pflügers Archiv*, 324(3), 205-216.
- Lepretre, P.M., Goosey-Tolfrey, V.L., Janssen, T.W., & Perret, C. (2016). Editorial: Rio, Tokyo Paralympic Games and Beyond: How to Prepare Athletes with Motor Disabilities for Peaking. *Frontiers in Physiology*, 7, 497.
- Moseley, A.M., Herbert, R.D., Sherrington, C., & Maher, C.G. (2002). Evidence for physiotherapy practice: a survey of the Physiotherapy Evidence Database (PEDro). *The Australian Journal of Physiotherapy*, 48(1), 43-49.
- Nybo, L., Secher, N.H., & Nielsen, B. (2002). Inadequate heat release from the human brain during prolonged exercise with hyperthermia. *The Journal of Physiology*, 545(2), 697-704.
- Price, M.J. (2016). Preparation of Paralympic Athletes; Environmental Concerns and Heat Acclimation. *Frontiers in Physiology*, 6, 415.
- Price, M.J., & Campbell, I.G. (1997). Thermoregulatory responses of paraplegic and able-bodied athletes at rest and during prolonged upper body exercise and passive recovery. *European Journal of Applied Physiology and Occupational Physiology*, 76(6), 552-560.
- Price, M.J., & Campbell, I.G. (1999). Thermoregulatory responses of spinal cord injured and able-bodied athletes to prolonged upper body exercise and recovery. *Spinal Cord*, 37(11), 772-779.
- Price, M.J., & Trbovich, M. (2018). Thermoregulation following spinal cord injury. *Handbook of Clinical Neurology*, 157, 799-820.
- Shafton, A.D., Kitchener, P., McKinley, M.J., & McAllen, R.M. (2014). Reflex control of rat tail sympathetic nerve activity by abdominal temperature. *Temperature*, 1(1), 37-41.
- Song, K., Wang, H., Kamm, G.B., Pohle, J., de Castro Reis, F., Heppenstall, P., Wende, H., & Siemens, J. (2016). The TRPM2 channel is a hypothalamic heat sensor that limits fever and can drive hypothermia. *Science*, 353(6306), 1393-1398.
- Theisen, D., Vanlandewijck, Y., Sturbois, X., & Francaux, M. (2001). Cutaneous vascular response and thermoregulation in individuals with paraplegia during sustained arm-cranking exercise. *International Journal of Sports Medicine*, 22(2), 97-102.
- Trbovich, M., Ortega, C., Schroeder, J., & Fredrickson, M. (2014). Effect of a cooling vest on core temperature in athletes with and without spinal cord injury. *Topics in Spinal Cord Injury Rehabilitation*, 20(1), 70-80.
- Van Duijnhoven, N.T., Janssen, T.W., Green, D.J., Minson, C.T., Hopman, M.T., & Thijssen, D.H. (2009). Effect of functional electrostimulation on impaired skin vasodilator responses to local heating in spinal cord injury. *Journal of Applied Physiology*, 106(4), 1065-1071.
- Veltmeijer, M.T., Pluim, B., Thijssen, D.H., Hopman, M.T., & Eijssvogels, T.M. (2014). Thermoregulatory responses in wheelchair tennis players: a pilot study. *Spinal Cord*, 52(5), 373-377.
- Walter, M., & Krassioukov, A.V. (2018). Autonomic Nervous System in Paralympic Athletes with Spinal Cord Injury. *Physical Medicine and Rehabilitation Clinics of North America*, 29(2), 245-266.
- Walters, T., Ryan, K., Tate, L., & Mason, P. (2000). Exercise in the heat is limited by a critical internal temperature. *Journal of Applied Physiology*, 89(2), 799-806.
- Zacharakis, E.D., Kounalakis, S.N., Nassis, G.P., & Geladas, N.D. (2013). Cardiovascular drift in trained paraplegic and able-bodied individuals during prolonged wheelchair exercise: effect of fluid replacement. *Applied Physiology, Nutrition, and Metabolism*, 38(4), 375-381.

Guidelines for Authors

Revised October 2017

*** Please use the bookmark function to navigate within the guidelines. ***

When preparing the final version of the manuscripts, either NEW or REVISED authors should strictly follow the guidelines. Manuscripts departing substantially from the guidelines will be returned to the authors for revision or, rejected.

1. UNIFORM REQUIREMENTS

1.1. Overview

The *Journal of Anthropology of Sport and Physical Education* (JASPE) applies the Creative Commons Attribution (CC BY) license to articles and other works it publishes.

There is no charge for submissions and no page charge for accepted manuscripts. However, if the manuscript contains graphics in color, note that printing in color is charged.

JASPE adopts a double-blind approach for peer reviewing in which the reviewer's name is always concealed from the submitting authors as well as the author(s)'s name from the selected reviewers.

JASPE honors six-weeks for an initial decision of manuscript submission.

Authors should submit the manuscripts as one Microsoft Word (.doc) file.

Manuscripts must be provided either in standard UK or US English or Montenegrin language. Chosen language standards should be consistent throughout the manuscripts.

Format the manuscript in A4 paper size; margins are 1 inch or 2.5 cm all around.

Type the whole manuscript double-spaced, justified alignment.

Use Times New Roman font, size eleven (11) point.

Number (Arabic numerals) the pages consecutively (centering at the bottom of each page), beginning with the title page as page 1 and ending with the Figure legend page.

Include line numbers (continuous) for the convenience of the reviewers.

Apart from chapter headings and sub-headings avoid any kind of formatting in the main text of the manuscripts.

1.2. Type & Length

JASPE publishes following types of papers:

Original scientific papers are the results of empirically- or theoretically-based scientific research, which employ scientific methods, and which report experimental or observational aspects of anthropology of sport and physical education from five major fields of anthropology: cultural, global, biological, linguistic and medical. Descriptive analyses or data inferences should include rigorous methodological structure as well as sound theory. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

☒ Open Submissions

☒ Indexed

☒ Peer Reviewed

Original scientific papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 30;
- Maximum combined total of 6 Tables/Figures.

Review papers should provide concise in-depth reviews of both established and new areas, based on a critical examination of the literature, analyzing the various approaches to a specific topic in all aspects of anthropology of sport and physical education from five major fields of anthropology: cultural, global, biological, linguistic and medical.

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Editorials are written or commissioned by the editors, but suggestions for possible topics and authors are welcome. It could be peer reviewed by two reviewers who may be external or by the Editorial Board.

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Short reports of experimental work, new methods, or a preliminary report can be accepted as two page papers. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

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Short reports should be:

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Peer review - fair review provides authors who feel their paper has been unfairly rejected (at any journal) the opportunity to share reviewer comments, explain their concerns, and have their paper reviewed for possible publication in JASPE.

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2. MANUSCRIPT STRUCTURE

2.1. Title Page

The first page of the manuscripts should be the title page, containing: title, type of publication, running head, authors, affiliations, corresponding author, and manuscript information. *See example:*

Analysis of Dietary Intake and Body Composition of Female Athletes over a Competitive Season

Original Scientific Paper

Diet and Body Composition of Female Athletes

Svetlana Nepocatych¹, Gytis Balilionis¹, Eric K. O'Neal²

¹Elon University, Department of Exercise Science¹, Elon, NC 27215

²University of North Alabama, Department of Health, Physical Education and Recreation, Florence, AL 35632

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2525 CB

Elon, NC 27244

United States

E-mail: snepocatych@elon.edu

Word count: 2,946

Word count: 4259

Abstract word count: 211

Number of Tables: 3

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Title should be short and informative and the recommended length is no more than 20 words. The title should be in Title Case, written in uppercase and lowercase letters (initial uppercase for all words except articles, conjunctions, short prepositions no longer than four letters etc.) so that first letters of the words in the title are capitalized. Exceptions are words like: “and”, “or”, “between” etc. The word following a colon (:) or a hyphen (-) in the title is always capitalized.

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Short running title should not exceed 50 characters including spaces.

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The form of an author's name is first name, middle initial(s), and last name. In one line list all authors with full names separated by a comma (and space). Avoid any abbreviations of academic or professional titles. If authors belong to different institutions, following a family name of the author there should be a number in superscript designating affiliation.

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Affiliation consists of the name of an institution, department, city, country/territory (in this order) to which the author(s) belong and to which the presented / submitted work should be attributed. List all affiliations (each in a separate line) in the order corresponding to the list of authors. Affiliations must be written in English, so carefully check the official English translation of the names of institutions and departments.

Only if there is more than one affiliation, should a number be given to each affiliation in order of appearance. This number should be written in superscript at the beginning of the line, separated from corresponding affiliation with a space. This number should also be put after corresponding name of the author, in superscript with no space in between.

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Corresponding author's name with full postal address in English and e-mail address should appear, after the affiliations. It is preferred that submitted address is institutional and not private. Corresponding author's name should include only initials of the first and middle names separated by a full stop (and a space) and the last name. Postal address should be written in the following line in sentence case. Parts of the address should be separated by a comma instead of a line break. E-mail (if possible) should be placed in the line following the postal address. Author should clearly state whether or not the e-mail should be published.

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The second page of the manuscripts should be the abstract and key words. It should be placed on second page of the manuscripts after the standard title written in upper and lower case letters, bold.

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Abstract

Results of the analysis of

Key words: *spatial memory, blind, transfer of learning, feedback*

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Starting from the third page of the manuscripts, it should be the main chapters. Depending on the type of publication main manuscript chapters may vary. The general outline is: Introduction, Methods, Results, Discussion, Acknowledgements (optional), Conflict of Interest (optional), and Title, Author's Affiliations, Abstract and Key words must be in English (for both each chosen language of full paper). However, this scheme may not be suitable for reviews or publications from some areas and authors should then adjust their chapters accordingly but use the general outline as much as possible.

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✓ *Table position of the research football team*

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2.4.2. Examples for Reference citations

One work by one author

- ✓ In one study (Reilly, 1997), soccer players
- ✓ In the study by Reilly (1997), soccer players
- ✓ In 1997, Reilly's study of soccer players

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- ✓ In one study (Duffield & Marino, 2007), soccer players
- ✓ In 2007, Duffield and Marino's study of soccer players

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- ✓ Subsequent citation: Bangsbo et al. (2008) stated that

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- ✓ Krstrup et al. (2003) studied
- ✓ In one study (Krstrup et al., 2003), soccer players

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- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

2.4.3. Examples for Reference list

Journal article (print):

- Nepocaty, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine*, 6(2), 57-65. doi: 10.26773/mjssm.2017.09.008
- Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology*, 100(6), 727-735. doi: 10.1007/s00421-007-0468-x
- Krstrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise*, 35(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

Journal article (online; electronic version of print source):

- Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies*, 50(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

- Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia*, 4. Retrieved November 15, 2005, from <http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html>

Conference paper:

- Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from <http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf>

Encyclopedia entry (print, with author):

- Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

- Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from <http://www.bartleby.com/65/et/ethnolog.html>

Thesis and dissertation:

- Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport*. Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from Routledge Handbook of Applied Sport Psychology ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

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Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See example:*

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Table footnotes should be written below the table.

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✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

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✓ ^aOne participant was diagnosed with heat illness and n = 19.^bn = 20.

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✓ *P<0.05, †p<0.01.

2.5.4. Table citation

In the text, tables should be cited as full words. *See example:*

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓in our results (Tables 1, 2 and 5)... (citing more tables at once)

2.6. Figures

On the last separate page of the main manuscript file, authors should place the legends of all the figures submitted separately.

All graphic materials should be of sufficient quality for print with a minimum resolution of 600 dpi. JASPE prefers TIFF, EPS and PNG formats.

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Figures and figure legends should be completely intelligible without reference to the text.

The price of printing in color is 50 EUR per page as printed in an issue of JASPE.

2.6.1. Figure legends

Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written below the figure, in sentence case. *See example:*

- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, * $p > 0.01$, † $p > 0.05$.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See example:*

- ✓ Figure 1
 - × figure 1
 - × Figure 1.
 - ✓ ...exhibit greater variance than the year before (Figure 2). Therefore...
 - ✓ ...as shown in Figures 1 and 3. (citing more figures at once)
 - ✓ ...result has shown (Figures 1-3) that... (citing more figures at once)
 - ✓in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See example:*

- ✓ Figure 1a
- ✓ ...in Figures 1a and b we can...
- ✓ ...data represent (Figures 1a-d)...

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

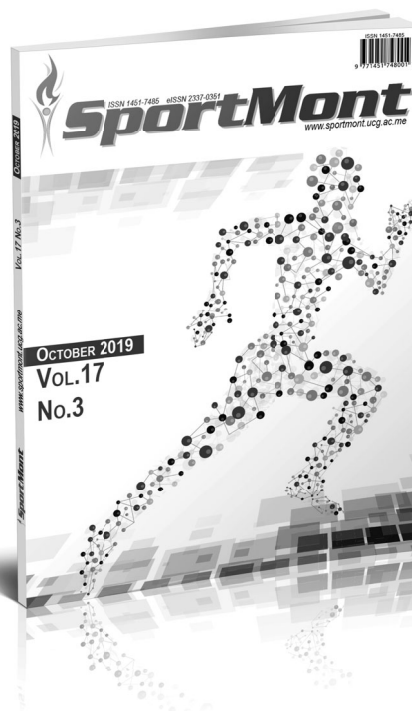
Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10°	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 °	× 10kg	× 12 : 2	× .056
Signs should be placed immediately preceding the relevant number.				
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age		
× 45 ± 3.4	× p < 0.01	× males > 30 years of age		

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. See example:

✓ First time appearing: *musculus biceps brachii*
Abbreviated: *m. biceps brachii*



ISSN 1451-7485

Journal Sport Mont (SM) is a print (ISSN 1451-7485) and electronic scientific journal (eISSN 2337-0351) aims to present easy access to the scientific knowledge for sport-conscious individuals using contemporary methods. The purpose is to minimize the problems like the delays in publishing process of the articles or to acquire previous issues by drawing advantage from electronic medium. Hence, it provides:

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- Peer review by expert, practicing researchers;
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Journal SM is published three times a year, in February, June and October of each year. Journal SM publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest.

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Publication date: Winter issue – February 2019
Summer issue – June 2019
Autumn issue – October 2019

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The University of Montenegro is the leading higher education and research institution in Montenegro. It is a public institution, established by the state, operating as a unique legal entity represented by the Rector. It is an integrated university organized on the model of the most European universities. Organizational units are competent for provision of study programmes, scientific-research and artistic work, use of allocated funds and membership in professional associations.

Since its foundation, the University of Montenegro has continuously been conducting reforms in the area of education and research, while since 2003 in line with the trends in EHEA. After adoption of the Bologna Declaration, University of Montenegro organized systematic preparation of documents aligned with it. Already in 2003, the experimental teaching programme started and today, all studies are organised in line with the Bologna principles. During the last two years systematic reforms of the University's study programmes have been conducted in order to harmonize domestic higher education system with European standards and market needs to highest extent.

The University of Montenegro has unique academic, business and development objectives. It comprises 19 faculties and two research institutes. The seat of the UoM is in Podgorica, the capital city, while university units are located in eight Montenegrin towns. The University support services and centers (advisory services, accounting department, international cooperation, career orientation) are located in the Rectorate.

Academic community of University of Montenegro is aware of the importance of its functioning for further development of the state and wider region. It has been so far, and will be in the future, the leader in processes of social and cultural changes, along with the economic development.

In the aspect of attaining its mission, University of Montenegro is oriented towards the priority social needs of the time in which it accomplishes its mission; open for all the students and staff exclusively based on their knowledge and abilities; dedicated to preservation of multicultural and multi-ethnic society in Montenegro; entrepreneurial in stimulating social and economic application of supreme achievements within the scope of its activities.

In 2015/16 there were a total of 1.192 employees at UoM, 845 of which were engaged in teaching. In the same year there were 20.236 students registered at all three cycles of studies.

Internationalization is high on the agenda of UoM priorities, thus it has participated in a number of international projects – over 50 projects funded under the Tempus programme, over 15 Erasmus Mundus Action 2 projects for student mobility, a number of projects under FP7 funding scheme or IPA supported projects, Erasmus + capacity building and International credit mobility projects and other.

For more information about University of Montenegro, please visit our website www.ucg.ac.me or send e-mail to pr.centar@ac.me.





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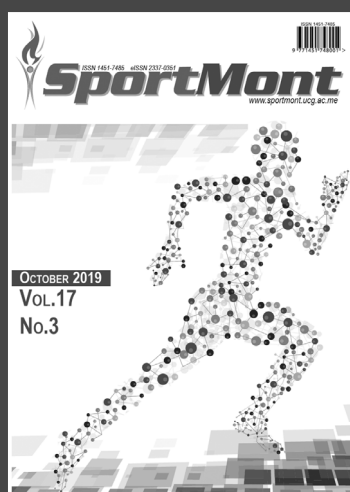
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Volume 17, 2019, 3 issues per year; Print ISSN: 1451-7485, Online ISSN: 2337-0351

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Volume 8, 2019, 2 issues per year; Print ISSN: 1800-8755, Online ISSN: 1800-8763

Montenegrin Journal of Sports Science and Medicine (MJSSM) is published biannually, in September and March of each year. MJSSM publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest. MJSSM covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

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Publication date: Spring issue – March 2019
Autumn issue – September 2019



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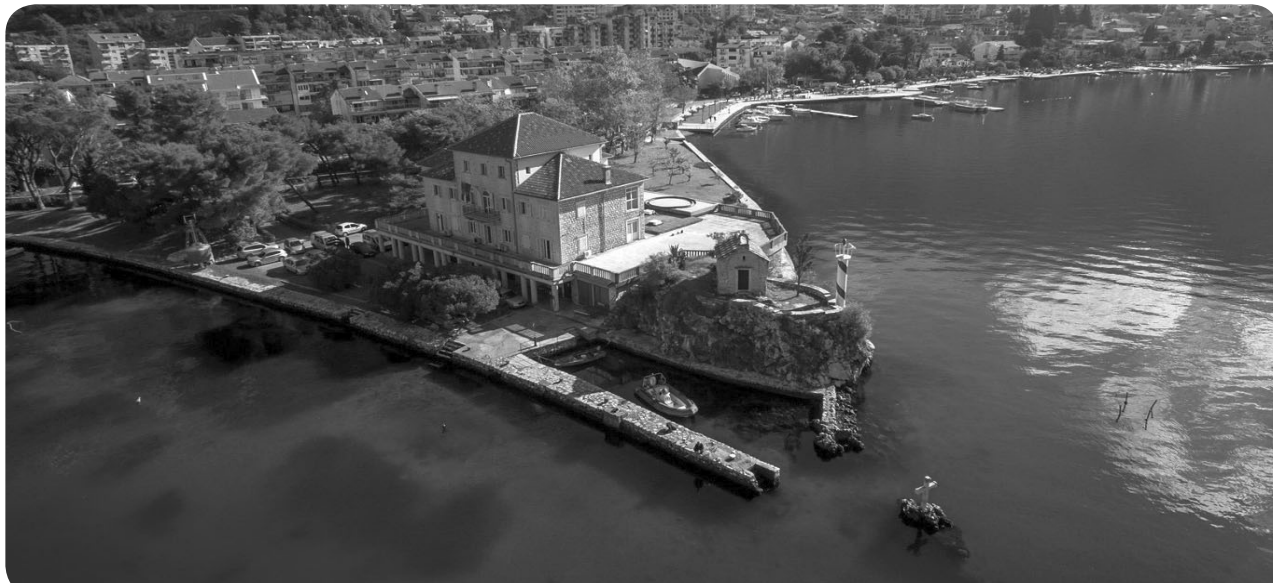
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Znanje i zdravlje!



Univerzitet Crne Gore

UNIVERZITET CRNE GORE INSTITUT ZA BIOLOGIJU MORA



University of Montenegro – Institute for marine biology is located in Kotor, Montenegro. Since its establishment in 1961, the Institute performed comprehensive research of the marine and coastal area, which has its wide impact to the environmental protection, pollution-prevention and practical application. Core competencies of the Institute are focused on research in the fields of marine conservation, ichthyology and marine fisheries, marine chemistry, aquaculture, plankton research, neuro and eco-physiology. The main research area is investigating and protection of Adriatic sea with special interest of South Adriatic area. Institute for marine biology have a wide range of international cooperation with Marine research institutions and Universities all over Mediterranean area through a numerous EU funded scientific projects.

All over the year Institute is looking to hire young students from the field of general biology, marine biology, marine chemistry, molecular biology or similar disciplines on voluntary basis to work with us. We need opportunity for international internship or MSc or PhD thesis that could be performed on Institute in our 5 different labs: Fisheries and ichthyology, Aquaculture, Marine chemistry, Plankton and sea water quality and Benthos and marine conservation.

Every year Institute organizes several summer schools and workshops for interested students, MSc and PhD candidates. From 01-05 July 2019 we will organize Summer school "Blue Growth: emerging technologies, trends and opportunities" in frame of InnoBlueGrowth Project which is financed by Interreg Med programme. Through the specific theme courses, workshops and working labs offered – covering different areas of the blue economy – the Summer School aims at encouraging young people involvement in blue economy sectors by offering high-quality technical knowledge and fostering their entrepreneurial spirit. The Summer School will facilitate fruitful exchanges and a stronger understanding among a variety of actors coming from different Mediterranean countries with diverse profiles, including representatives from the academia, the public and private sectors, but also potential funders and investors. These activities will count on specific team building activities for participants as well to reinforce interpersonal skills and foster cohesion among blue academia and sectors.

If you are interested, apply on the following link: <https://www.ucg.ac.me/objava/blog/1221/objava/45392-ljetnja-skola-plavi-rast-nove-tehnologije-trendovi-i-mogucnosti>

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Faculty of Economics

University of Montenegro

The Faculty of Economics celebrated its 57th anniversary this year, and it is the oldest higher education institution in the country. Since its establishment, 8,630 students graduated at our Faculty.

Today, Faculty of Economics is a largely interdisciplinary institution, characterized by expressed dynamism in its work. Employees at the Faculty are dedicated to constant improvements and enhancements, all in accordance with the needs brought by the changes.

We provide our students with the best theoretical and practical knowledge, enabling them to develop critical spirit in approaching economic phenomena and solving concrete problems in daily work. From September 2017, at the Faculty, the new generation will start a 3 + 2 + 3 study, which will improve the quality of studying.



Development of Faculty of Economics in the coming period will follow the vision of development of the University of Montenegro, pursuing full achievement of its mission

Comprehensive literature, contemporary authors and works have always been imperative in creation of new academic directions at Faculty of Economics, which will form the basis of our future.

Faculty and its employees are dedicated to developing interest in strengthening the entrepreneurial initiative, creative and interdisciplinary approach among young people, using modern teaching and research methods. In this regard, the Faculty has modern textbooks and adequate IT technology, which supports the objectives set.





UNIVERSITY OF MONTENEGRO FACULTY OF MECHANICAL ENGINEERING Podgorica

www.ucg.ac.me/mf

Mechanical engineering studies in Montenegro started during the school year 1970/71. On April 15th, within the Technical Faculty, the Department of Mechanical Engineering was formed. The Department of Mechanical Engineering of the Technical Faculty was transformed in 1978 into the Faculty of Mechanical Engineering, within the University "Veljko Vlahović". Since 1992 the Faculty of Mechanical Engineering is an autonomous University unit of the University of Montenegro. It is situated in Podgorica.

The University of Montenegro is the only state university in the country, and the Faculty of Mechanical Engineering is the only faculty in Montenegro from the field of mechanical engineering.

Activities of the Faculty of Mechanical Engineering can be divided into three fields: teaching, scientific-research work and professional work.

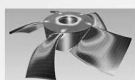
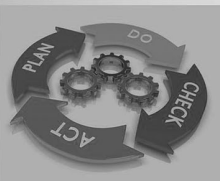
Two study programmes were accredited within the Faculty of Mechanical Engineering:

- Academic study programme MECHANICAL ENGINEERING
- Academic study programme ROAD TRAFFIC

The study programmes are realised according to the Bologna system of studies in accordance to the formula 3+2+3.

On the study program Mechanical Engineering it is possible to study next modules:

- Mechanical Engineering – Production
- Applied Mechanics and Construction
- Energetics
- Energy Efficiency
- Mechatronics
- Quality



At the Faculty of Mechanical Engineering, as organisational units, there are centres and laboratories through which scientific-research and professional work is done:

- Centre for Energetics
- Centre for Vehicles
- Centre for Quality
- Centre for Construction Mechanics
- Centre for Traffic and Mechanical Engineering Expertise
- Centre for transport machines and metal constructions
- 3D Centre
- Didactic Centre – Centre for Automation and Mechanomics training
- European Information and Innovation Centre
- Cooperation Training Centre
- Laboratory for Metal Testing
- Laboratory for Turbulent Flow Studies
- Laboratory for Vehicle Testing
- Laboratory for Attesting of Devices on the Technical Examination Line



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ISSN 2536-569X
COBISS.CG-ID 33826832

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